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Lin et al.

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(54) **OPTICAL IMAGING LENS ASSEMBLY,
IMAGE CAPTURING DEVICE AND
ELECTRONIC DEVICE**

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Taichung (TW)

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G02B 9/62 (2006.01)

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(2013.01); **G02B 27/0025** (2013.01)

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G02B 27/0025; **G02B 9/60**

USPC 359/713-714, 756-757, 759-760
See application file for complete search history.

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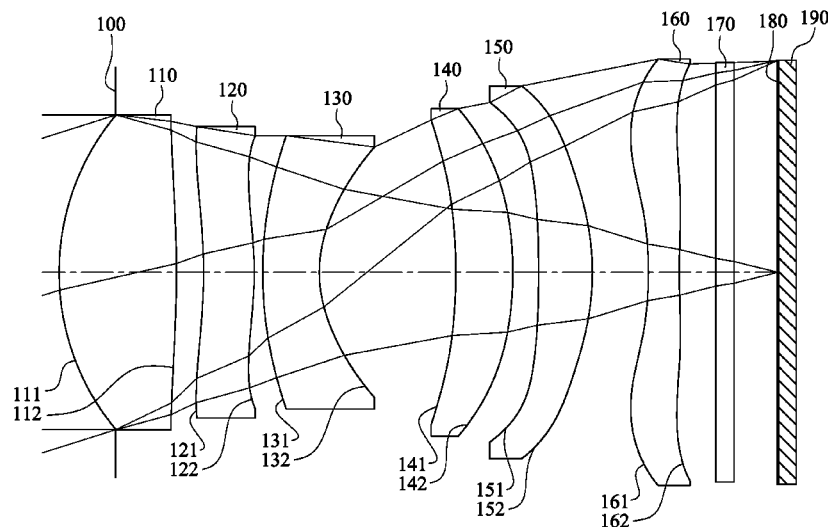
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Rodack, LLP

(57) **ABSTRACT**

An optical imaging lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element and a sixth lens element. The first lens element has positive refractive power. The second lens element has positive refractive power. The third lens element has negative refractive power. The fourth lens element has an object-side surface and an image-side surface being aspheric. The fifth lens element has an object-side surface and an image-side surface being aspheric. The sixth lens element has an object-side surface and an image-side surface being aspheric, wherein at least one of the object-side surface and the image-side surface of the sixth lens element includes at least one inflection point.

23 Claims, 29 Drawing Sheets



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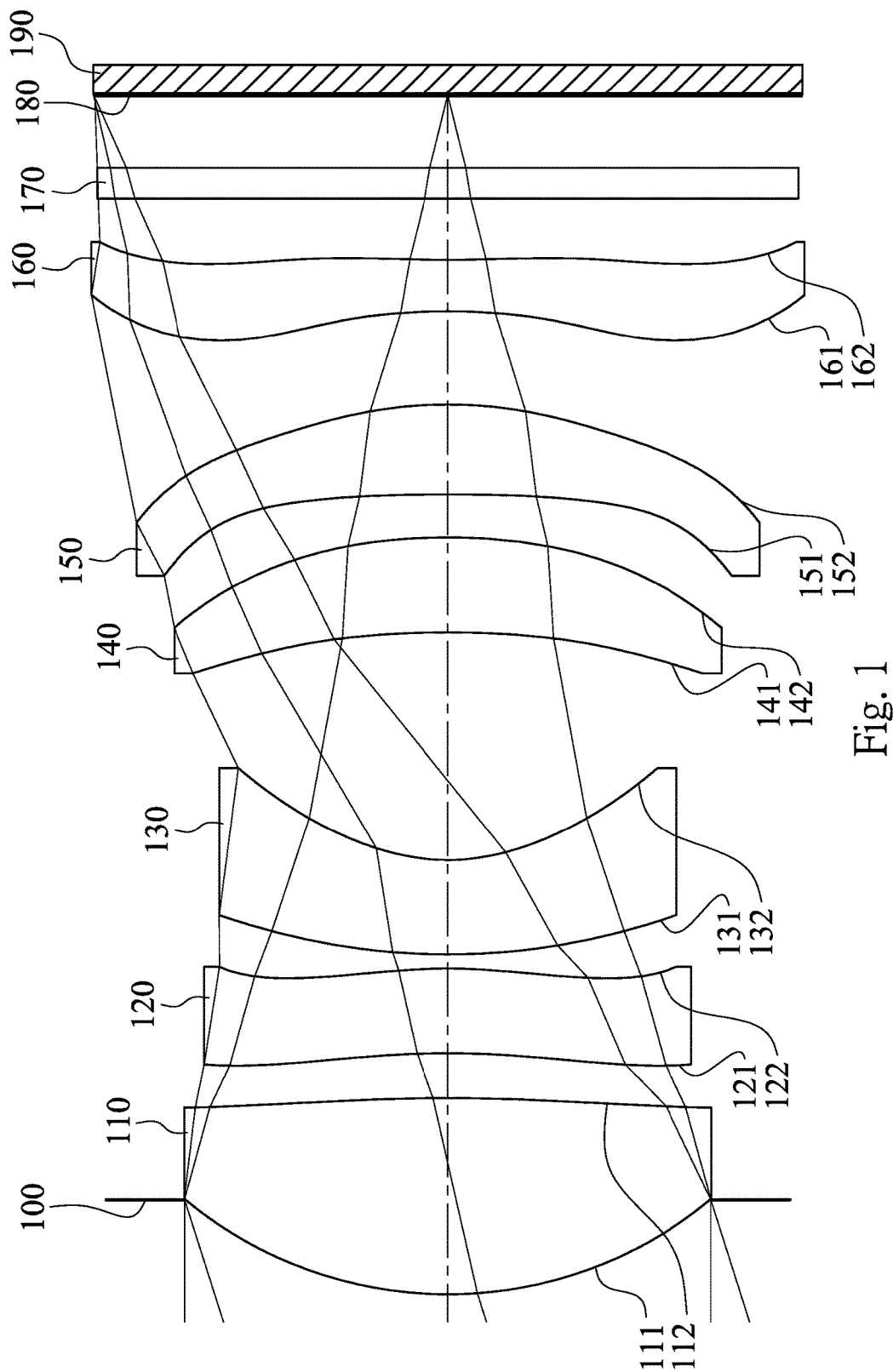


Fig. 1

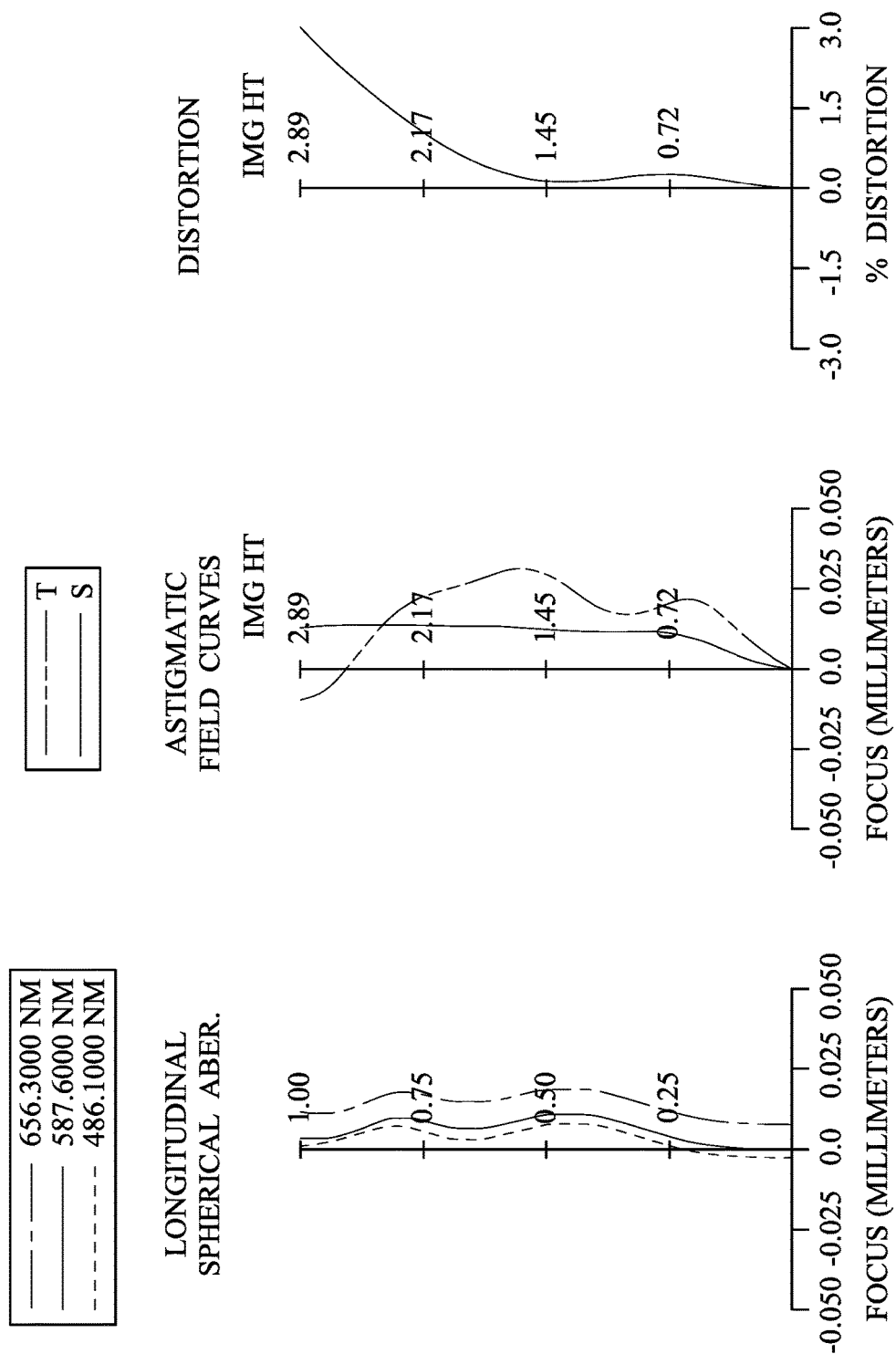
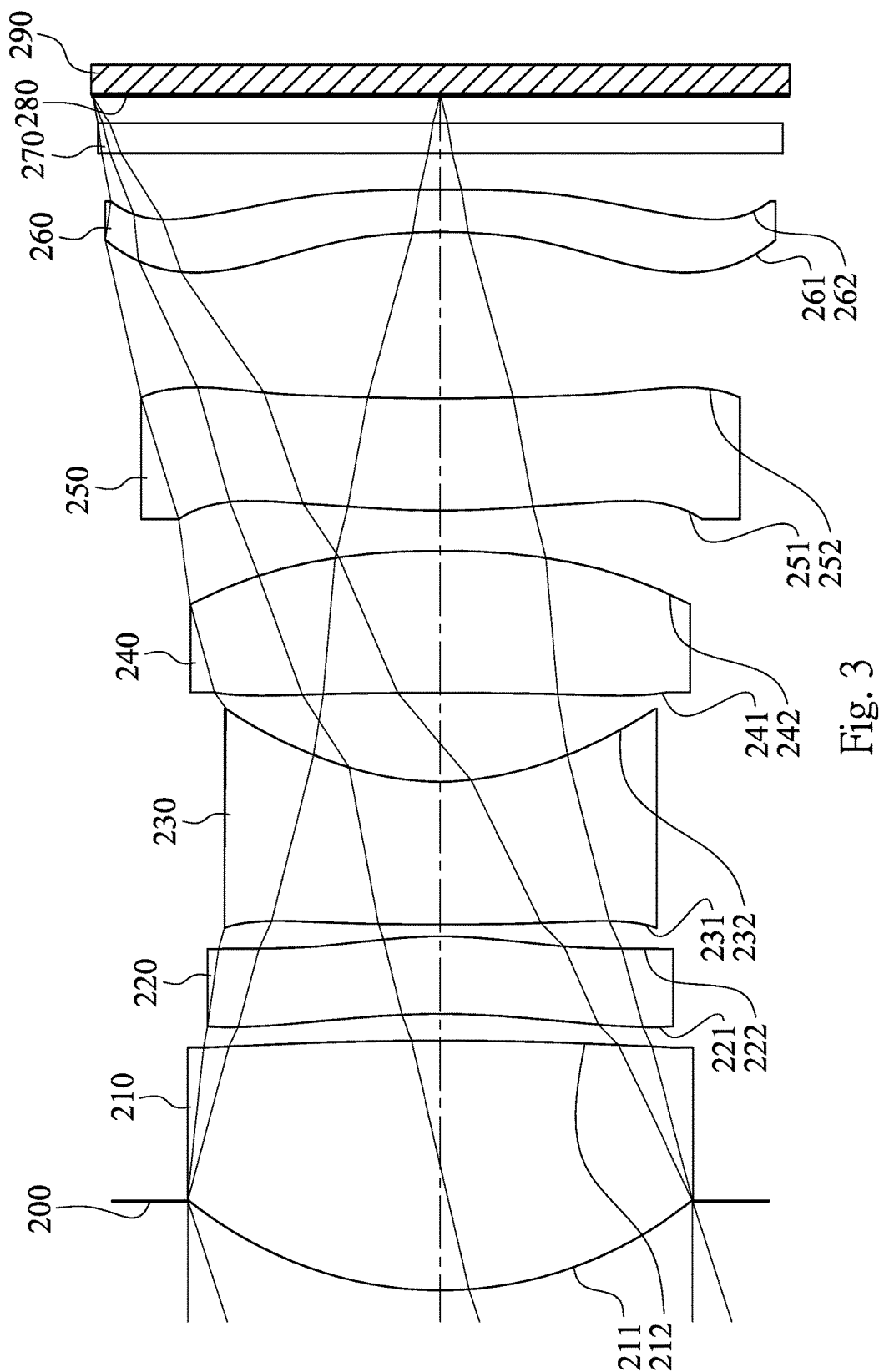


Fig. 2



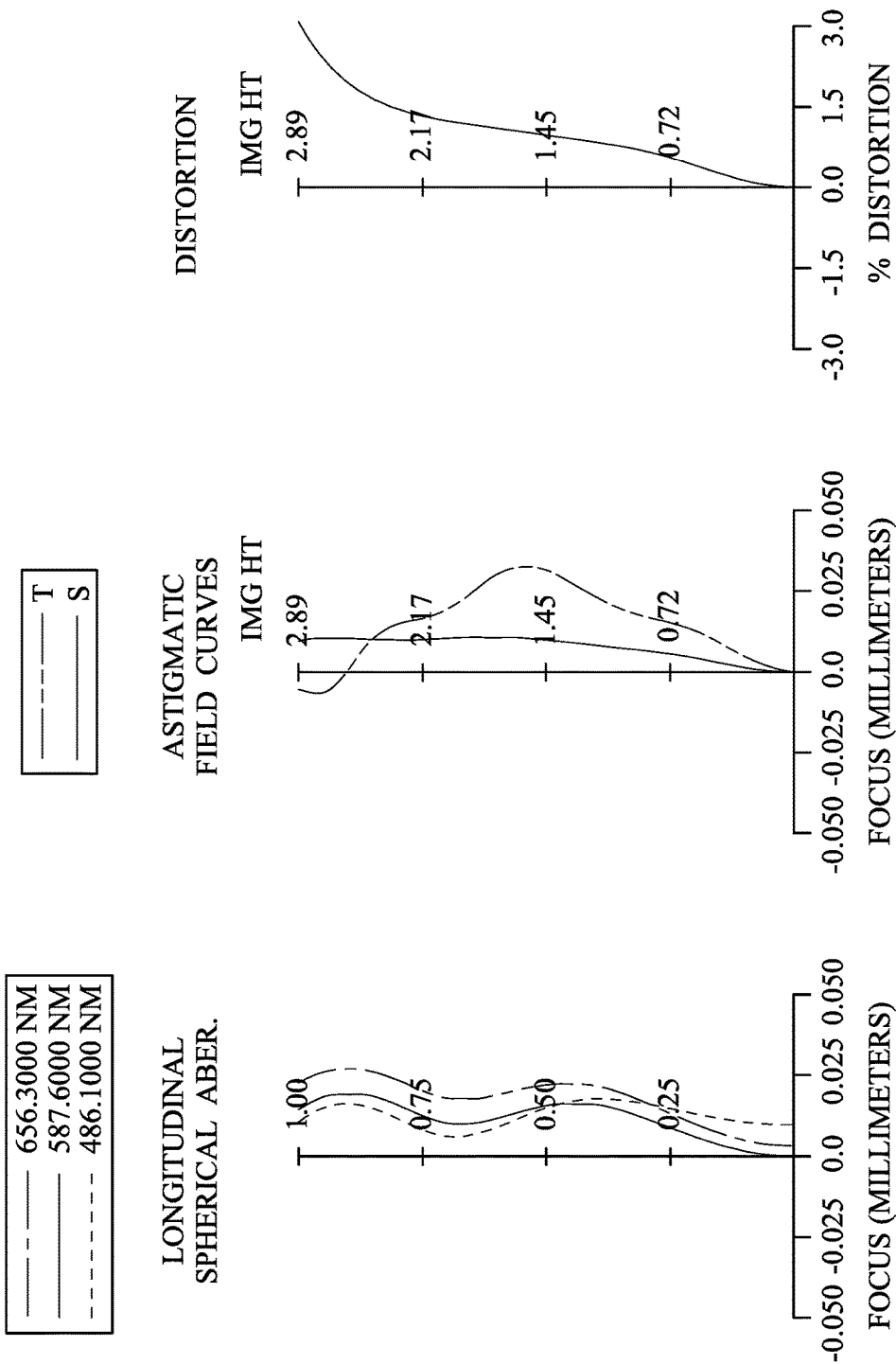


Fig. 4

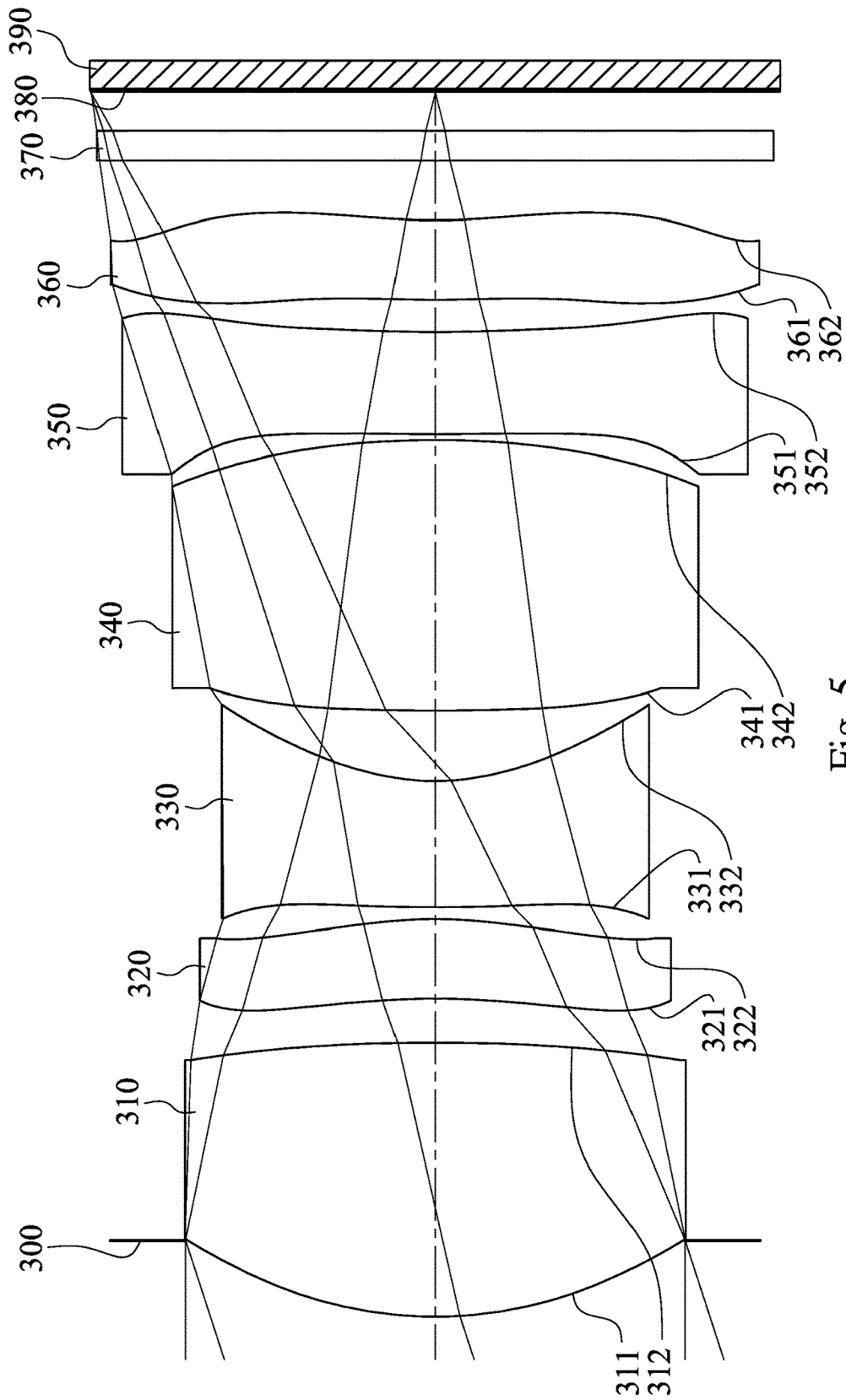


Fig. 5

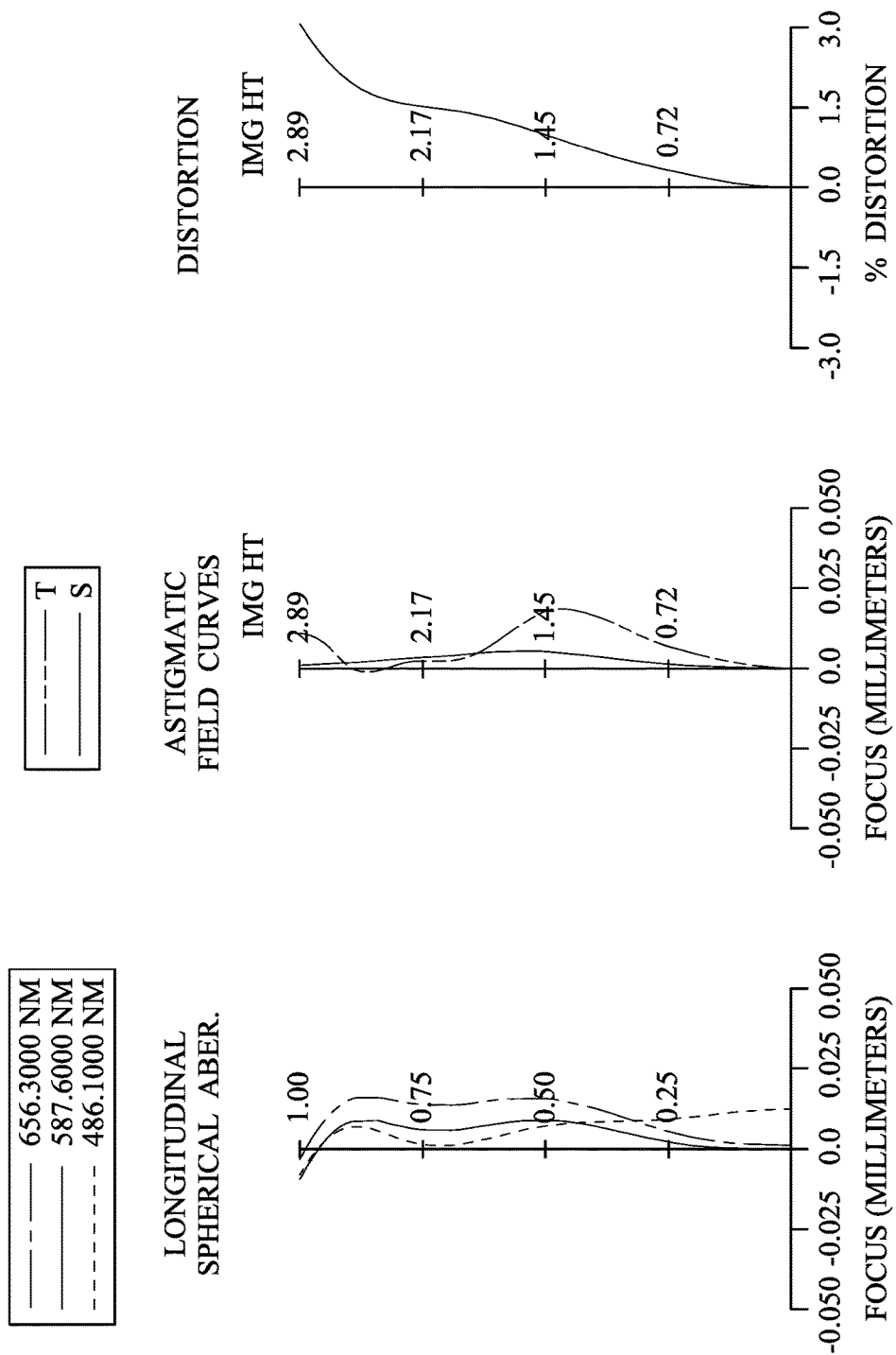


Fig. 6

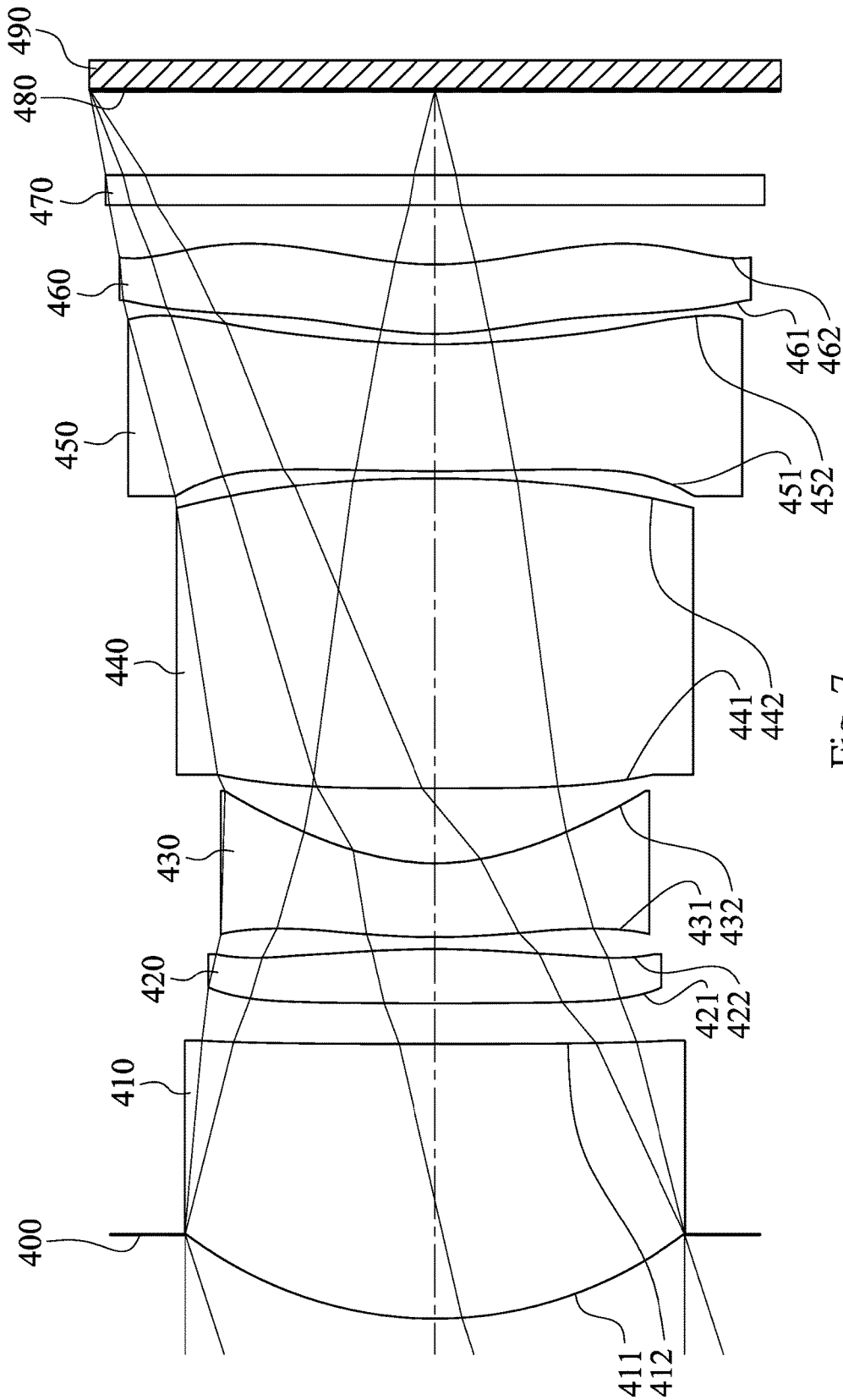


Fig. 7

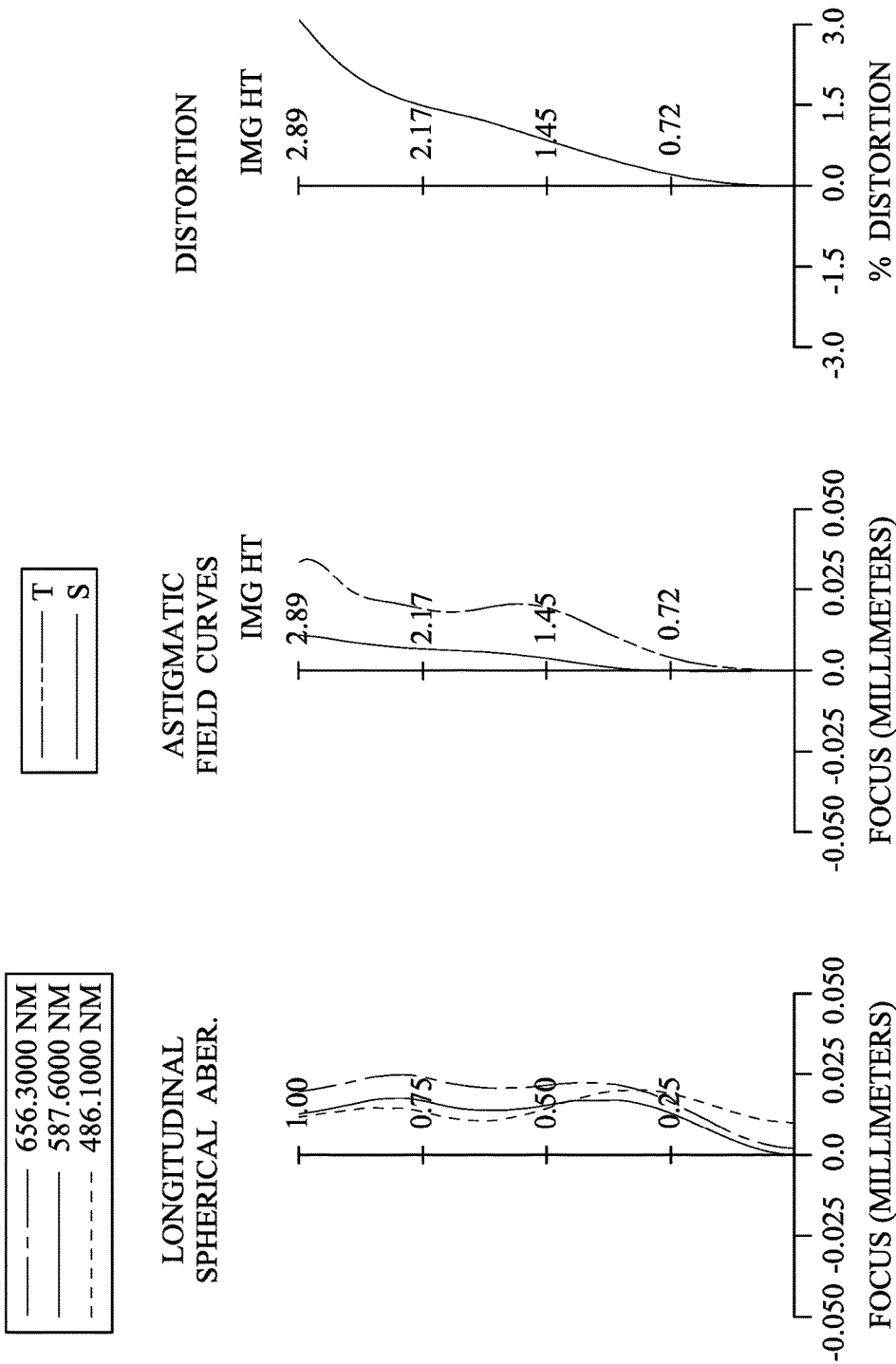


Fig. 8

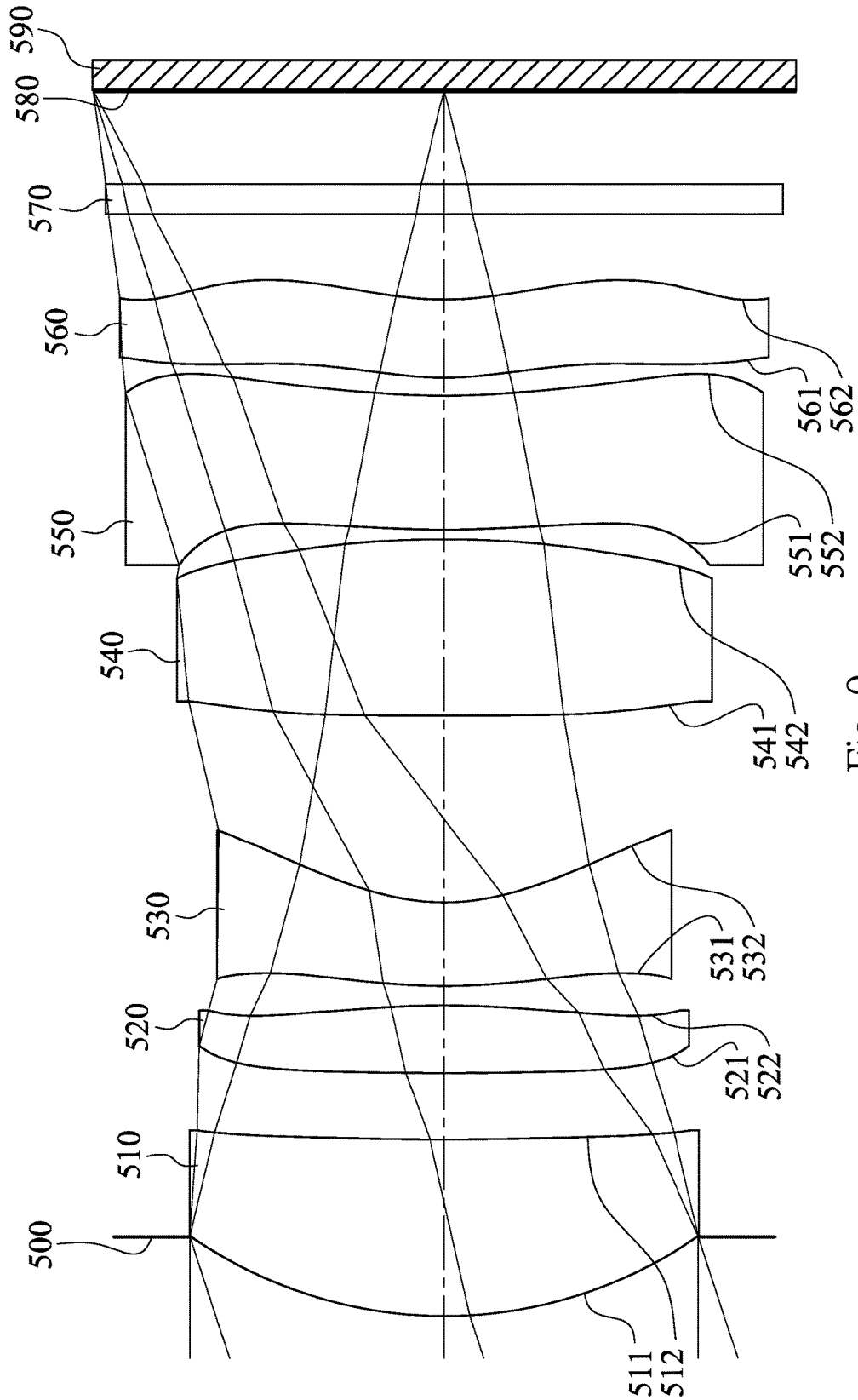


Fig. 9

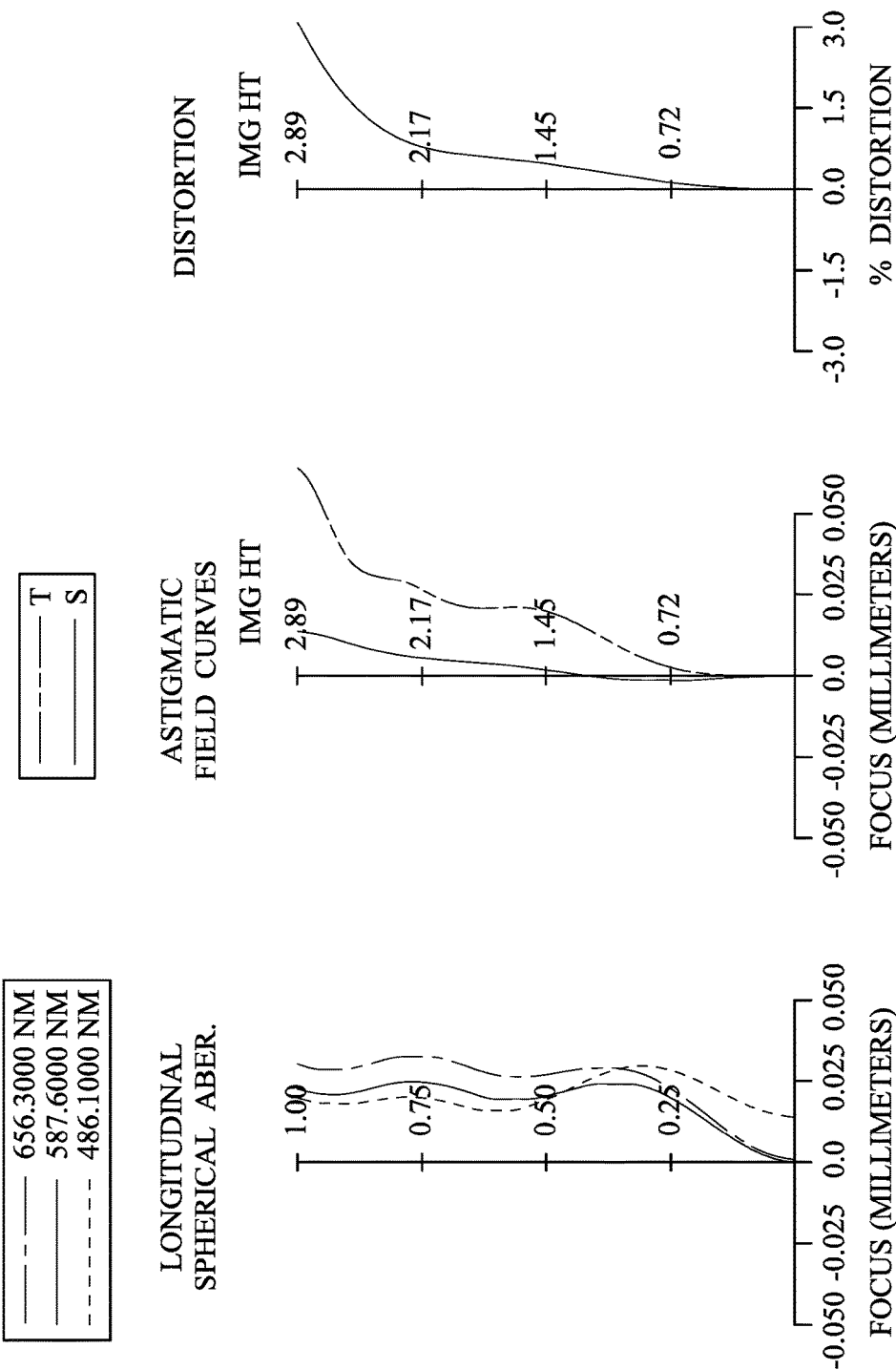


Fig. 10

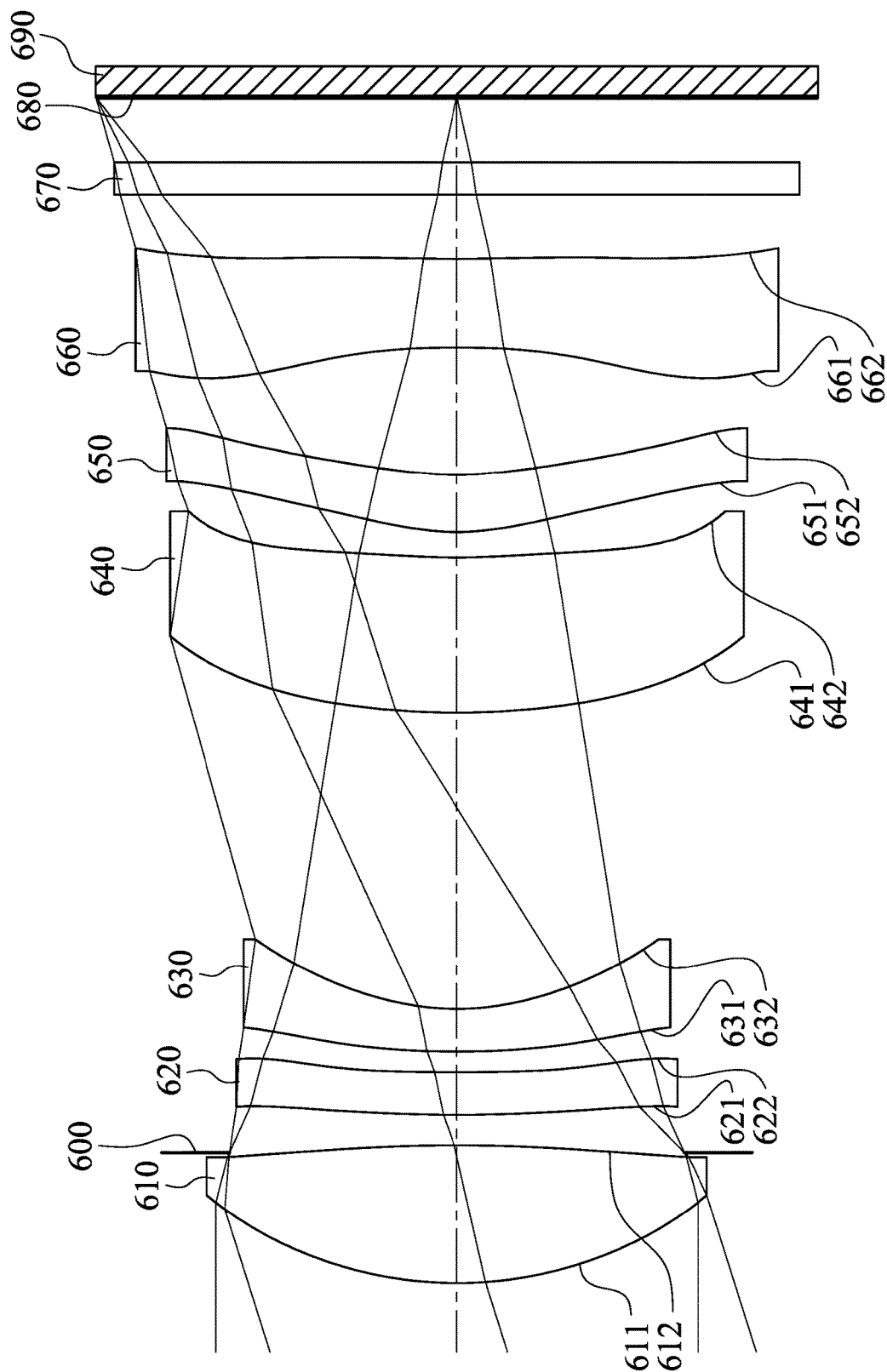


Fig. 11

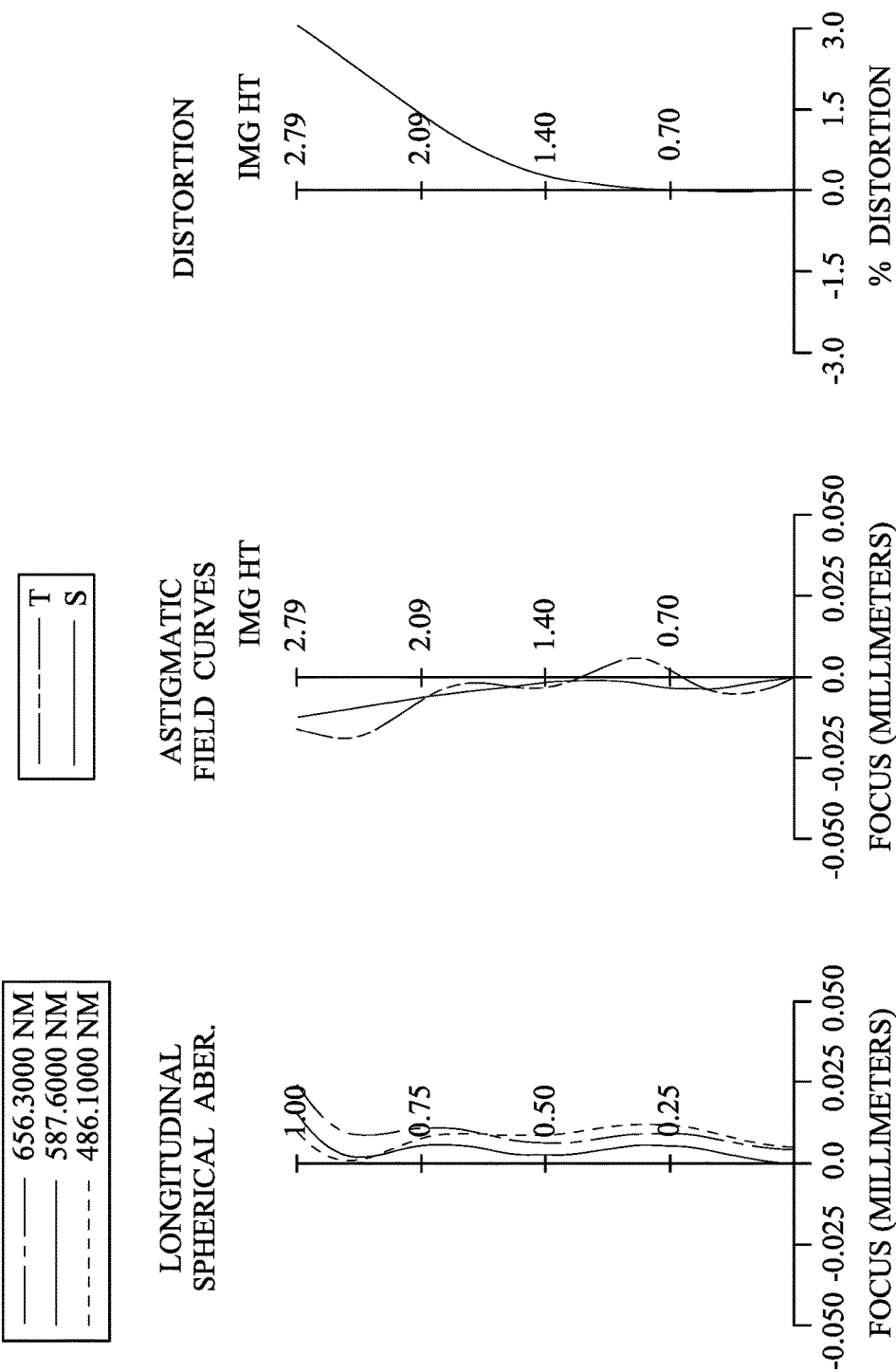


Fig. 12

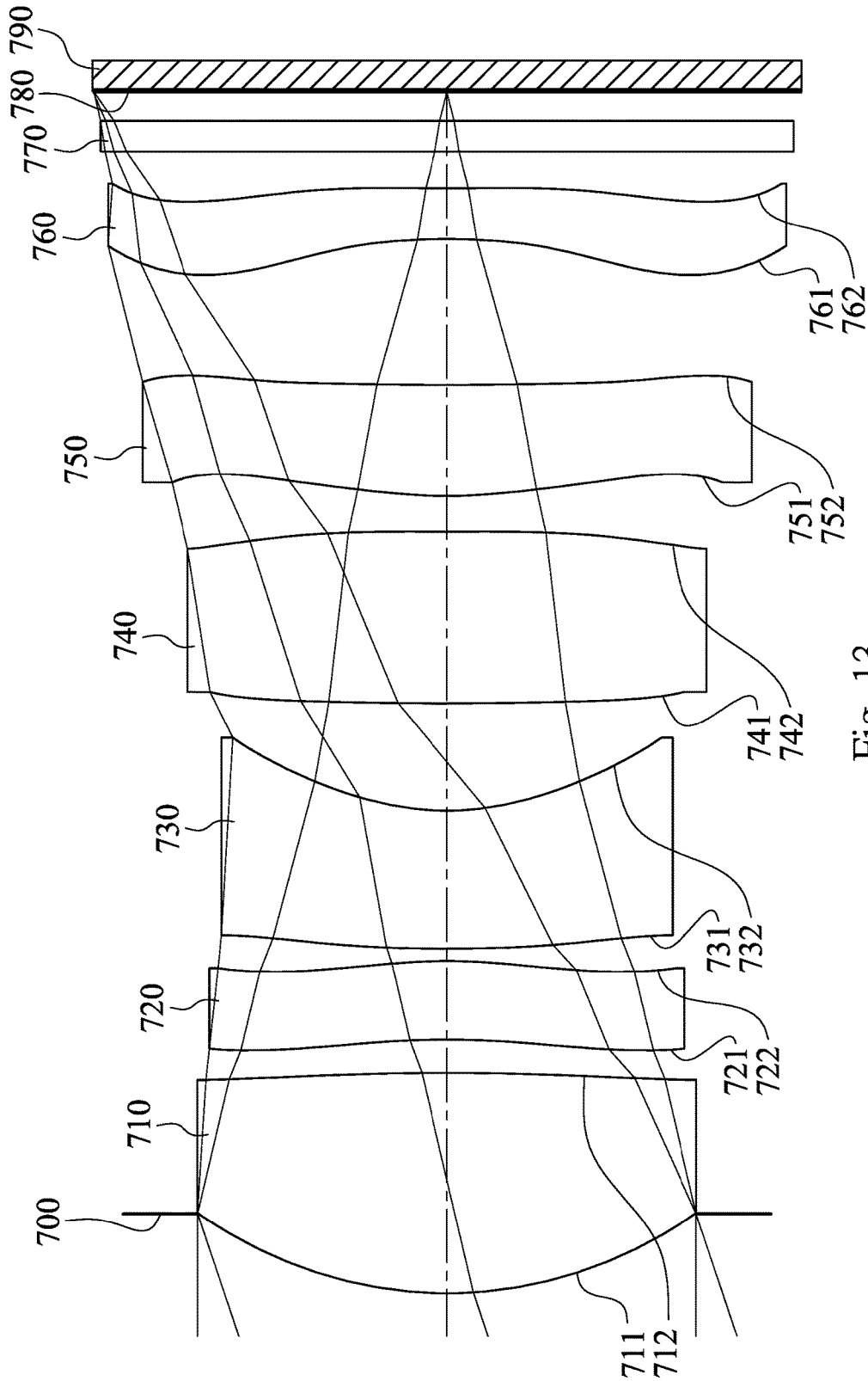
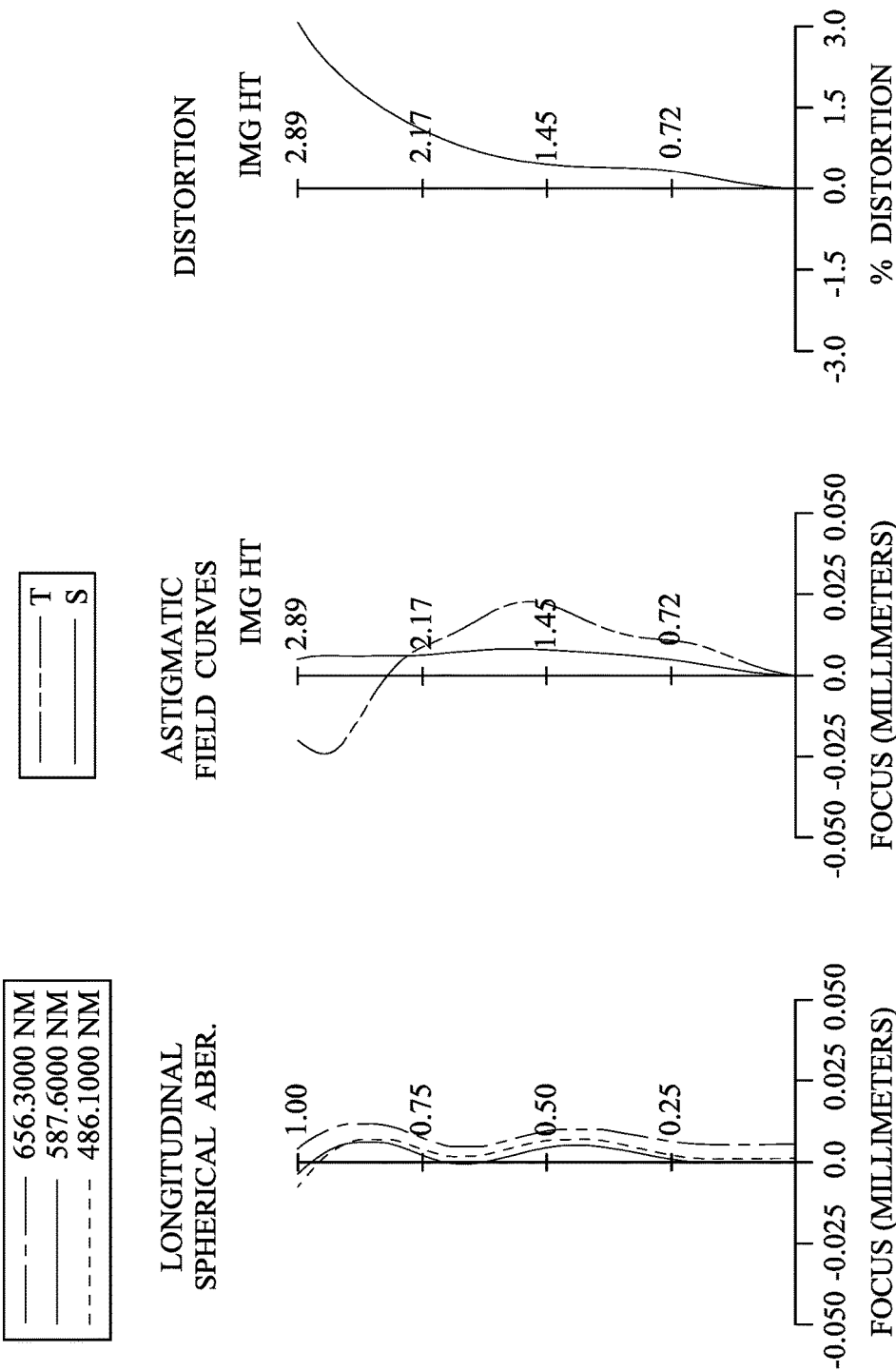


Fig. 13



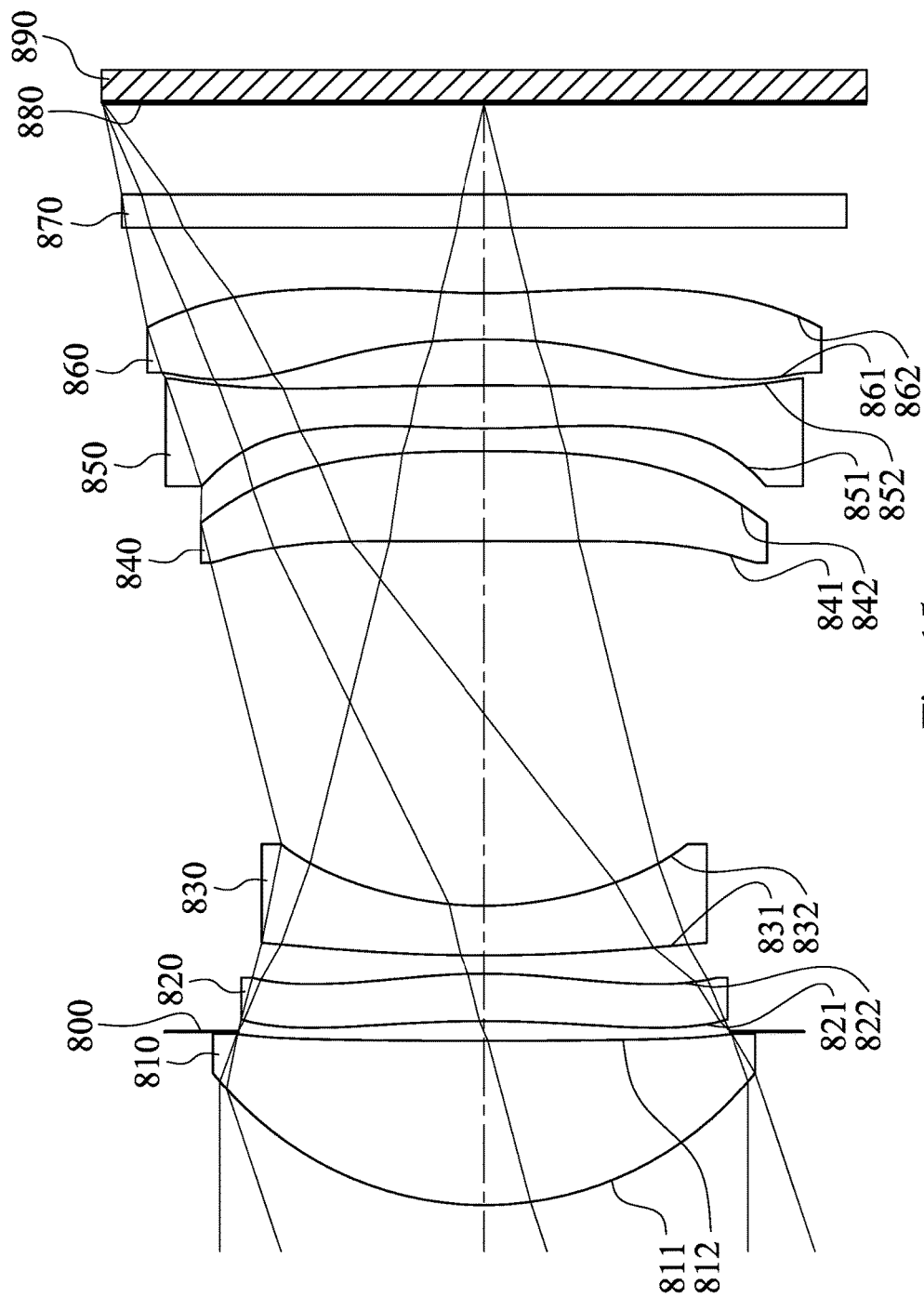


Fig. 15

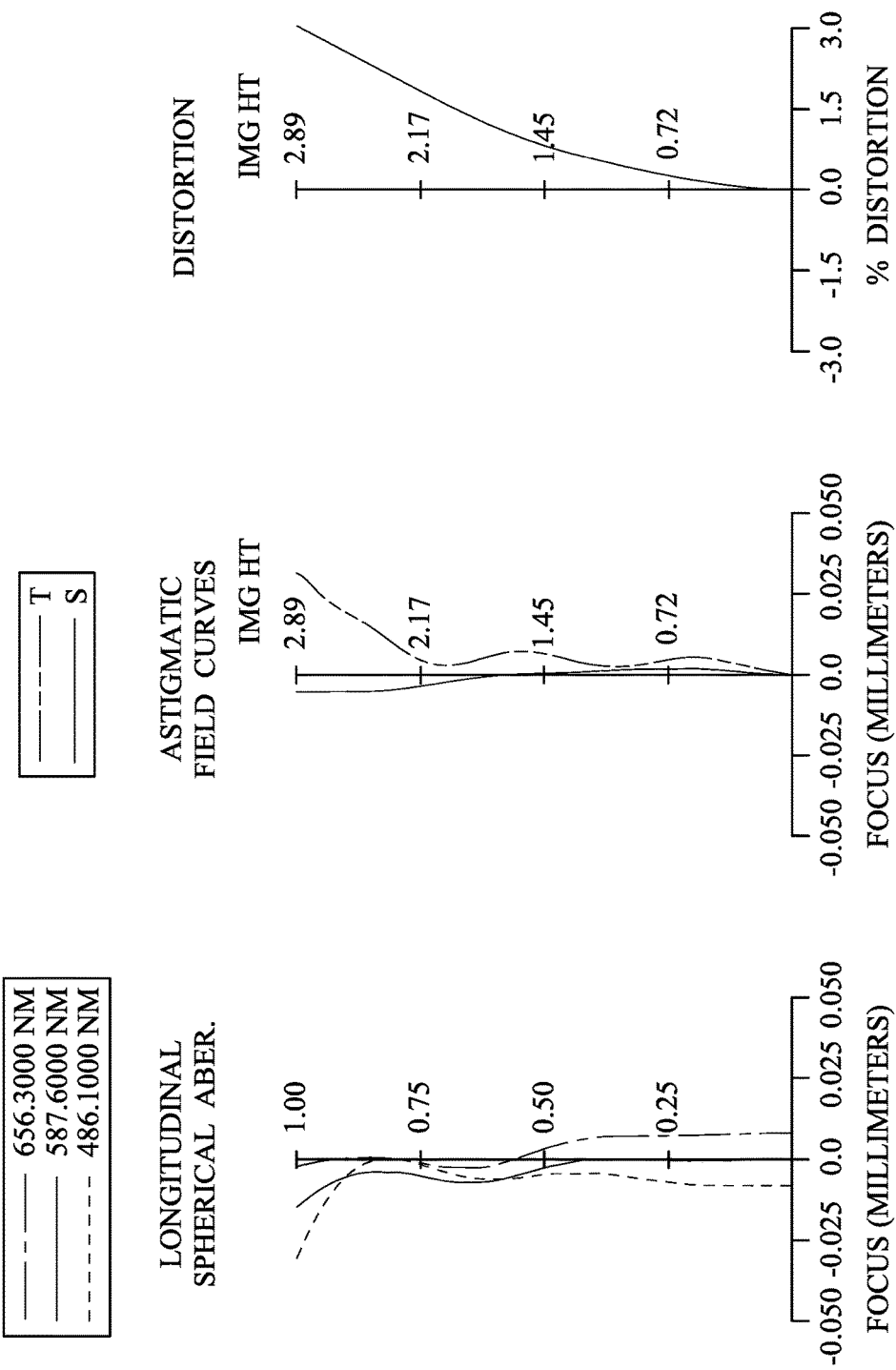


Fig. 16

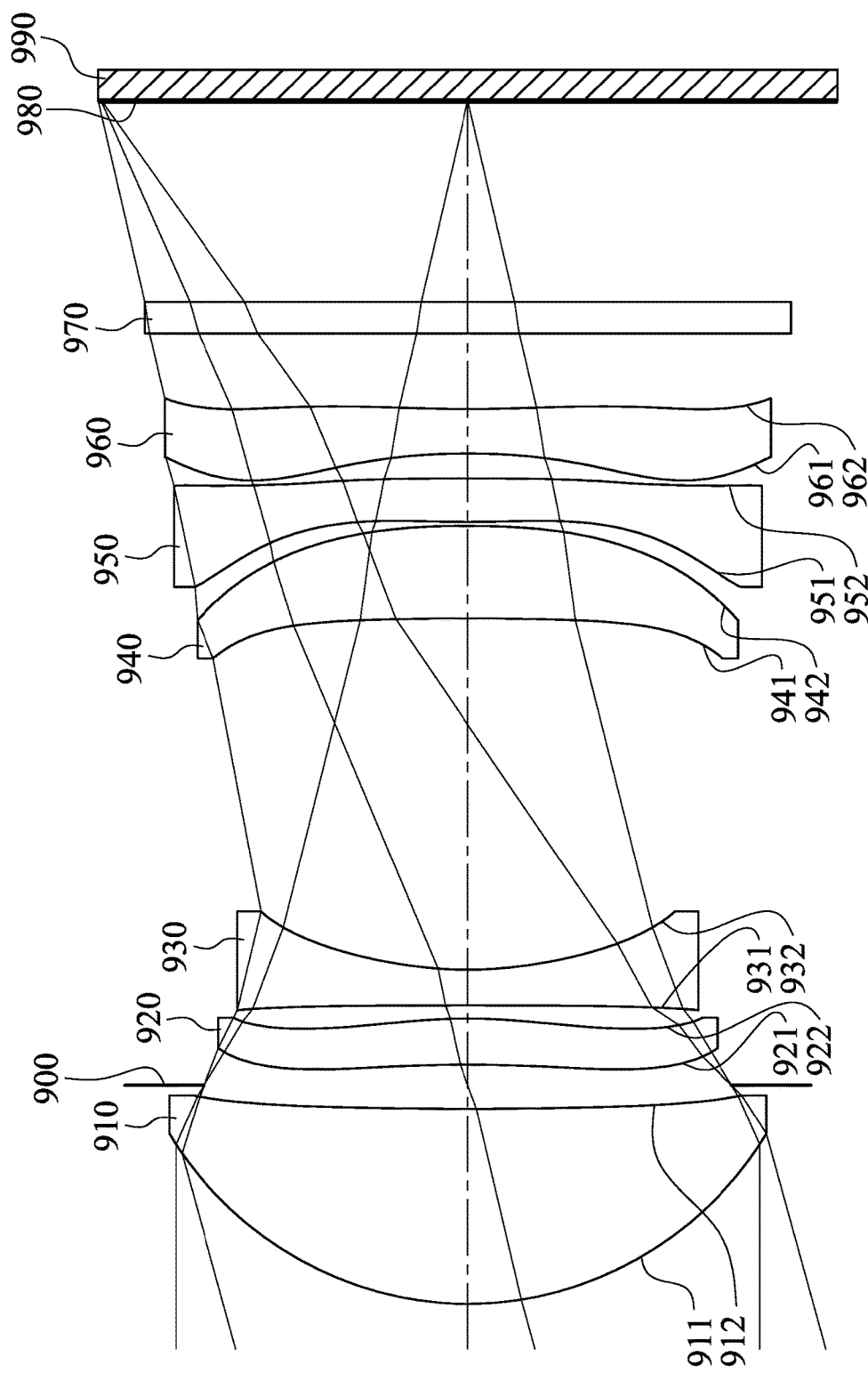


Fig. 17

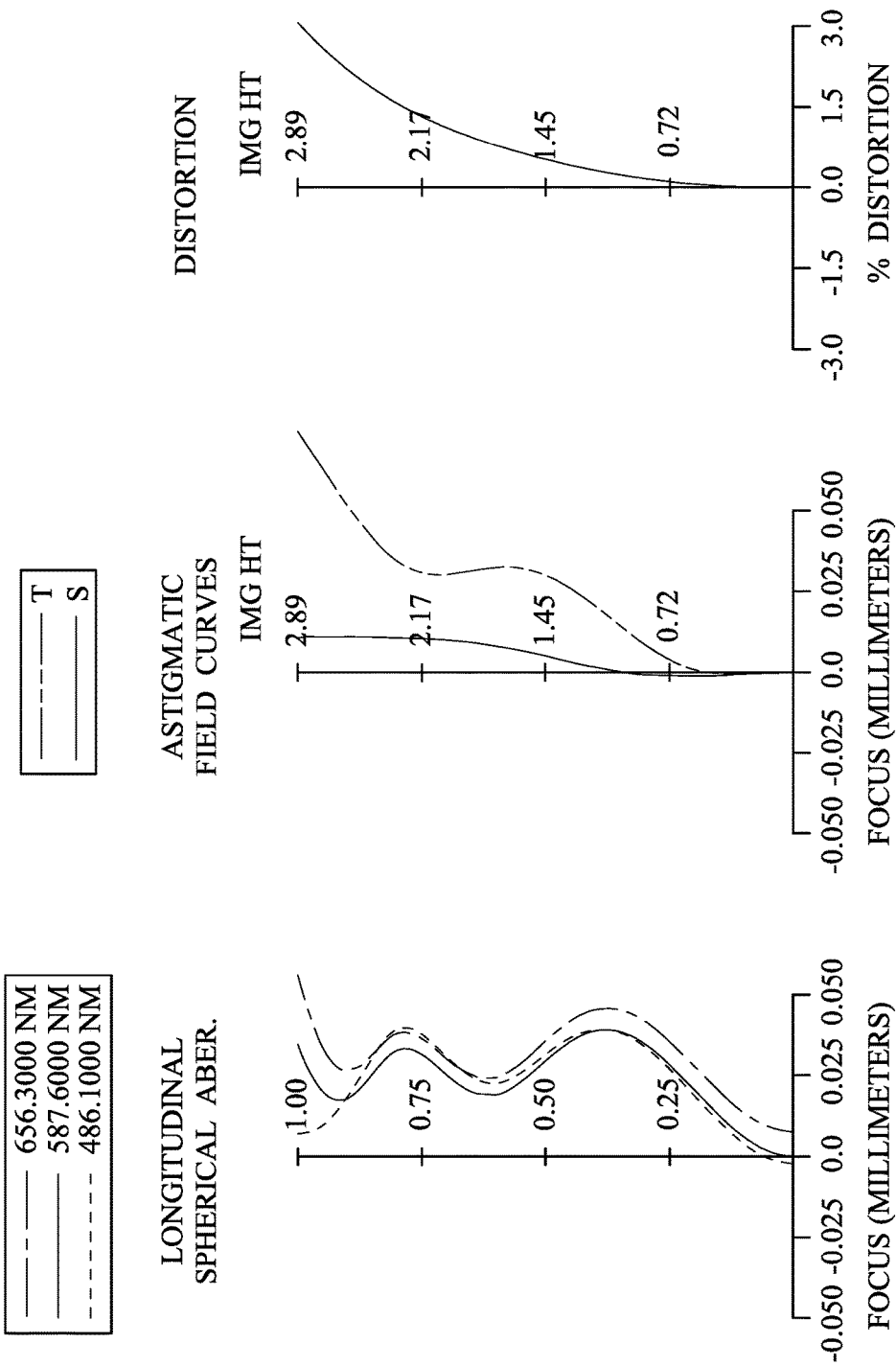


Fig. 18

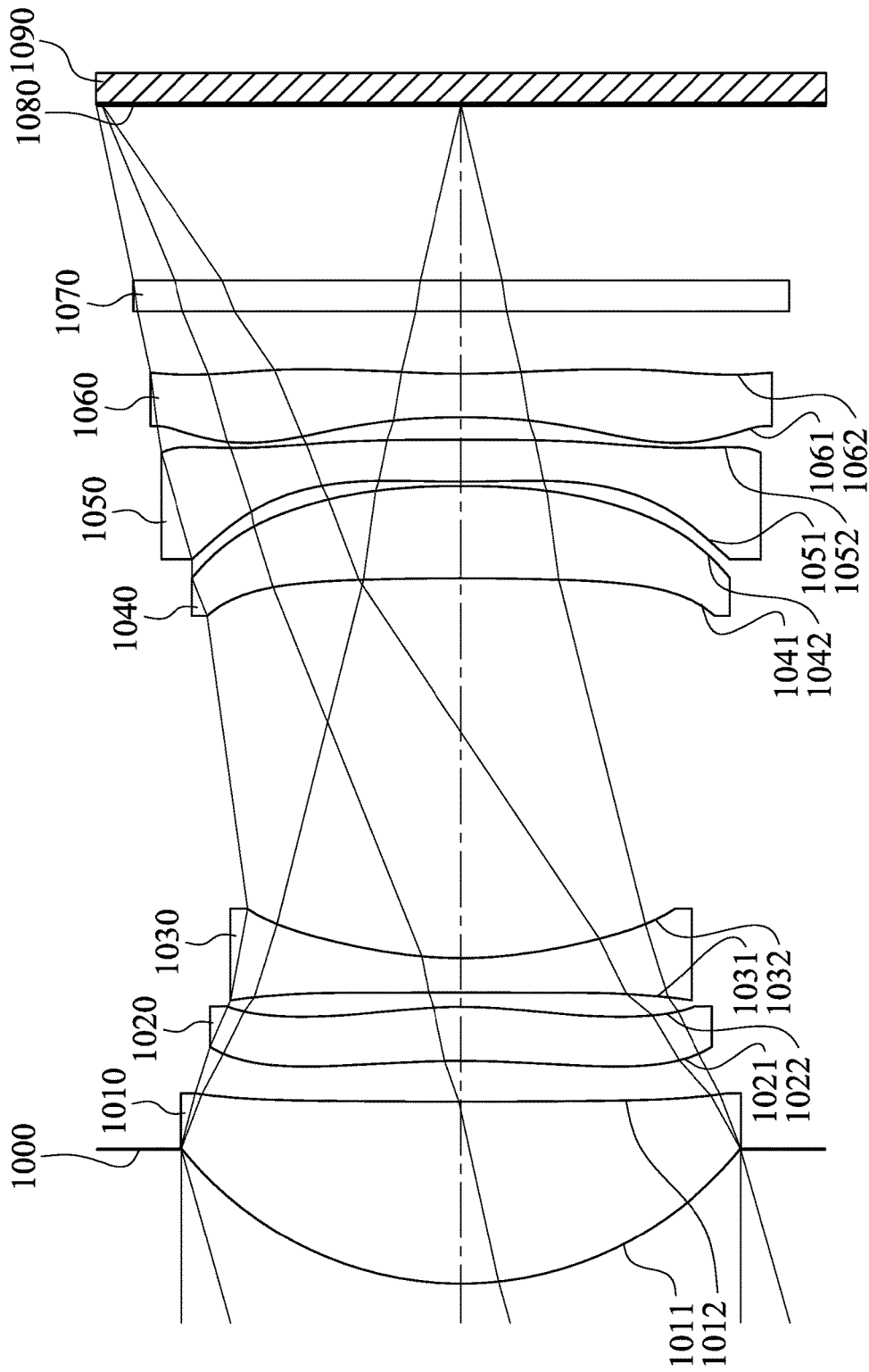


Fig. 19

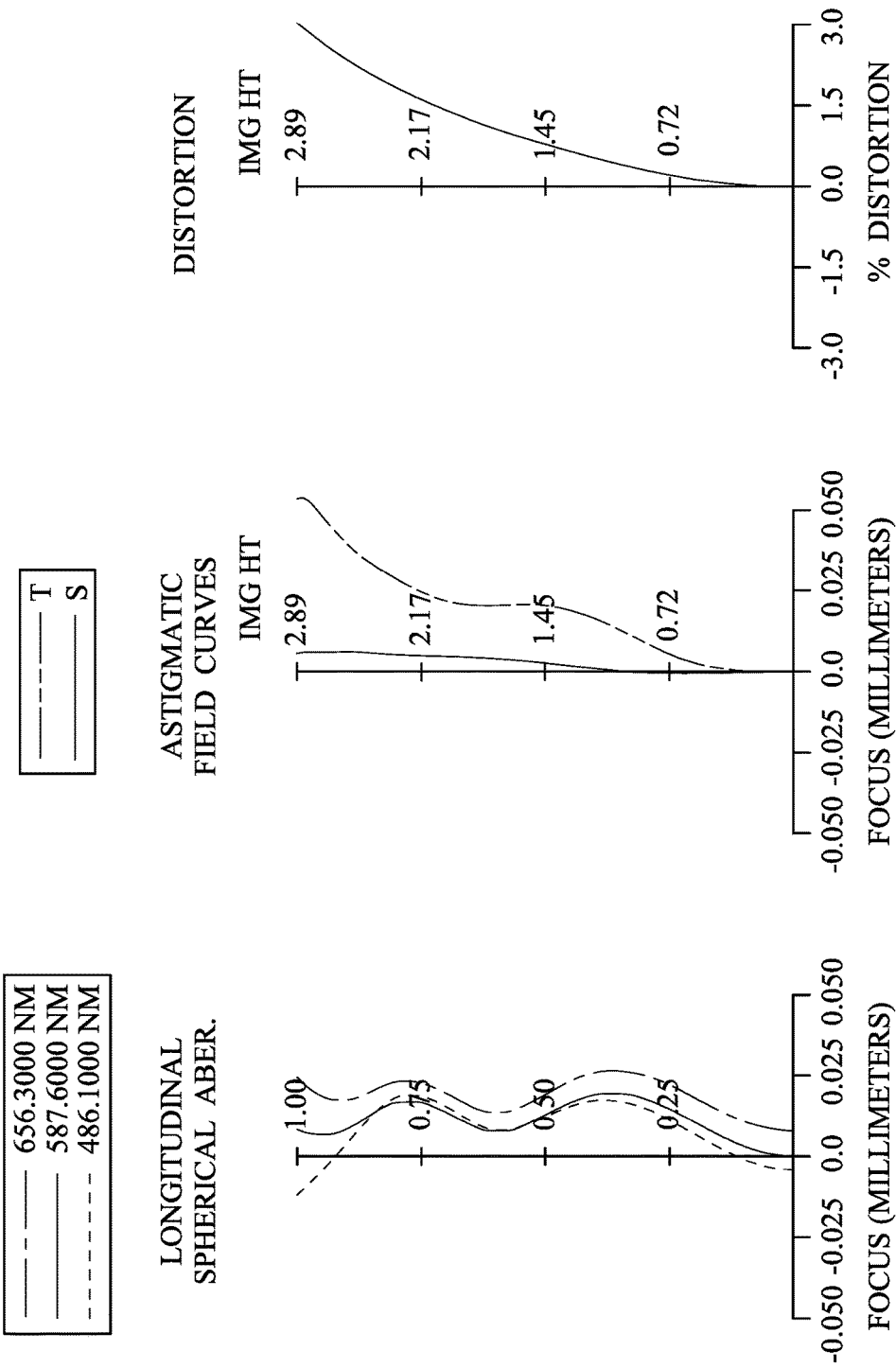
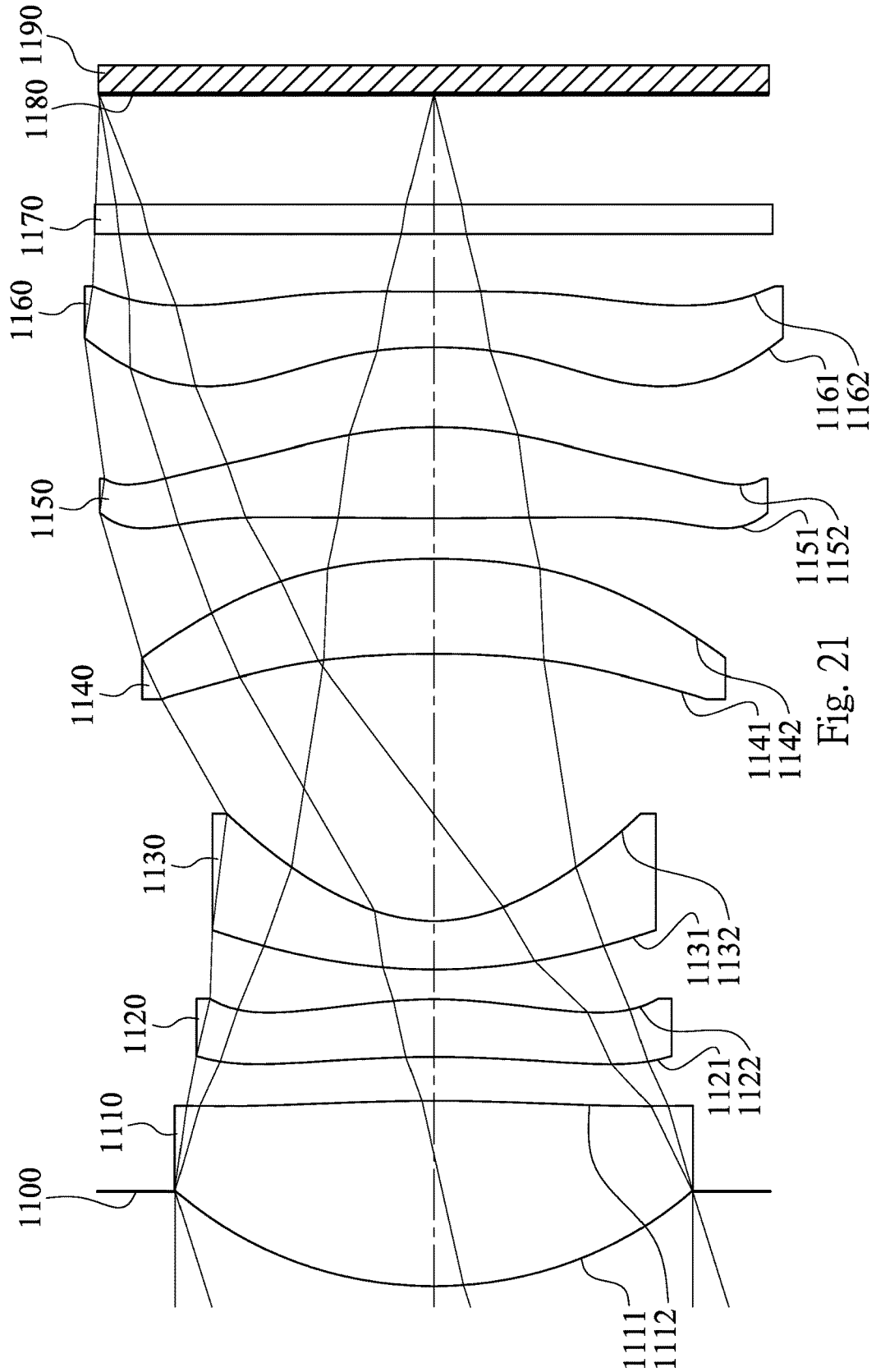
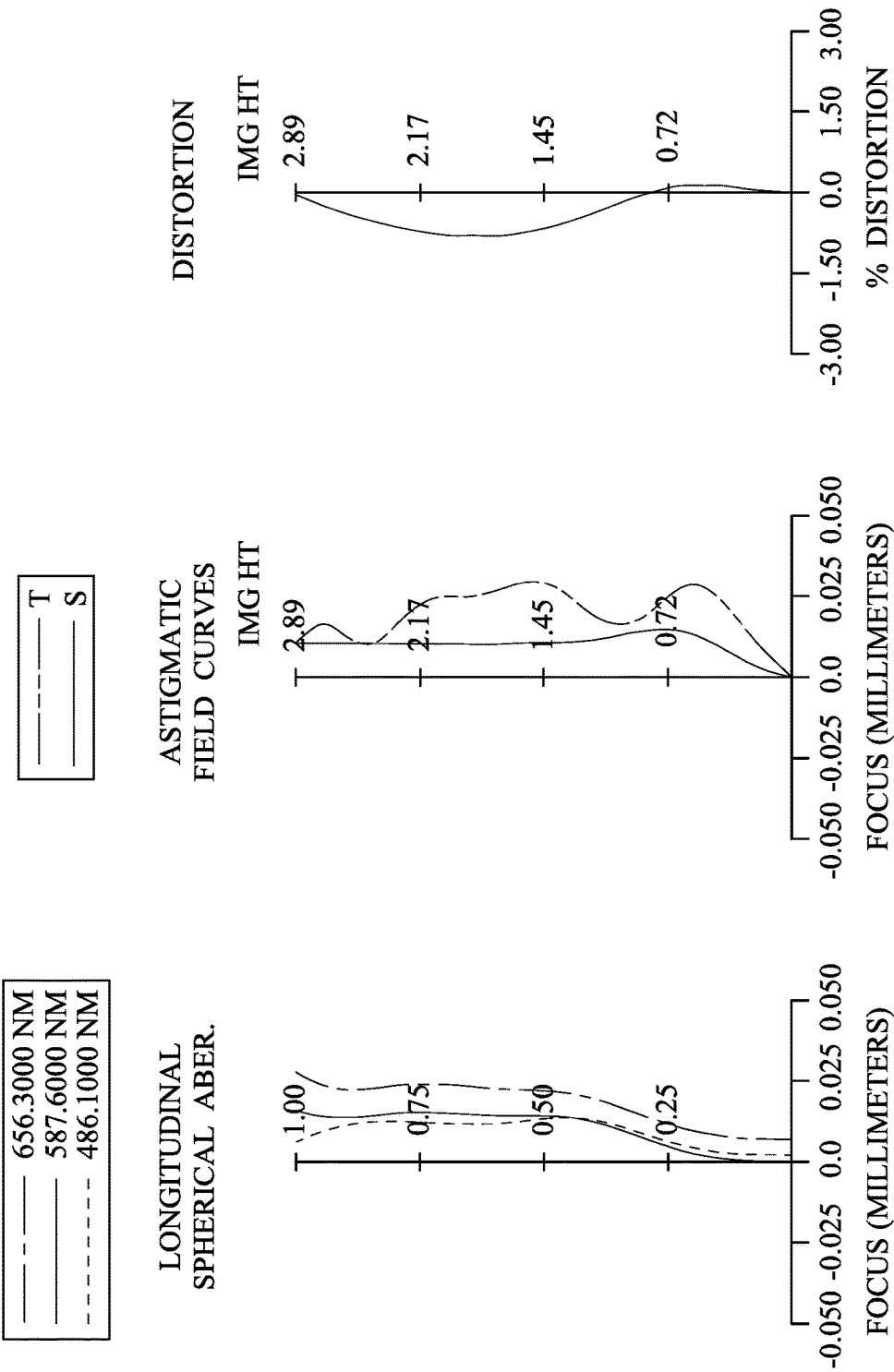


Fig. 20





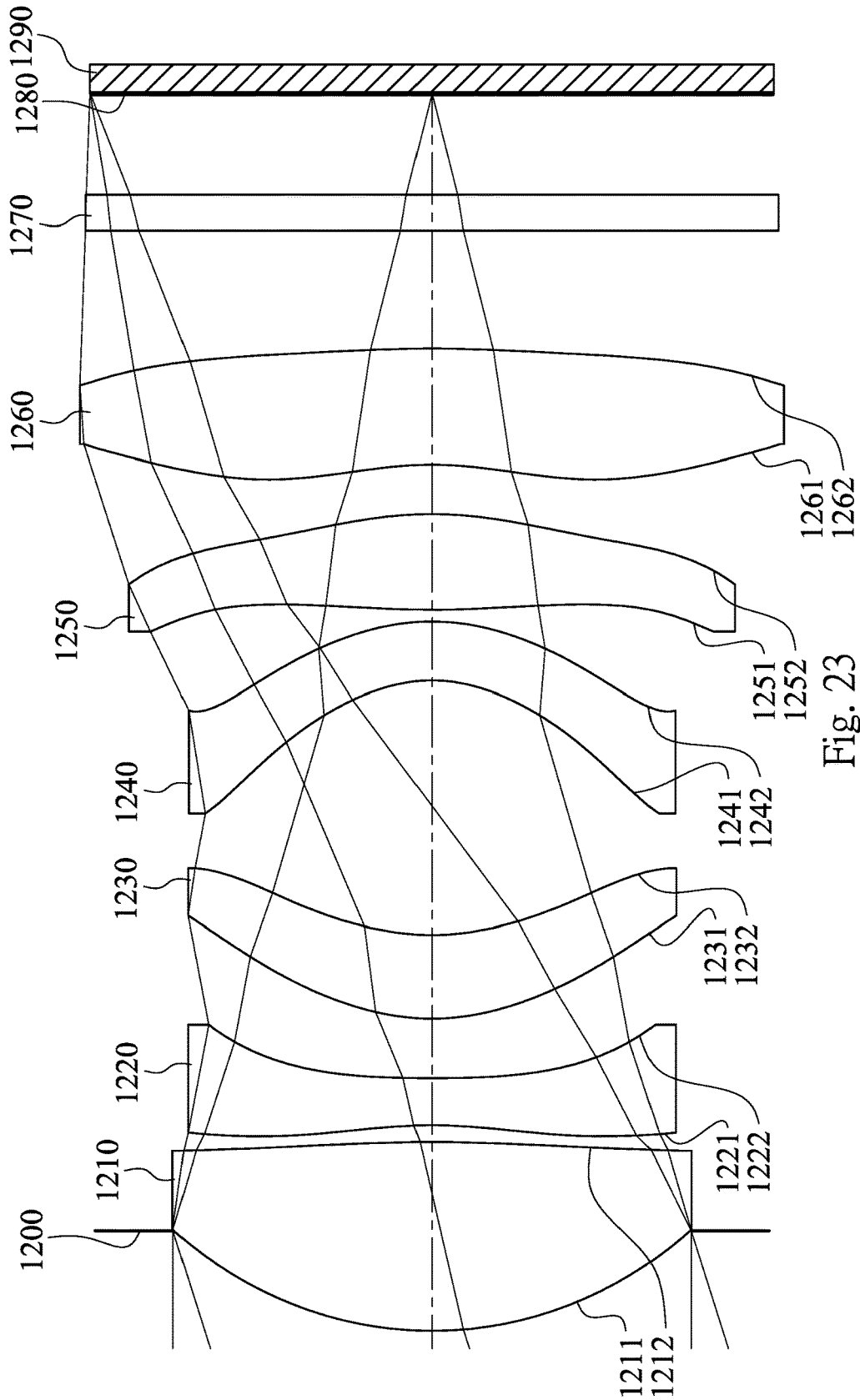


Fig. 23

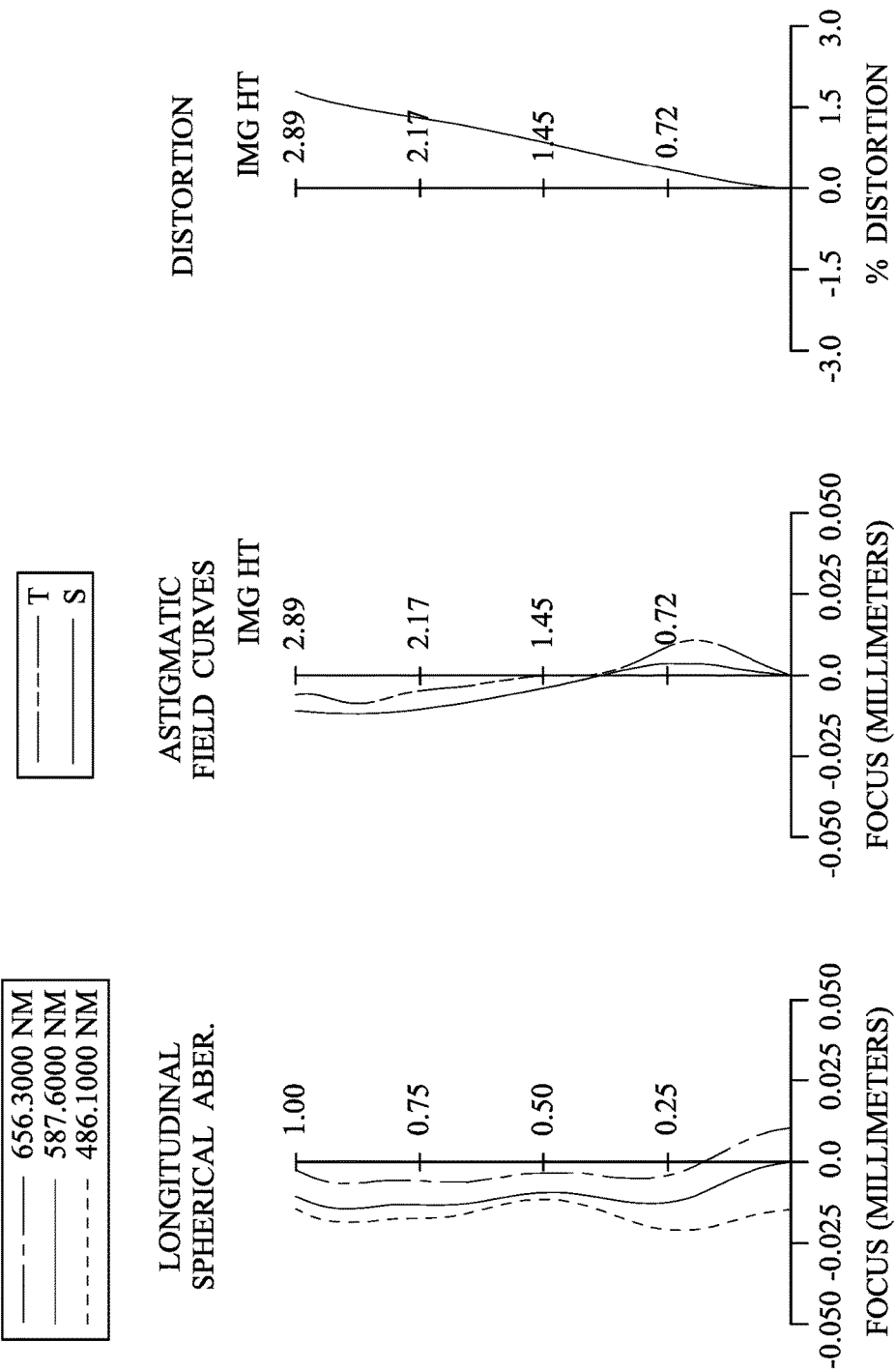


Fig. 24

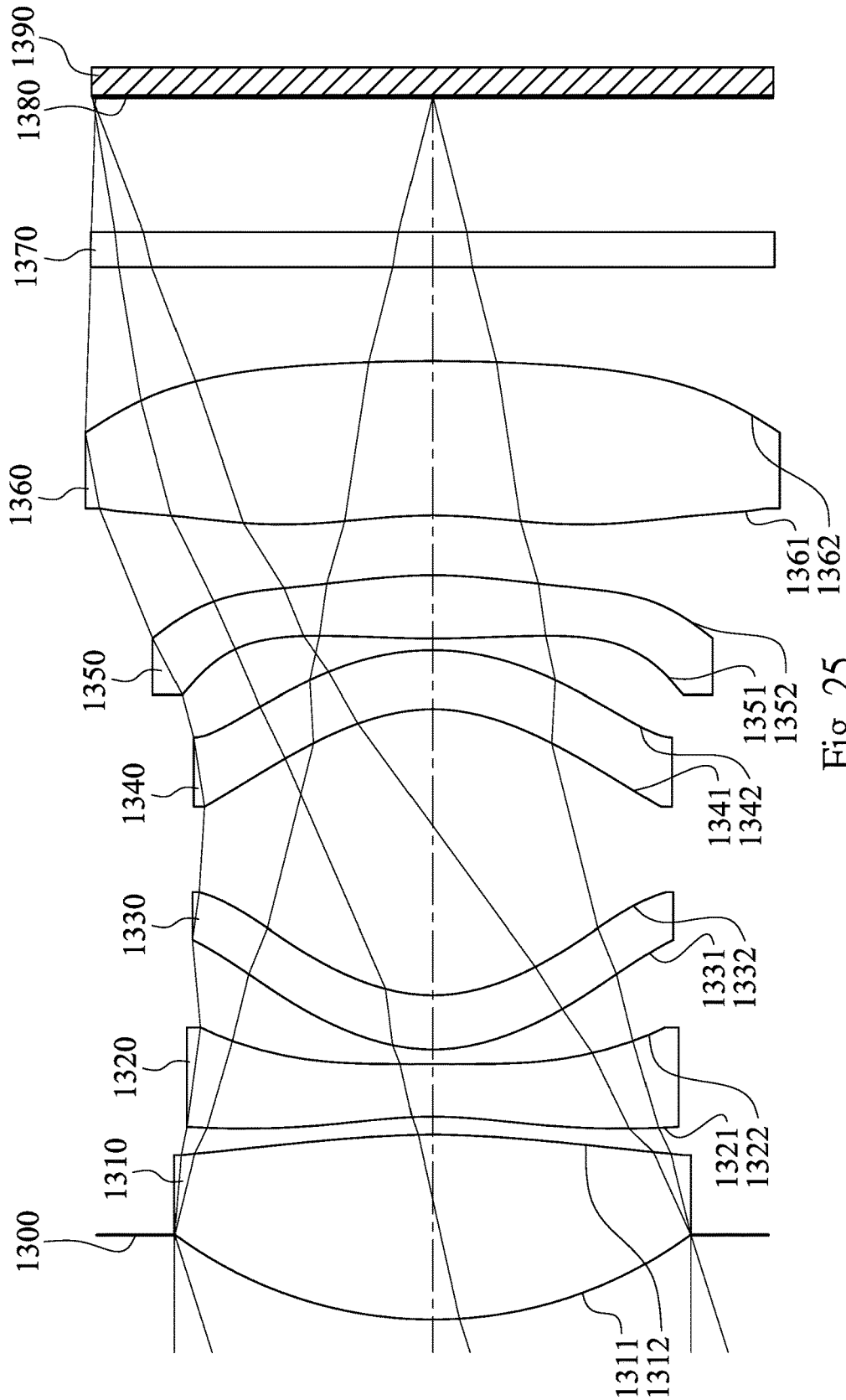


Fig. 25

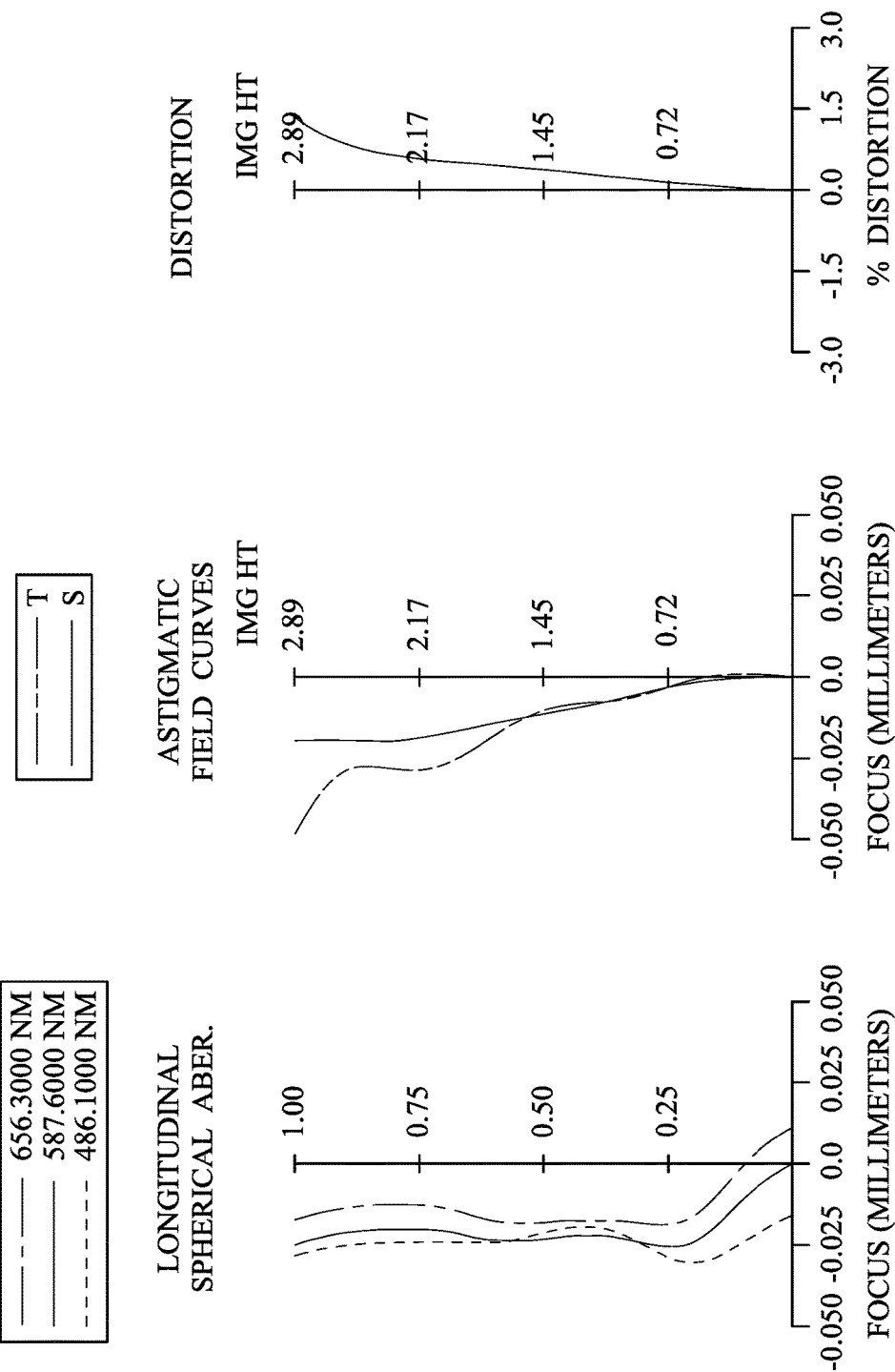


Fig. 26

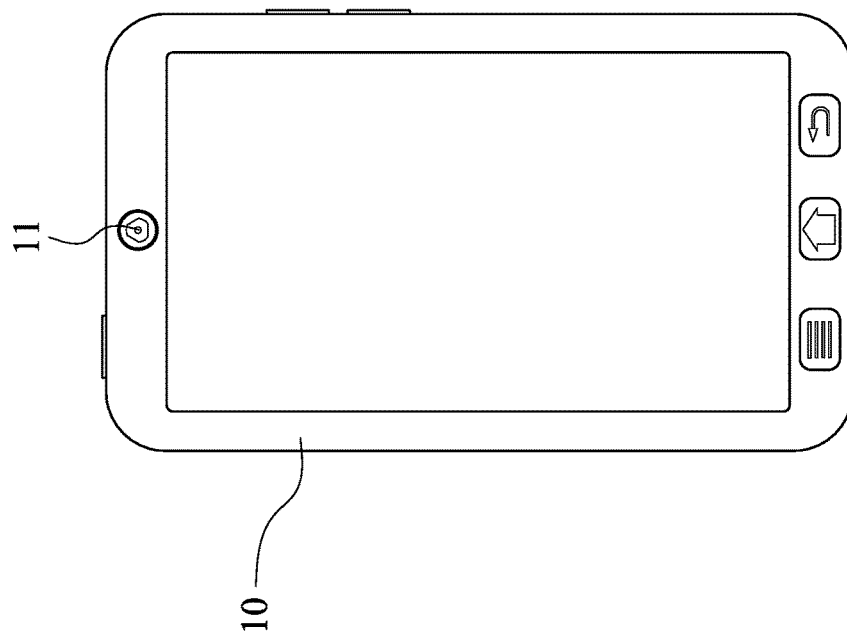


Fig. 27

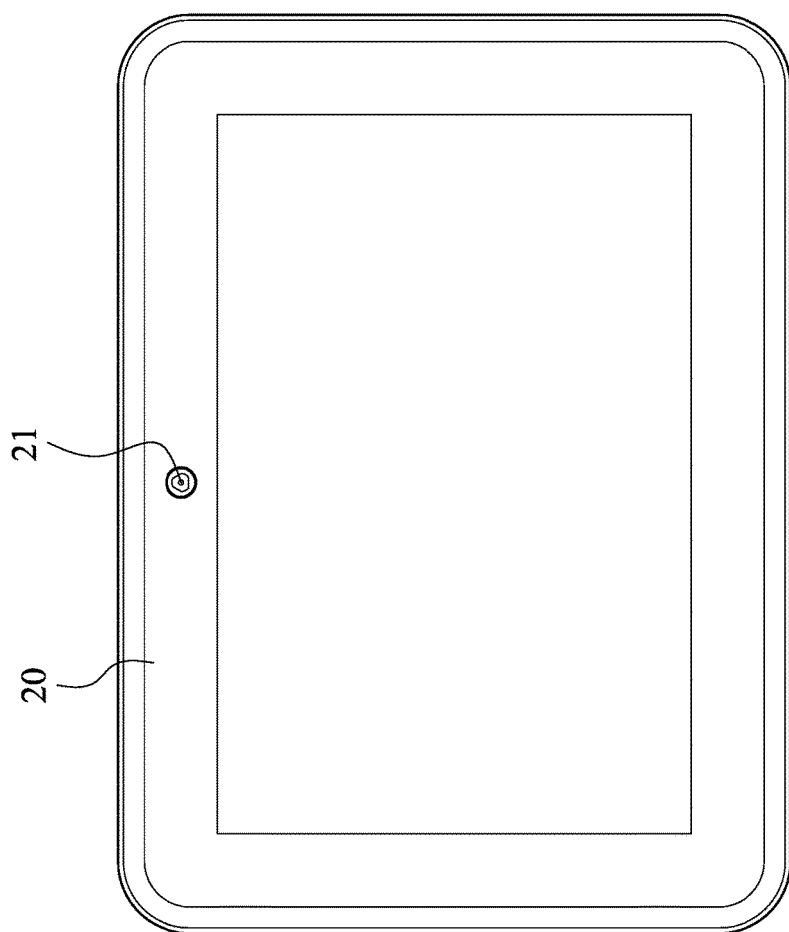


Fig. 28

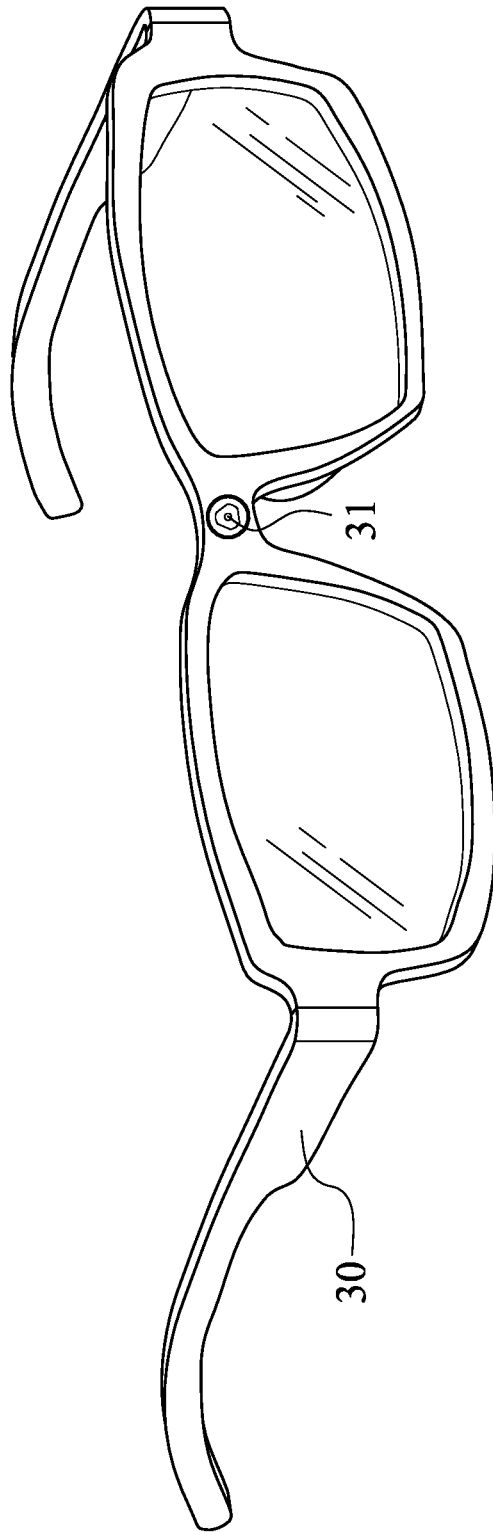


Fig. 29

1

OPTICAL IMAGING LENS ASSEMBLY, IMAGE CAPTURING DEVICE AND ELECTRONIC DEVICE

RELATED APPLICATIONS

This application claims priority to Taiwan Application Serial Number 105111883, filed Apr. 15, 2016, which is herein incorporated by reference.

BACKGROUND

Technical Field

The present disclosure relates to an optical imaging lens assembly and an image capturing device. More particularly, the present disclosure relates to an optical imaging lens assembly and an image capturing device with a compact size applicable to electronic devices.

Description of Related Art

With the popularity of optical lens assembly applications, utilizing optical lens assemblies in various intelligent electronic devices, automobile devices, recognition devices, entertainment devices, sport devices and household intelligent assistance systems is becoming a major trend in developments of future technologies. For obtaining extensive experiences in utilizations of the optical lens assemblies, intelligent devices with one or more optical lens assemblies are the market mainstream, and various lens systems with different features are developed in response to different demands.

In the conventional compact lens assemblies, the imaging effect and the image quality has been relatively limited, and the various image applications are hard to achieve. While conventional imaging systems with high imaging quality usually adopt multi-element lens configuration of spherical glass lens elements, the size of the imaging systems would be too large and cumbersome. Also, the costs of the imaging systems would be too expensive for applications in various devices and products. Hence, optical lens assemblies cannot satisfy the current trend of the technological developments.

SUMMARY

According to one aspect of the present disclosure, an optical imaging lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element and a sixth lens element. The first lens element has positive refractive power. The second lens element has positive refractive power. The third lens element has negative refractive power. The fourth lens element has an object-side surface and an image-side surface being aspheric. The fifth lens element has an object-side surface and an image-side surface being aspheric. The sixth lens element has an object-side surface and an image-side surface being aspheric, wherein at least one of the object-side surface and the image-side surface of the sixth lens element includes at least one inflection point. The optical imaging lens assembly has a total of six lens elements. When a focal length of the first lens element is f_1 , a focal length of the second lens element is f_2 , a focal length of the optical imaging lens assembly is f , a curvature radius of an object-side surface of the third lens element is R_5 , a curvature radius of an image-side surface of the third lens element is

2

R_6 , a central thickness of the third lens element is CT_3 , an axial distance between the image-side surface of the sixth lens element and an image surface is BL , an axial distance between an object-side surface of the first lens element and the image-side surface of the sixth lens element is TD , and a maximum image height of the optical imaging lens assembly is $ImgH$, the following conditions are satisfied:

$$|f_1| < |f_2|;$$

$$-10.0 < (f/R_5) - (f/R_6) + (f \times CT_3 / (R_5 \times R_6)) < -1.7;$$

$$0 < BL/TD < 0.45; \text{ and}$$

$$0.10 < ImgH/f < 0.50.$$

According to another aspect of the present disclosure, an image capturing device includes the optical imaging lens assembly of the aforementioned aspect and an image sensor, wherein the image sensor is disposed on the image surface of the optical imaging lens assembly.

According to yet another aspect of the present disclosure, an electronic device includes the image capturing device of the aforementioned aspect.

According to further another aspect of the present disclosure, an optical imaging lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element and a sixth lens element. The first lens element has positive refractive power. The third lens element has an object-side surface and an image-side surface being aspheric. The fourth lens element has an object-side surface and an image-side surface being aspheric. The fifth lens element has an object-side surface and an image-side surface being aspheric. The sixth lens element has an object-side surface and an image-side surface being aspheric, wherein at least one of the object-side surface and the image-side surface of the sixth lens element includes at least one inflection point. The optical imaging lens assembly has a total of six lens elements, and there is an air space between every two lens elements of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element that are adjacent to each other. At least three lens elements of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element are made of plastic materials. When an entrance pupil diameter of the optical imaging lens assembly is EPD , a maximum image height of the optical imaging lens assembly is $ImgH$, an axial distance between the image-side surface of the sixth lens element and an image surface is BL , an axial distance between an object-side surface of the first lens element and the image-side surface of the sixth lens element is TD , a refractive index of the first lens element is N_1 , a refractive index of the second lens element is N_2 , a refractive index of the third lens element is N_3 , a refractive index of the fourth lens element is N_4 , a refractive index of the fifth lens element is N_5 , and a refractive index of the sixth lens element is N_6 , a maximum of N_1, N_2, N_3, N_4, N_5 and N_6 is N_{max} , the following conditions are satisfied:

$$1.25 < EPD/ImgH < 2.0;$$

$$0 < BL/TD < 0.45; \text{ and}$$

$$1.60 < N_{max} < 1.72.$$

According to still another aspect of the present disclosure, an image capturing device includes the optical imaging lens

3

assembly of the aforementioned aspect and an image sensor, wherein the image sensor is disposed on the image surface of the optical imaging lens assembly.

According to yet another aspect of the present disclosure, an electronic device includes the image capturing device of the aforementioned aspect.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure;

FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 1st embodiment;

FIG. 3 is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure;

FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment;

FIG. 5 is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure;

FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment;

FIG. 7 is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure;

FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 4th embodiment;

FIG. 9 is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure;

FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 5th embodiment;

FIG. 11 is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure;

FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment;

FIG. 13 is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure;

FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment;

FIG. 15 is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure;

FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment;

FIG. 17 is a schematic view of an image capturing device according to the 9th embodiment of the present disclosure;

FIG. 18 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 9th embodiment;

FIG. 19 is a schematic view of an image capturing device according to the 10th embodiment of the present disclosure;

FIG. 20 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 10th embodiment;

FIG. 21 is a schematic view of an image capturing device according to the 11th embodiment of the present disclosure;

FIG. 22 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 11th embodiment;

FIG. 23 is a schematic view of an image capturing device according to the 12th embodiment of the present disclosure;

4

FIG. 24 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 12th embodiment;

FIG. 25 is a schematic view of an image capturing device according to the 13th embodiment of the present disclosure;

FIG. 26 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 13th embodiment;

FIG. 27 is a schematic view of an electronic device according to the 14th embodiment of the present disclosure;

FIG. 28 is a schematic view of an electronic device according to the 15th embodiment of the present disclosure; and

FIG. 29 is a schematic view of an electronic device according to the 16th embodiment of the present disclosure.

DETAILED DESCRIPTION

An optical imaging lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element and a sixth lens element, wherein the optical imaging lens assembly has a total of six lens elements.

According to the optical imaging lens assembly of the present disclosure, there is an air space between every two lens elements of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element that are adjacent to each other. That is, each of the first through sixth lens elements is a single and non-cemented lens element, and every two lens elements adjacent to each other are not cemented, and there is a space between the two lens elements. Moreover, the manufacturing process of the cemented lenses is more complex than the non-cemented lenses. In other words, of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element of the optical imaging lens assembly, there is a space in a paraxial region between every pair of lens elements that are adjacent to each other. In particular, a cementing surface of one lens element and a cementing surface of the following lens element need to have accurate curvature to ensure these two lens elements will be highly cemented. However, during the cementing process, those two lens elements might not be highly cemented due to displacements and it is thereby not favorable for image quality of the optical imaging lens assembly. Therefore, according to the optical imaging lens assembly of the present disclosure, having an air space in a paraxial region between every two of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element that are adjacent to each other of the present disclosure avoids the problem generated by the cemented lens elements.

The first lens element with positive refractive power has an object-side surface being convex. Therefore, it is favorable for providing the main converging ability of the incident light in the optical imaging lens assembly, so that the photographing range can be controlled effectively so as to avoid the excessive total track length of the optical imaging lens assembly.

The second lens element can have positive refractive power so as to distribute the converging ability of the incident light of the first lens element. Therefore, excessive aberrations can be avoided with a proper curvature of the

lens element, and the sensitivity to the object distance of the optical imaging lens assembly can be reduced.

The third lens element can have negative refractive power, so that the size of image points converged from incident light with different wavelengths can be reduced, and aberrations of the object side and the image side of the optical imaging lens assembly can be balanced.

The fourth lens element can have an object-side surface being concave and an image-side surface being convex. Therefore, it is favorable for enhancing the ability for correcting the coma aberration.

The sixth lens element can have negative refractive power, and at least one of the object-side surface and the image-side surface of the sixth lens element can be concave in a paraxial region thereof and include at least one convex shape in an off-axial region thereof. Therefore, it is favorable for positioning the principal point closer to the object side so as to reduce the back focal length and avoid the excessive total track length, and it is also favorable for correcting the off-axial aberration so as to enhancing the image quality. Furthermore, at least one of the object-side surface and the image-side surface of the sixth lens element includes at least one inflection point, so that it is favorable for correcting off-axial aberrations so as to obtain high image quality with a large aperture.

When an axial distance between the image-side surface of the sixth lens element and an image surface is BL, and an axial distance between an object-side surface of the first lens element and the image-side surface of the sixth lens element is TD, the following condition is satisfied: $0 < BL/TD < 0.45$. Therefore, it is favorable for obtaining the compact size of the optical imaging lens assembly by effectively controlling the back focal length thereof. Preferably, the following condition can be satisfied: $0 < BL/TD < 0.25$.

When a focal length of the first lens element is f_1 , and a focal length of the second lens element is f_2 , the following condition is satisfied: $|f_1| < |f_2|$. Therefore, it is favorable for utilizing the telephoto characteristic by enhancing the light converging ability on the object side.

When a focal length of the optical imaging lens assembly is f , a curvature radius of an object-side surface of the third lens element is R_5 , a curvature radius of an image-side surface of the third lens element is R_6 , and a central thickness of the third lens element is CT_3 , the following condition is satisfied: $-10.0 < (f/R_5) - (f/R_6) + (f \times CT_3 / (R_5 \times R_6)) < -1.7$. Therefore, it is favorable for balancing aberrations of the object side and the image side of the optical imaging lens assembly so as to enhance the sharpness and clarity of the image. Preferably, the following condition can be satisfied: $-6.0 < (f/R_5) - (f/R_6) + (f \times CT_3 / (R_5 \times R_6)) < -2.3$.

When the focal length of the optical imaging lens assembly is f , and a maximum image height of the optical imaging lens assembly is $ImgH$, the following condition is satisfied: $0.10 < ImgH/f < 0.50$. Therefore, it is favorable for obtaining the telephoto structure and suppressing the formation of distortion effectively, so that the optical imaging lens assembly can be utilized widely. Preferably, the following condition can be satisfied: $0.10 < ImgH/f < 0.40$.

When an entrance pupil diameter of the optical imaging lens assembly is EPD, and the maximum image height of the optical imaging lens assembly is $ImgH$, the following condition is satisfied: $1.25 < EPD/ImgH < 2.0$. Therefore, it is favorable for obtaining the sufficient image illumination by effectively controlling the light into the optical imaging lens assembly. Preferably, the following condition can be satisfied: $1.35 < EPD/ImgH < 2.0$.

When a refractive index of the first lens element is N_1 , a refractive index of the second lens element is N_2 , a refractive index of the third lens element is N_3 , a refractive index of the fourth lens element is N_4 , a refractive index of the fifth lens element is N_5 , a refractive index of the sixth lens element is N_6 , and a maximum of N_1, N_2, N_3, N_4, N_5 and N_6 is N_{max} , the following condition is satisfied: $1.60 < N_{max} < 1.72$. Therefore, it is favorable for configuring the lens elements of the entire optical imaging lens assembly so as to provide the balancing ability of aberrations.

When a central thickness of the first lens element is CT_1 , and a central thickness of the second lens element is CT_2 , the following condition is satisfied: $1.0 < CT_1/CT_2 < 7.0$. Therefore, it is favorable for strengthening the optical imaging lens assembly against possible hazards of the external environment along with an easier maintenance so as to increase the durability thereof.

The optical imaging lens assembly can further include an aperture stop disposed between an imaged object and the second lens element. When an axial distance between the aperture stop and the image surface is SL, and an axial distance between the object-side surface of the first lens element and the image surface is TL, the following condition is satisfied: $0.78 < SL/TL < 1.10$. Therefore, it is favorable for balancing the position of the aperture stop and configuring the compact size of the optical imaging lens assembly and the image illumination.

The aperture stop can be located along an optical axis between a center of the object-side surface of the first lens element on the optical axis and a location of a maximum effective radius of the object-side surface of the first lens element projected on the optical axis. Therefore, both of the compact size and high illumination of the optical imaging lens assembly can be obtained within limited space.

When an Abbe number of the third lens element is V_3 , the following condition is satisfied: $V_3 < 25.0$. Therefore, the chromatic aberration of the optical imaging lens assembly can be corrected for condensed image points converged from incident light with different wavelengths.

When half of a maximum field of view of the optical imaging lens assembly is HFOV, the following condition is satisfied: $0.20 < \tan(2 \times HFOV) < 1.0$. Therefore, it is favorable for controlling the photographing range effectively so as to enhance the identification ability and broaden the field of application thereof. Preferably, the following condition can be satisfied: $0.20 < \tan(2 \times HFOV) < 0.85$.

When the curvature radius of the image-side surface of the third lens element is R_6 , and the focal length of the optical imaging lens assembly is f , the following condition is satisfied: $0 < R_6/f < 0.45$. Therefore, it is favorable for controlling the angle of the exiting light from the third lens element and providing the sufficient divergent ability to the third lens element so as to balance the light convergence on the object side of the optical imaging lens assembly.

When a curvature radius of the object-side surface of the sixth lens element is R_{11} , and a curvature radius of the image-side surface of the sixth lens element is R_{12} , the following condition is satisfied: $|R_{11}/R_{12}| < 1.30$. Therefore, it is favorable for adjusting curved images and correcting the Petzval surface so as to increase the clarity in the off-axial region of image.

When the curvature radius of the object-side surface of the third lens element is R_5 , and the curvature radius of the image-side surface of the third lens element is R_6 , the following condition is satisfied: $0.50 < (R_5 + R_6)/(R_5 - R_6) < 2.50$. Therefore, the shape of the third lens element can be controlled so as to improve the light divergent ability on

the image side; by such arrangement, the optical imaging lens assembly can be formed into a symmetrical structure for enhancing the image quality.

When the focal length of the optical imaging lens assembly is f , and a focal length of the third lens element is f_3 , the following condition is satisfied: $-4.0 < f/f_3 < -1.5$. Therefore, it is favorable for balancing the light divergent ability of the optical imaging lens assembly and providing the telephoto characteristic with the configuration of the first lens element and the second lens element.

When the focal length of the optical imaging lens assembly is f , and the focal length of the first lens element is f_1 , the following condition is satisfied: $1.0 < f/f_1 < 3.0$. Therefore, it is favorable for controlling the refractive power of the first lens element so as to avoid insufficient convergence of light by weaker refractive power, or excessive spherical aberration by stronger refractive power.

At least three surfaces of object-side surfaces and image-side surfaces of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element include at least one inflection point thereon. Therefore, aberrations in the off-axial region can be corrected effectively.

When an axial distance between the first lens element and the second lens element is T_{12} , an axial distance between the second lens element and the third lens element is T_{23} , an axial distance between the third lens element and the fourth lens element is T_{34} , an axial distance between the fourth lens element and the fifth lens element is T_{45} , and an axial distance between the fifth lens element and the sixth lens element is T_{56} , the following condition is satisfied: $2.0 < (T_{34} + T_{56}) / (T_{12} + T_{23} + T_{45}) < 10.0$. Therefore, it is favorable for assembling the optical imaging lens assembly by controlling the space between any two lens elements adjacent to each other so as to balance the space arrangement of the optical imaging lens assembly.

When a maximum effective radius of the object-side surface of the first lens element is Y_{11} , and a maximum effective radius of the image-side surface of the sixth lens element is Y_{62} , the following condition is satisfied: $0.70 < Y_{11}/Y_{62} < 1.10$. Therefore, it is favorable for increasing the symmetry of two ends of the optical imaging lens assembly, and ensuring the optical imaging lens assembly with sufficient image illumination.

When an axial distance between the object-side surface of the first lens element and the image surface is TL , and the focal length of the optical imaging lens assembly is f , the following condition is satisfied: $TL/f \leq 1.20$. Therefore, it is favorable for forming the compact size optical imaging lens assembly with a telephoto structure.

When the focal length of the second lens element is f_2 , a focal length of the third lens element is f_3 , a focal length of the fourth lens element is f_4 , a focal length of the fifth lens element is f_5 , and a focal length of the sixth lens element is f_6 , the following conditions are satisfied: $|f_3| < |f_2|$; $|f_3| < |f_4|$; $|f_3| < |f_5|$; and $|f_3| < |f_6|$. Therefore, it is favorable for strengthening the light controlling ability of the middle part of the optical imaging lens assembly, so that the refractive power of the object side and the image side thereof can be balanced for applying to different demands.

When a central thickness of the first lens element is CT_1 , a central thickness of the second lens element is CT_2 , the central thickness of the third lens element is CT_3 , a central thickness of the fourth lens element is CT_4 , a central thickness of the fifth lens element is CT_5 , and a central thickness of the sixth lens element is CT_6 , CT_1 is a maximum among CT_1 , CT_2 , CT_3 , CT_4 , CT_5 and CT_6 .

Therefore, it is favorable for increasing the resistance against the environment hazards of the optical imaging lens assembly so as to increase the adaptability against the collision impact and temperature variation for different applications.

When an axial distance between the object-side surface of the first lens element and the image surface is TL , and the maximum image height of the optical imaging lens assembly is $ImgH$, the following condition is satisfied: $2.2 < TL/ImgH < 4.0$. Therefore, it is favorable for adjusting the ratio of the total track length and the image height of the optical imaging lens assembly for various applications.

When the focal length of the optical imaging lens assembly is f , and the entrance pupil diameter of the optical imaging lens assembly is EPD , the following condition is satisfied: $1.25 < f/EPD < 2.30$. Therefore, it is favorable for increasing the image illumination by increasing the coverage of imaging fields of the optical imaging lens assembly.

When an axial distance between the first lens element and the second lens element is T_{12} , an axial distance between the second lens element and the third lens element is T_{23} , an axial distance between the third lens element and the fourth lens element is T_{34} , an axial distance between the fourth lens element and the fifth lens element is T_{45} , and an axial distance between the fifth lens element and the sixth lens element is T_{56} , T_{34} is a maximum among T_{12} , T_{23} , T_{34} , T_{45} and T_{56} . Therefore, it is favorable for enhancing the symmetry of the optical imaging lens assembly by balancing the space arrangement thereof.

According to the optical imaging lens assembly of the present disclosure, the lens elements thereof can be made of glass or plastic materials. When the lens elements are made of glass materials, the distribution of the refractive power of the optical imaging lens assembly may be more flexible to design. When the lens elements are made of plastic materials, manufacturing costs can be effectively reduced. Furthermore, surfaces of each lens element can be arranged to be aspheric, since the aspheric surface of the lens element is easy to form a shape other than a spherical surface so as to have more controllable variables for eliminating aberrations thereof, and to further decrease the required amount of lens elements in the optical imaging lens assembly. Therefore, the total track length of the optical imaging lens assembly can also be reduced.

According to the optical imaging lens assembly of the present disclosure, each of an object-side surface and an image-side surface has a paraxial region and an off-axial region. The paraxial region refers to the region of the surface where light rays travel close to an optical axis, and the off-axial region refers to the region of the surface away from the paraxial region. Particularly, when the lens element has a convex surface, it indicates that the surface can be convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface can be concave in the paraxial region thereof.

According to the optical imaging lens assembly of the present disclosure, the refractive power or the focal length of a lens element being positive or negative may refer to the refractive power or the focal length in a paraxial region of the lens element.

According to the optical imaging lens assembly of the present disclosure, the optical imaging lens assembly can include at least one stop, such as an aperture stop, a glare stop or a field stop. Said glare stop or said field stop is for eliminating the stray light and thereby improving the image resolution thereof.

According to the optical imaging lens assembly of the present disclosure, the image surface of the optical imaging lens assembly, based on the corresponding image sensor, can be flat or curved. In particular, the image surface can be a curved surface being concave facing towards the object side.

According to the optical imaging lens assembly of the present disclosure, an aperture stop can be configured as a middle stop. A middle stop disposed between the first lens element and the image surface is favorable for enlarging the field of view of the optical imaging lens assembly and thereby provides a wider field of view for the same.

According to the optical imaging lens assembly of the present disclosure, the optical imaging lens assembly can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, smart TVs, surveillance systems, motion sensing input devices, driving recording systems, rearview camera systems, and wearable devices.

According to the present disclosure, an image capturing device is provided. The image capturing device includes the aforementioned optical imaging lens assembly and an image sensor, wherein the image sensor is disposed on the image side of the aforementioned optical imaging lens assembly, that is, the image sensor can be disposed on or near the image surface of the aforementioned optical imaging lens assembly. By the arrangement of the aforementioned optical imaging lens assembly including the first lens element with positive refractive power, it is favorable providing the main converging ability of the incident light in the optical imaging lens assembly, so that the photographing range can be controlled effectively so as to avoid the excessive total track length of the optical imaging lens assembly. With the sixth lens element including the inflection point on the surface, it is favorable for correcting off-axial aberrations so as to obtain high quality under the large aperture characteristic. Preferably, the image capturing device can further include a barrel member, a holder member or a combination thereof.

According to the present disclosure, an electronic device is provided, which includes the aforementioned image capturing device. Therefore, image quality of the electronic device can be improved. Preferably, the electronic device can further include but not limited to a control unit, a display, a storage unit, a random access memory unit (RAM) or a combination thereof.

According to the above description of the present disclosure, the following 1st-16th specific embodiments are provided for further explanation.

1st Embodiment

FIG. 1 is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure. FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 1st embodiment. In FIG. 1, the image capturing device includes an optical imaging lens assembly (its reference numeral is omitted) and an image sensor 190. The optical imaging lens assembly includes, in order from an object side to an image side, an aperture stop 100, a first lens element 110, a second lens element 120, a third lens element 130, a fourth lens element 140, a fifth lens element 150, a sixth lens element 160, an IR-cut filter 170 and an image surface 180, wherein the image sensor 190 is disposed on the image surface 180 of the optical imaging lens

assembly. The optical imaging lens assembly has a total of six lens elements (110-160), and there is an air space between every two lens elements of the first lens element 110, the second lens element 120, the third lens element 130, the fourth lens element 140, the fifth lens element 150 and the sixth lens element 160 that are adjacent to each other.

The first lens element 110 with positive refractive power has an object-side surface 111 being convex and an image-side surface 112 being convex. The first lens element 110 is made of a plastic material, and has the object-side surface 111 and the image-side surface 112 being both aspheric. Furthermore, the image-side surface 112 of the first lens element 110 includes at least one inflection point.

The second lens element 120 with positive refractive power has an object-side surface 121 being concave and an image-side surface 122 being convex. The second lens element 120 is made of a plastic material, and has the object-side surface 121 and the image-side surface 122 being both aspheric. Furthermore, both of the object-side surface 121 and the image-side surface 122 of the second lens element 120 include at least one inflection point.

The third lens element 130 with negative refractive power has an object-side surface 131 being convex and an image-side surface 132 being concave. The third lens element 130 is made of a plastic material, and has the object-side surface 131 and the image-side surface 132 being both aspheric. Furthermore, the object-side surface 131 of the third lens element 130 includes at least one inflection point.

The fourth lens element 140 with positive refractive power has an object-side surface 141 being concave and an image-side surface 142 being convex. The fourth lens element 140 is made of a plastic material, and has the object-side surface 141 and the image-side surface 142 being both aspheric.

The fifth lens element 150 with positive refractive power has an object-side surface 151 being concave and an image-side surface 152 being convex. The fifth lens element 150 is made of a plastic material, and has the object-side surface 151 and the image-side surface 152 being both aspheric.

The sixth lens element 160 with negative refractive power has an object-side surface 161 being concave in a paraxial region thereof and an image-side surface 162 being concave in a paraxial region thereof. The sixth lens element 160 is made of a plastic material, and has the object-side surface 161 and the image-side surface 162 being both aspheric. Furthermore, the object-side surface 161 of the sixth lens element 160 includes at least one convex shape in an off-axial region thereof, the image-side surface 162 of the sixth lens element 160 includes at least one convex shape in an off-axial region thereof, and both of the object-side surface 161 and the image-side surface 162 of the sixth lens element 160 include at least one inflection point.

The aperture stop 100 is located along an optical axis between a center of the object-side surface 111 of the first lens element 110 on the optical axis and a location of a maximum effective radius of the object-side surface 111 of the first lens element 110 projected on the optical axis.

The IR-cut filter 170 is made of a glass material and located between the sixth lens element 160 and the image surface 180, and will not affect the focal length of the optical imaging lens assembly.

The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

11

$$X(Y) = (Y^2/R)/(1 + \sqrt{1 - (1+k) \times (Y/R)^2}) + \sum_i (Ai) \times (Y^i),$$

where,

X is the relative distance between a point on the aspheric surface spaced at a distance Y from the optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

Y is the vertical distance from the point on the aspheric surface to the optical axis;

R is the curvature radius;

k is the conic coefficient; and

Ai is the i-th aspheric coefficient.

In the optical imaging lens assembly according to the 1st embodiment, when a focal length of the optical imaging lens assembly is f, an f-number of the optical imaging lens assembly is Fno, and half of a maximum field of view of the optical imaging lens assembly is HFOV, these parameters have the following values: f=8.84 mm; Fno=2.06; and HFOV=17.6 degrees.

In the optical imaging lens assembly according to the 1st embodiment, when a refractive index of the first lens element **110** is N1, a refractive index of the second lens element **120** is N2, a refractive index of the third lens element **130** is N3, a refractive index of the fourth lens element **140** is N4, a refractive index of the fifth lens element **150** is N5, a refractive index of the sixth lens element **160** is N6, and a maximum of N1, N2, N3, N4, N5 and N6 is Nmax (according to the 1st embodiment, the maximum is N3), the following condition is satisfied: Nmax=1.660.

In the optical imaging lens assembly according to the 1st embodiment, when an Abbe number of the third lens element **130** is V3, the following condition is satisfied: V3=20.4.

In the optical imaging lens assembly according to the 1st embodiment, when a central thickness of the first lens element **110** is CT1, and a central thickness of the second lens element **120** is CT2, the following condition is satisfied: CT1/CT2=2.32.

In the optical imaging lens assembly according to the 1st embodiment, when an axial distance between the first lens element **110** and the second lens element **120** is T12, an axial distance between the second lens element **120** and the third lens element **130** is T23, an axial distance between the third lens element **130** and the fourth lens element **140** is T34, an axial distance between the fourth lens element **140** and the fifth lens element **150** is T45, and an axial distance between the fifth lens element **150** and the sixth lens element **160** is T56, the following condition is satisfied: (T34+T56)/(T12+T23+T45)=3.14.

In the optical imaging lens assembly according to the 1st embodiment, when a curvature radius of the image-side surface **132** of the third lens element **130** is R6, and the focal length of the optical imaging lens assembly is f, the following condition is satisfied: R6/f=0.19.

In the optical imaging lens assembly according to the 1st embodiment, when the curvature radius of the object-side surface **131** of the third lens element **130** is R5, a curvature radius of the image-side surface **132** of the third lens element **130** is R6, the following condition is satisfied: (R5+R6)/(R5-R6)=1.75.

In the optical imaging lens assembly according to the 1st embodiment, when a curvature radius of the object-side surface **161** of the sixth lens element **160** is R11, and a

12

curvature radius of the image-side surface **162** of the sixth lens element **160** is R12, the following condition is satisfied: |R11/R12|=0.63.

In the optical imaging lens assembly according to the 1st embodiment, when the focal length of the optical imaging lens assembly is f, and a focal length of the first lens element **110** is f1, the following condition is satisfied: f/f1=1.68.

In the optical imaging lens assembly according to the 1st embodiment, when the focal length of the optical imaging lens assembly is f, and a focal length of the third lens element **130** is f3, the following condition is satisfied: f/f3=-2.30.

In the optical imaging lens assembly according to the 1st embodiment, when the focal length of the optical imaging lens assembly is f, the curvature radius of the object-side surface **131** of the third lens element **130** is R5, the curvature radius of the image-side surface **132** of the third lens element **130** is R6, and a central thickness of the third lens element **130** is CT3, the following condition is satisfied: (f/R5)-(f/R6)+(f×CT3/(R5×R6))=-3.12.

In the optical imaging lens assembly according to the 1st embodiment, when a maximum effective radius of the object-side surface **111** of the first lens element **110** is Y11, and a maximum effective radius of the image-side surface **162** of the sixth lens element **160** is Y62, the following condition is satisfied: Y11/Y62=0.76.

In the optical imaging lens assembly according to the 1st embodiment, when half of a maximum field of view of the optical imaging lens assembly is HFOV, the following condition is satisfied: tan(2×HFOV)=0.71.

In the optical imaging lens assembly according to the 1st embodiment, when an axial distance between the aperture stop **100** and the image surface **180** is SL, and an axial distance between the object-side surface **111** of the first lens element **110** and the image surface **180** is TL, the following condition is satisfied: SL/TL=0.92.

In the optical imaging lens assembly according to the 1st embodiment, when an axial distance between the image-side surface **162** of the sixth lens element **160** and an image surface **180** is BL, and an axial distance between an object-side surface **111** of the first lens element **110** and the image-side surface **162** of the sixth lens element **160** is TD, the following condition is satisfied: BL/TD=0.16.

In the optical imaging lens assembly according to the 1st embodiment, when the focal length of the optical imaging lens assembly is f, and an entrance pupil diameter of the optical imaging lens assembly is EPD, the following condition is satisfied: f/EPD=2.06.

In the optical imaging lens assembly according to the 1st embodiment, when the entrance pupil diameter of the optical imaging lens assembly is EPD, and a maximum image height of the optical imaging lens assembly is ImgH (half of a diagonal length of an effective photosensitive area of the image sensor **190**), the following condition is satisfied: EPD/ImgH=1.48.

In the optical imaging lens assembly according to the 1st embodiment, when an axial distance between the object-side surface **111** of the first lens element **110** and the image surface **180** is TL, and the maximum image height of the optical imaging lens assembly is ImgH, the following condition is satisfied: TL/ImgH=3.39.

In the optical imaging lens assembly according to the 1st embodiment, when the maximum image height of the optical imaging lens assembly is ImgH, and the focal length of the optical imaging lens assembly is f, the following condition is satisfied: ImgH/f=0.33.

In the optical imaging lens assembly according to the 1st embodiment, when the axial distance between the object-side surface **111** of the first lens element **110** and the image surface **180** is TL, and the focal length of the optical imaging lens assembly is f, the following condition is satisfied: $TL/f=1.11$.

In the optical imaging lens assembly according to the 1st embodiment, when the focal length of the first lens element **110** is f1, a focal length of the second lens element **120** is f2, the focal length of the third lens element **130** is f3, a focal length of the fourth lens element **140** is f4, a focal length of the fifth lens element **150** is f5, and a focal length of the sixth lens element **160** is f6, the following conditions are satisfied: $|f1|<|f2|$; $|f3|<|f2|$; $|f3|<|f4|$; $|f3|<|f5|$; and $|f3|<|f6|$.

In the optical imaging lens assembly according to the 1st embodiment, when the central thickness of the first lens element **110** is CT1, the central thickness of the second lens element **120** is CT2, the central thickness of the third lens element **130** is CT3, a central thickness of the fourth lens element **140** is CT4, a central thickness of the fifth lens element **150** is CT5, and a central thickness of the sixth lens element **160** is CT6, CT1 is a maximum among CT1, CT2, CT3, CT4, CT5 and CT6.

In the optical imaging lens assembly according to the 1st embodiment, when the axial distance between the first lens element **110** and the second lens element **120** is T12, the axial distance between the second lens element **120** and the third lens element **130** is T23, the axial distance between the third lens element **130** and the fourth lens element **140** is T34, the axial distance between the fourth lens element **140** and the fifth lens element **150** is T45, the axial distance between the fifth lens element **150** and the sixth lens element **160** is T56, and T34 is a maximum among T12, T23, T34, T45 and T56.

The detailed optical data of the 1st embodiment are shown in Table 1 and the aspheric surface data are shown in Table 2 below.

TABLE 1

1st Embodiment							
f = 8.84 mm, Fno = 2.06, HFOV = 17.6 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.770				
2	Lens 1	3.386 ASP	1.600	Plastic	1.544	55.9	5.25
3		-15.206 ASP	0.369				
4	Lens 2	-7.436 ASP	0.691	Plastic	1.639	23.3	16.62
5		-4.532 ASP	0.116				
6	Lens 3	6.319 ASP	0.770	Plastic	1.660	20.4	-3.84
7		1.720 ASP	1.860				
8	Lens 4	-8.426 ASP	0.778	Plastic	1.639	23.3	19.89
9		-5.248 ASP	0.351				
10	Lens 5	-28.389 ASP	0.733	Plastic	1.544	55.9	7.84
11		-3.739 ASP	0.764				
12	Lens 6	-6.139 ASP	0.420	Plastic	1.515	56.5	-7.24
13		9.687 ASP	0.500				
14	IR-cut filter	Plano	0.250	Glass	1.517	64.2	—
15		Plano	0.597				
16	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line).

TABLE 2

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	8.6138E-03	4.1097E+01	-5.9947E+01	-2.7957E+01	-2.0074E+00	-1.1345E+00
A4 =	2.3021E-04	2.8945E-03	-1.4134E-02	2.4031E-02	2.6499E-02	-2.6953E-02
A6 =	-5.2824E-05	3.2183E-03	1.8565E-02	-5.5954E-03	-2.2342E-02	1.2891E-02
A8 =	7.7831E-05	-1.3479E-03	-9.1350E-03	2.6970E-03	1.0755E-02	-4.1413E-03
A10 =	-1.9980E-05	2.4749E-04	2.4680E-03	-1.2090E-03	-3.2868E-03	9.5536E-04
A12 =	1.6017E-06	-1.5101E-05	-3.2979E-04	3.8569E-04	6.1492E-04	-9.9580E-05
A14 =			1.6722E-05	-4.4520E-05	-5.1833E-05	
Surface #	8	9	10	11	12	13
k =	-7.6213E+00	-4.3868E+01	-8.6898E+01	7.8643E-01	1.9169E+00	8.7890E+00
A4 =	-1.1887E-02	-5.9339E-02	-3.9410E-02	-1.1801E-02	-4.5210E-02	-5.6202E-02
A6 =	2.0663E-03	3.3881E-02	2.7684E-02	1.4268E-02	3.0565E-02	1.8784E-02
A8 =	-1.9106E-04	-1.6041E-02	-1.1345E-02	-2.5604E-03	-7.1295E-03	-3.4194E-03
A10 =	2.5400E-05	4.6560E-03	2.2039E-03	-2.8577E-04	9.3689E-04	4.0686E-04
A12 =	-2.4212E-06	-7.9052E-04	-2.3623E-04	1.4197E-04	-7.1385E-05	-3.4449E-05
A14 =		7.1981E-05	1.3807E-05	-1.6492E-05	2.9405E-06	2.0294E-06
A16 =		-2.6867E-06	-3.6334E-07	6.5982E-07	-4.9555E-08	-6.0823E-08

In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-16 represent the surfaces sequentially arranged from the object side to the image side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A4-A16 represent the aspheric coefficients ranging from the 4th order to the 16th order. The tables presented below for each embodiment correspond to schematic parameter and aberration curves of each embodiment, and term definitions of the tables are the same as those in Table 1 and Table 2 of the 1st embodiment. Therefore, an explanation in this regard will not be provided again.

Furthermore, according to the 1st embodiment, a number of inflection points of each of the object-side surfaces (111, 121, 131, 141, 151, 161) and the image-side surfaces (112, 122, 132, 142, 152, 162) of the first lens element **110**, the second lens element **120**, the third lens element **130**, the fourth lens element **140**, the fifth lens element **150** and the sixth lens element **160** are listed in the following Table, wherein the number of the inflection points of each surface is a number of the inflection points located between an axial point of the surface and a position of a maximum effective radius of the surface.

Lens element	First		Second		Third	
Surface	111	112	121	122	131	132
Number of reflection point(s)	0	2	1	1	1	0
Lens element	Fourth		Fifth		Sixth	
Surface	141	142	151	152	161	162
Number of reflection point(s)	0	0	0	0	1	2

2nd Embodiment

FIG. 3 is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure. FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment. In FIG. 3, the image

capturing device includes an optical imaging lens assembly (its reference numeral is omitted) and an image sensor **290**. The optical imaging lens assembly includes, in order from an object side to an image side, an aperture stop **200**, a first lens element **210**, a second lens element **220**, a third lens element **230**, a fourth lens element **240**, a fifth lens element **250**, a sixth lens element **260**, an IR-cut filter **270** and an image surface **280**, wherein the image sensor **290** is disposed on the image surface **280** of the optical imaging lens assembly. The optical imaging lens assembly has a total of six lens elements (**210-260**), and there is an air space between every two lens elements of the first lens element **210**, the second lens element **220**, the third lens element **230**, the fourth lens element **240**, the fifth lens element **250** and the sixth lens element **260** that are adjacent to each other.

The first lens element **210** with positive refractive power has an object-side surface **211** being convex and an image-side surface **212** being convex. The first lens element **210** is made of a plastic material, and has the object-side surface **211** and the image-side surface **212** being both aspheric. Furthermore, the image-side surface **212** of the first lens element **210** includes at least one inflection point.

The second lens element **220** with positive refractive power has an object-side surface **221** being concave and an image-side surface **222** being convex. The second lens element **220** is made of a plastic material, and has the object-side surface **221** and the image-side surface **222** being both aspheric. Furthermore, both of the object-side surface **221** and the image-side surface **222** of the second lens element **220** include at least one inflection point.

The third lens element **230** with negative refractive power has an object-side surface **231** being convex and an image-side surface **232** being concave. The third lens element **230** is made of a plastic material, and has the object-side surface **231** and the image-side surface **232** being both aspheric. Furthermore, the object-side surface **231** of the third lens element **230** includes at least one inflection point.

The fourth lens element **240** with positive refractive power has an object-side surface **241** being concave and an image-side surface **242** being convex. The fourth lens element **240** is made of a plastic material, and has the object-side surface **241** and the image-side surface **242** being both aspheric. Furthermore, both of the object-side surface **241**

and the image-side surface **242** of the fourth lens element **240** include at least one inflection point.

The fifth lens element **250** with positive refractive power has an object-side surface **251** being convex and an image-side surface **252** being concave. The fifth lens element **250** is made of a plastic material, and has the object-side surface **251** and the image-side surface **252** being both aspheric. Furthermore, both of the object-side surface **251** and the image-side surface **252** of the fifth lens element **250** include at least one inflection point.

The sixth lens element **260** with negative refractive power has an object-side surface **261** being concave in a paraxial region thereof and an image-side surface **262** being convex in a paraxial region thereof. The sixth lens element **260** is made of a plastic material, and has the object-side surface **261** and the image-side surface **262** being both aspheric. Furthermore, the object-side surface **261** of the sixth lens element **260** includes at least one convex shape in an off-axial region thereof, and both of the object-side surface **261** and the image-side surface **262** of the sixth lens element **260** include at least one inflection point.

The aperture stop **200** is located along an optical axis between a center of the object-side surface **211** of the first lens element **210** on the optical axis and a location of a maximum effective radius of the object-side surface **211** of the first lens element **210** projected on the optical axis.

The IR-cut filter **270** is made of a glass material and located between the sixth lens element **260** and the image surface **280**, and will not affect the focal length of the optical imaging lens assembly.

The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric surface data are shown in Table 4 below.

TABLE 3

2nd Embodiment							
f = 8.61 mm, Fno = 2.06, HFOV = 18.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.740				
2	Lens 1	3.262 ASP	2.072	Plastic	1.544	55.9	5.62
3		-37.805 ASP	0.221				
4	Lens 2	-7.027 ASP	0.643	Plastic	1.583	30.2	13.41
5		-3.826 ASP	0.100				
6	Lens 3	122.570 ASP	1.184	Plastic	1.660	20.4	-3.52
7		2.273 ASP	0.736				
8	Lens 4	-71.824 ASP	1.181	Plastic	1.639	23.3	14.34
9		-8.178 ASP	0.334				
10	Lens 5	6.632 ASP	0.927	Plastic	1.544	55.9	20.95
11		15.095 ASP	1.384				
12	Lens 6	-5.816 ASP	0.350	Plastic	1.515	56.5	-12.95
13		-46.555 ASP	0.300				
14	IR-cut filter	Plano	0.250	Glass	1.517	64.2	—
15		Plano	0.240				
16	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line).

TABLE 4

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-3.1257E-02	9.0000E+01	-6.0070E+01	-2.0175E+01	-9.0000E+01	-1.1379E+00
A4 =	-2.8001E-05	-3.5479E-03	-1.4440E-02	2.7965E-02	2.8689E-02	-2.6713E-02
A6 =	-6.5218E-05	2.9064E-03	1.8370E-02	-5.5236E-03	-2.2599E-02	1.2987E-02
A8 =	2.2454E-05	-1.3773E-03	-9.0414E-03	2.1520E-03	1.0025E-02	-4.0285E-03
A10 =	-6.8038E-06	3.1348E-04	2.4589E-03	-1.2847E-03	-3.4614E-03	7.4954E-04

TABLE 4-continued

Aspheric Coefficients						
A12 =	3.8956E-07	-2.4286E-05	-3.3823E-04	3.5839E-04	6.5223E-04	-5.1977E-05
A14 =			1.8876E-05	-3.8723E-05	-5.4078E-05	
Surface #	8	9	10	11	12	13
k =	-9.0000E+01	-9.0000E+01	-2.7743E+01	-9.0000E+01	2.7534E+00	-9.0000E+01
A4 =	-7.5320E-03	-5.2412E-02	-3.7551E-02	-2.0008E-02	-5.3672E-02	-4.7078E-02
A6 =	3.5141E-03	3.3245E-02	2.8680E-02	1.3945E-02	3.1409E-02	1.7479E-02
A8 =	-4.2878E-04	-1.5927E-02	-1.1324E-02	-2.5014E-03	-7.0359E-03	-3.3427E-03
A10 =	1.8116E-05	4.6479E-03	2.1909E-03	-2.8245E-04	9.3463E-04	3.9573E-04
A12 =	1.3500E-05	-7.9699E-04	-2.3979E-04	1.4128E-04	-7.2032E-05	-3.0902E-05
A14 =		7.1594E-05	1.3569E-05	-1.6771E-05	2.9306E-06	2.1490E-06
A16 =		-2.3265E-06	-2.4975E-07	6.7391E-07	-4.2257E-08	-9.2542E-08

In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following conditions:

2nd Embodiment			
f [mm]	8.61	f/f3	-2.45
Fno	2.06	(f/R5) - (f/R6) + (f × CT3/(R5 × R6))	-3.68
HFOV [deg.]	18.0	Y11/Y62	0.77
Nmax	1.660	tan(2 × HFOV)	0.73
V3	20.4	SL/TL	0.93
CT1/CT2	3.22	BL/TD	0.09
(T34 + T56)/(T12 + T23 + T45)	3.24	f/EPD	2.06
R6/f	0.26	EPD/ImgH	1.45
(R5 + R6)/(R5 - R6)	1.04	TL/ImgH	3.43
R11/R12	0.12	ImgH/f	0.34
f/fl	1.53	TL/f	1.15

Furthermore, in the optical imaging lens assembly according to the 2nd embodiment, when a focal length of the first lens element **210** is f1, a focal length of the second lens element **220** is f2, a focal length of the third lens element **230** is f3, a focal length of the fourth lens element **240** is f4, a focal length of the fifth lens element **250** is f5, and a focal length of the sixth lens element **260** is f6, the following conditions are satisfied: $f1 < |f2|$; $|f3| < |f2|$; $|f3| < |f4|$; $|f3| < |f5|$; and $|f3| < |f6|$.

In the optical imaging lens assembly according to the 2nd embodiment, when a central thickness of the first lens element **210** is CT1, a central thickness of the second lens element **220** is CT2, a central thickness of the third lens element **230** is CT3, a central thickness of the fourth lens element **240** is CT4, a central thickness of the fifth lens element **250** is CT5, and a central thickness of the sixth lens element **260** is CT6, CT1 is a maximum among CT1, CT2, CT3, CT4, CT5 and CT6.

According to the 2nd embodiment, a number of inflection points of each of the object-side surfaces (211, 221, 231, 241, 251, 261) and the image-side surfaces (212, 222, 232, 242, 252, 262) of the first lens element **210**, the second lens element **220**, the third lens element **230**, the fourth lens element **240**, the fifth lens element **250** and the sixth lens element **260** are listed in the following Table, wherein the number of the inflection points of each surface is a number

of the inflection points located between an axial point of the surface and a position of a maximum effective radius of the surface.

Lens element	First		Second		Third	
Surface	211	212	221	222	231	232
Number of reflection point(s)	0	1	1	2	1	0

-continued

Lens element	Fourth		Fifth		Sixth	
Surface	241	242	251	252	261	262
Number of reflection point(s)	1	1	1	1	1	1

3rd Embodiment

FIG. 5 is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure. FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment. In FIG. 5, the image capturing device includes an optical imaging lens assembly (its reference numeral is omitted) and an image sensor **390**. The optical imaging lens assembly includes, in order from an object side to an image side, an aperture stop **300**, a first lens element **310**, a second lens element **320**, a third lens element **330**, a fourth lens element **340**, a fifth lens element **350**, a sixth lens element **360**, an IR-cut filter **370** and an image surface **380**, wherein the image sensor **390** is disposed on the image surface **380** of the optical imaging lens assembly. The optical imaging lens assembly has a total of

six lens elements (310-360), and there is an air space between every two lens elements of the first lens element 310, the second lens element 320, the third lens element 330, the fourth lens element 340, the fifth lens element 350 and the sixth lens element 360 that are adjacent to each other.

The first lens element 310 with positive refractive power has an object-side surface 311 being convex and an image-side surface 312 being convex. The first lens element 310 is made of a plastic material, and has the object-side surface 311 and the image-side surface 312 being both aspheric.

The second lens element 320 with positive refractive power has an object-side surface 321 being concave and an image-side surface 322 being convex. The second lens element 320 is made of a plastic material, and has the object-side surface 321 and the image-side surface 322 being both aspheric. Furthermore, both of the object-side surface 321 and the image-side surface 322 of the second lens element 320 include at least one inflection point.

The third lens element 330 with negative refractive power has an object-side surface 331 being convex and an image-side surface 332 being concave. The third lens element 330 is made of a plastic material, and has the object-side surface 331 and the image-side surface 332 being both aspheric. Furthermore, the object-side surface 331 of the third lens element 330 includes at least one inflection point.

The fourth lens element 340 with positive refractive power has an object-side surface 341 being convex and an image-side surface 342 being convex. The fourth lens element 340 is made of a plastic material, and has the object-side surface 341 and the image-side surface 342 being both aspheric. Furthermore, the image-side surface 342 of the fourth lens element 340 includes at least one inflection point.

The fifth lens element 350 with negative refractive power has an object-side surface 351 being convex and an image-side surface 352 being concave. The fifth lens element 350 is made of a plastic material, and has the object-side surface 351 and the image-side surface 352 being both aspheric. Furthermore, both of the object-side surface 351 and the image-side surface 352 of the fifth lens element 350 include at least one inflection point.

The sixth lens element 360 with negative refractive power has an object-side surface 361 being convex in a paraxial region thereof and an image-side surface 362 being concave in a paraxial region thereof. The sixth lens element 360 is made of a plastic material, and has the object-side surface 361 and the image-side surface 362 being both aspheric. Furthermore, the image-side surface 362 of the sixth lens element 360 includes at least one convex shape in an off-axial region thereof, and both of the object-side surface 361 and the image-side surface 362 of the sixth lens element 360 include at least one inflection point.

The aperture stop 300 is located along an optical axis between a center of the object-side surface 311 of the first lens element 310 on the optical axis and a location of a maximum effective radius of the object-side surface 311 of the first lens element 310 projected on the optical axis.

The IR-cut filter 370 is made of a glass material and located between the sixth lens element 360 and the image surface 380, and will not affect the focal length of the optical imaging lens assembly.

The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

TABLE 5

3rd Embodiment								
f = 8.63 mm, Fno = 2.06, HFOV = 18.0 deg.								
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Ape. Stop	Plano		-0.640				
2	Lens 1	3.473	ASP	2.300	Plastic	1.544	55.9	5.63
3		-19.872	ASP	0.370				
4	Lens 2	-6.910	ASP	0.666	Plastic	1.544	55.9	14.25
5		-3.777	ASP	0.100				
6	Lens 3	17.185	ASP	1.061	Plastic	1.660	20.4	-3.70
7		2.085	ASP	0.593				
8	Lens 4	24.930	ASP	2.267	Plastic	1.639	23.3	11.05
9		-9.492	ASP	0.049				
10	Lens 5	17.721	ASP	0.859	Plastic	1.544	55.9	-52.27
11		10.729	ASP	0.271				
12	Lens 6	8.938	ASP	0.667	Plastic	1.515	56.5	-24.64
13		5.110	ASP	0.500				
14	IR-cut filter	Plano		0.250	Glass	1.517	64.2	—
15		Plano		0.342				
16	Image	Plano		—				

Reference wavelength is 587.6 nm (d-line).

TABLE 6

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-1.5589E-01	7.6817E+01	-5.1277E+01	-1.8012E+01	-9.0000E+01	-1.0105E+00
A4 =	-6.5866E-04	-4.3804E-03	-1.3921E-02	1.8068E-02	1.3584E-02	-2.5471E-02
A6 =	-1.1264E-04	3.0152E-03	1.7833E-02	-7.4542E-03	-2.3595E-02	9.2955E-03
A8 =	-1.8601E-05	-1.3542E-03	-8.8032E-03	2.6306E-03	9.8476E-03	-3.6478E-03
A10 =	-4.0845E-06	3.2575E-04	2.5532E-03	-1.0731E-03	-3.3212E-03	9.1094E-04
A12 =	-2.1278E-07	-2.6868E-05	-3.3934E-04	3.5687E-04	7.3769E-04	-8.5147E-05

TABLE 6-continued

Aspheric Coefficients						
A14 =			1.7425E-05	-4.0065E-05	-7.4035E-05	
Surface #	8	9	10	11	12	13
k =	8.3530E+01	-9.0000E+01	-7.7236E+00	-5.7951E+01	-4.6351E+01	-5.8199E+00
A4 =	3.5914E-03	-4.3782E-02	-3.9535E-02	-2.5758E-02	-6.7973E-02	-4.9176E-02
A6 =	2.4154E-03	3.3141E-02	2.6957E-02	1.5356E-02	3.1310E-02	1.5070E-02
A8 =	-4.2762E-04	-1.6125E-02	-1.1274E-02	-2.4330E-03	-7.0514E-03	-3.3436E-03
A10 =	-2.3328E-05	4.6329E-03	2.2209E-03	-2.8785E-04	9.3470E-04	4.1065E-04
A12 =	1.3365E-05	-7.8948E-04	-2.3898E-04	1.3967E-04	-7.1881E-05	-3.0047E-05
A14 =		7.3402E-05	1.3363E-05	-1.6873E-05	2.9242E-06	2.1401E-06
A16 =		-2.7923E-06	-1.9029E-07	6.9976E-07	-4.7716E-08	-1.0067E-07

15

In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following conditions:

3rd Embodiment			
f [mm]	8.63	f/f3	-2.33
Fno	2.06	(f/R5) - (f/R6) + (f × CT3/(R5 × R6))	-3.38
HFOV [deg.]	18.0	Y1/Y62	0.77
Nmax	1.660	tan (2 × HFOV)	0.73
V3	20.4	SL/TL	0.94
CT1/CT2	3.45	BL/TD	0.12
(T34 + T56)/(T12 + T23 + T45)	1.66	f/EPD	2.06
R6/f	0.24	EPD/ImgH	1.45
(R5 + R6)/(R5 - R6)	1.28	TL/ImgH	3.56
R11/R12	1.75	ImgH/f	0.34
f/fl	1.53	TL/f	1.19

20

between the fifth lens element **350** and the sixth lens element **360** is T56, and T34 is a maximum among T12, T23, T34, T45 and T56.

According to the 3rd embodiment, a number of inflection points of each of the object-side surfaces (311, 321, 331, 341, 351, 361) and the image-side surfaces (312, 322, 332, 342, 352, 362) of the first lens element **310**, the second lens element **320**, the third lens element **330**, the fourth lens element **340**, the fifth lens element **350** and the sixth lens element **360** are listed in the following Table, wherein the

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Furthermore, in the optical imaging lens assembly according to the 3rd embodiment, when a focal length of the first lens element **310** is f1, a focal length of the second lens element **320** is f2, a focal length of the third lens element **330** is f3, a focal length of the fourth lens element **340** is f4, a focal length of the fifth lens element **350** is f5, and a focal length of the sixth lens element **360** is f6, the following conditions are satisfied: |f1|<|f2|; |f3|<|f2|; |f3|<|f4|; |f3|<|f5|; and |f3|<|f6|.

In the optical imaging lens assembly according to the 3rd embodiment, when a central thickness of the first lens element **310** is CT1, a central thickness of the second lens element **320** is CT2, a central thickness of the third lens element **330** is CT3, a central thickness of the fourth lens element **340** is CT4, a central thickness of the fifth lens element **350** is CT5, and a central thickness of the sixth lens element **360** is CT6, CT1 is a maximum among CT1, CT2, CT3, CT4, CT5 and CT6.

In the optical imaging lens assembly according to the 3rd embodiment, when the axial distance between the first lens element **310** and the second lens element **320** is T12, the axial distance between the second lens element **320** and the third lens element **330** is T23, the axial distance between the third lens element **330** and the fourth lens element **340** is T34, the axial distance between the fourth lens element **340** and the fifth lens element **350** is T45, the axial distance

45

number of the inflection points of each surface is a number of the inflection points located between an axial point of the surface and a position of a maximum effective radius of the surface.

Lens element	First		Second		Third	
Surface	311	312	321	322	331	332
Number of reflection point(s)	0	0	1	1	1	0

Lens element	Fourth		Fifth		Sixth	
Surface	341	342	351	352	361	362
Number of reflection point(s)	0	1	1	1	2	2

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4th Embodiment

FIG. 7 is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure. FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 4th embodiment. In FIG. 7, the image capturing device includes an optical imaging lens assembly (its reference numeral is omitted) and an image sensor **490**. The optical imaging lens assembly includes, in order from

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an object side to an image side, an aperture stop **400**, a first lens element **410**, a second lens element **420**, a third lens element **430**, a fourth lens element **440**, a fifth lens element **450**, a sixth lens element **460**, an IR-cut filter **470** and an image surface **480**, wherein the image sensor **490** is disposed on the image surface **480** of the optical imaging lens assembly. The optical imaging lens assembly has a total of six lens elements (**410-460**), and there is an air space between every two lens elements of the first lens element **410**, the second lens element **420**, the third lens element **430**, the fourth lens element **440**, the fifth lens element **450** and the sixth lens element **460** that are adjacent to each other.

The first lens element **410** with positive refractive power has an object-side surface **411** being convex and an image-side surface **412** being concave. The first lens element **410** is made of a plastic material, and has the object-side surface **411** and the image-side surface **412** being both aspheric. Furthermore, the image-side surface **412** of the first lens element **410** includes at least one inflection point.

The second lens element **420** with positive refractive power has an object-side surface **421** being convex and an image-side surface **422** being concave. The second lens element **420** is made of a plastic material, and has the object-side surface **421** and the image-side surface **422** being both aspheric. Furthermore, the image-side surface **422** of the second lens element **420** includes at least one inflection point.

The third lens element **430** with negative refractive power has an object-side surface **431** being convex and an image-side surface **432** being concave. The third lens element **430** is made of a plastic material, and has the object-side surface **431** and the image-side surface **432** being both aspheric. Furthermore, the object-side surface **431** of the third lens element **430** includes at least one inflection point.

The fourth lens element **440** with positive refractive power has an object-side surface **441** being convex and an

image-side surface **442** being convex. The fourth lens element **440** is made of a plastic material, and has the object-side surface **441** and the image-side surface **442** being both aspheric. Furthermore, the image-side surface **442** of the fourth lens element **440** includes at least one inflection point.

The fifth lens element **450** with negative refractive power has an object-side surface **451** being convex and an image-side surface **452** being concave. The fifth lens element **450** is made of a plastic material, and has the object-side surface **451** and the image-side surface **452** being both aspheric. Furthermore, both of the object-side surface **451** and the image-side surface **452** of the fifth lens element **450** include at least one inflection point.

The sixth lens element **460** with positive refractive power has an object-side surface **461** being convex in a paraxial region thereof and an image-side surface **462** being concave in a paraxial region thereof. The sixth lens element **460** is made of a plastic material, and has the object-side surface **461** and the image-side surface **462** being both aspheric. Furthermore, the image-side surface **462** of the sixth lens element **460** includes at least one convex shape in an off-axial region thereof, and both of the object-side surface **461** and the image-side surface **462** of the sixth lens element **460** include at least one inflection point.

The aperture stop **400** is located along an optical axis between a center of the object-side surface **411** of the first lens element **410** on the optical axis and a location of a maximum effective radius of the object-side surface **411** of the first lens element **410** projected on the optical axis.

The IR-cut filter **470** is made of a glass material and located between the sixth lens element **460** and the image surface **480**, and will not affect the focal length of the optical imaging lens assembly.

The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

TABLE 7

4th Embodiment								
f = 8.61 mm, Fno = 2.06, HFOV = 18.0 deg.								
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Ape. Stop	Plano		-0.703				
2	Lens 1	3.395	ASP	2.300	Plastic	1.544	55.9	6.45
3		80.347	ASP	0.340				
4	Lens 2	46.512	ASP	0.459	Plastic	1.544	55.9	12.44
5		-7.885	ASP	0.100				
6	Lens 3	6.770	ASP	0.616	Plastic	1.660	20.4	-4.68
7		2.045	ASP	0.627				
8	Lens 4	69.706	ASP	2.600	Plastic	1.639	23.3	21.26
9		-16.624	ASP	0.060				
10	Lens 5	13.914	ASP	1.060	Plastic	1.544	55.9	-18.24
11		5.635	ASP	0.088				
12	Lens 6	2.676	ASP	0.581	Plastic	1.515	56.5	50.35
13		2.765	ASP	0.500				
14	IR-cut filter	Plano		0.250	Glass	1.517	64.2	—
15		Plano		0.712				
16	Image	Plano		—				

Reference wavelength is 587.6 nm (d-line).

TABLE 8

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-4.6989E-02	-9.0000E+01	9.0000E+01	-2.3427E+01	-5.5637E+01	-1.1073E+00
A4 =	-1.9272E-04	-5.1585E-03	-1.3431E-02	1.9000E-02	1.4854E-02	-2.6999E-02
A6 =	5.0780E-05	4.1040E-03	1.6429E-02	-7.8883E-03	-2.3171E-02	7.7340E-03
A8 =	-1.7062E-05	-1.4531E-03	-8.3945E-03	2.4296E-03	9.8004E-03	-2.9567E-03
A10 =	3.0222E-06	3.1167E-04	2.5639E-03	-9.6718E-04	-3.2311E-03	8.8252E-04
A12 =	-3.7389E-07	-2.9850E-05	-3.6479E-04	3.9084E-04	7.7916E-04	-9.8879E-05
A14 =			1.9873E-05	-5.0488E-05	-8.6036E-05	
Surface #	8	9	10	11	12	13
k =	6.5344E+01	-1.7240E+01	2.2123E+01	-1.1969E+01	-3.8857E+00	-4.3803E+00
A4 =	5.6613E-03	-3.8398E-02	-4.0049E-02	-2.9595E-02	-8.4729E-02	-5.3584E-02
A6 =	2.1365E-03	3.3502E-02	2.6663E-02	1.5153E-02	3.2236E-02	1.4781E-02
A8 =	-4.9661E-04	-1.6261E-02	-1.1134E-02	-2.3590E-03	-7.0223E-03	-3.2426E-03
A10 =	-3.7982E-05	4.6229E-03	2.2282E-03	-2.8799E-04	9.3553E-04	4.2088E-04
A12 =	1.8855E-05	-7.8719E-04	-2.4267E-04	1.3880E-04	-7.1733E-05	-3.0191E-05
A14 =		7.3904E-05	1.3002E-05	-1.6931E-05	2.9228E-06	2.0335E-06
A16 =		-2.8373E-06	-7.3115E-08	7.1414E-07	-5.7730E-08	-1.1026E-07

In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following conditions:

4th Embodiment			
f [mm]	8.61	f/f3	-1.84
Fno	2.06	(f/R5) - (f/R6) + (f × CT3)/(R5 × R6)	-2.56
HFOV [deg.]	18.0	Y11/Y62	0.79
Nmax	1.660	tan(2 × HFOV)	0.73
V3	20.4	SL/TL	0.93
CT1/CT2	5.01	BL/TD	0.17
(T34 + T56)/(T12 + T23 + T45)	1.43	f/EPD	2.06
R6/f	0.24	EPD/ImgH	1.45
(R5 + R6)/(R5 - R6)	1.87	TL/ImgH	3.56
R11/R12	0.97	ImgH/f	0.34
f/f1	1.33	TL/f	1.20

Furthermore, in the optical imaging lens assembly according to the 4th embodiment, when a focal length of the first lens element **410** is f1, a focal length of the second lens element **420** is f2, a focal length of the third lens element **430** is f3, a focal length of the fourth lens element **440** is f4, a focal length of the fifth lens element **450** is f5, and a focal length of the sixth lens element **460** is f6, the following conditions are satisfied: |f1|<|f2|; |f3|<|f2|; |f3|<|f4|; |f3|<|f5|; and |f3|<|f6|.

In the optical imaging lens assembly according to the 4th embodiment, when the axial distance between the first lens element **410** and the second lens element **420** is T12, the axial distance between the second lens element **420** and the third lens element **430** is T23, the axial distance between the third lens element **430** and the fourth lens element **440** is T34, the axial distance between the fourth lens element **440** and the fifth lens element **450** is T45, the axial distance between the fifth lens element **450** and the sixth lens element **460** is T56, and T34 is a maximum among T12, T23, T34, T45 and T56.

According to the 4th embodiment, a number of inflection points of each of the object-side surfaces (411, 421, 431, 441, 451, 461) and the image-side surfaces (412, 422, 432, 442, 452, 462) of the first lens element **410**, the second lens element **420**, the third lens element **430**, the fourth lens element **440**, the fifth lens element **450** and the sixth lens element **460** are listed in the following Table, wherein the number of the inflection points of each surface is a number of the inflection points located between an axial point of the

surface and a position of a maximum effective radius of the surface.

Lens element		First		Second		Third	
Surface	411	412	421	422	431	432	
Number of reflection point(s)	0	1	0	1	1	0	
Lens element		Fourth		Fifth		Sixth	
Surface	441	442	451	452	461	462	
Number of reflection point(s)	0	1	1	1	3	2	

5th Embodiment

FIG. 9 is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure. FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device

according to the 5th embodiment. In FIG. 9, the image capturing device includes an optical imaging lens assembly (its reference numeral is omitted) and an image sensor 590. The optical imaging lens assembly includes, in order from an object side to an image side, an aperture stop 500, a first lens element 510, a second lens element 520, a third lens element 530, a fourth lens element 540, a fifth lens element 550, a sixth lens element 560, an IR-cut filter 570 and an image surface 580, wherein the image sensor 590 is disposed on the image surface 580 of the optical imaging lens assembly. The optical imaging lens assembly has a total of six lens elements (510-560), and there is an air space between every two lens elements of the first lens element 510, the second lens element 520, the third lens element 530, the fourth lens element 540, the fifth lens element 550 and the sixth lens element 560 that are adjacent to each other.

The first lens element 510 with positive refractive power has an object-side surface 511 being convex and an image-side surface 512 being concave. The first lens element 510 is made of a plastic material, and has the object-side surface 511 and the image-side surface 512 being both aspheric.

The second lens element 520 with positive refractive power has an object-side surface 521 being convex and an image-side surface 522 being convex. The second lens element 520 is made of a plastic material, and has the object-side surface 521 and the image-side surface 522 being both aspheric. Furthermore, the image-side surface 522 of the second lens element 520 includes at least one inflection point.

The third lens element 530 with negative refractive power has an object-side surface 531 being convex and an image-side surface 532 being concave. The third lens element 530 is made of a plastic material, and has the object-side surface 531 and the image-side surface 532 being both aspheric. Furthermore, both of the object-side surface 531 and the image-side surface 532 of the third lens element 530 include at least one inflection point.

The fourth lens element 540 with positive refractive power has an object-side surface 541 being convex and an

image-side surface 542 being convex. The fourth lens element 540 is made of a plastic material, and has the object-side surface 541 and the image-side surface 542 being both aspheric. Furthermore, the object-side surface 541 of the fourth lens element 540 includes at least one inflection point.

The fifth lens element 550 with negative refractive power has an object-side surface 551 being convex and an image-side surface 552 being concave. The fifth lens element 550 is made of a plastic material, and has the object-side surface 551 and the image-side surface 552 being both aspheric. Furthermore, both of the object-side surface 551 and the image-side surface 552 of the fifth lens element 550 include at least one inflection point.

The sixth lens element 560 with negative refractive power has an object-side surface 561 being convex in a paraxial region thereof and an image-side surface 562 being concave in a paraxial region thereof. The sixth lens element 560 is made of a plastic material, and has the object-side surface 561 and the image-side surface 562 being both aspheric. Furthermore, the image-side surface 562 of the sixth lens element 560 includes at least one convex shape in an off-axial region thereof, and both of the object-side surface 561 and the image-side surface 562 of the sixth lens element 560 include at least one inflection point.

The aperture stop 500 is located along an optical axis between a center of the object-side surface 511 of the first lens element 510 on the optical axis and a location of a maximum effective radius of the object-side surface 511 of the first lens element 510 projected on the optical axis.

The IR-cut filter 570 is made of a glass material and located between the sixth lens element 560 and the image surface 580, and will not affect the focal length of the optical imaging lens assembly.

The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

TABLE 9

5th Embodiment								
f = 8.61 mm, Fno = 2.06, HFOV = 18.0 deg.								
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Ape. Stop	Plano		-0.650				
2	Lens 1	3.614	ASP	1.450	Plastic	1.544	55.9	7.60
3		24.670	ASP	0.550				
4	Lens 2	28.397	ASP	0.556	Plastic	1.544	55.9	10.10
5		-6.761	ASP	0.160				
6	Lens 3	5.469	ASP	0.686	Plastic	1.660	20.4	-4.69
7		1.878	ASP	1.540				
8	Lens 4	2934.681	ASP	1.450	Plastic	1.639	23.3	18.58
9		-11.913	ASP	0.080				
10	Lens 5	7.566	ASP	1.100	Plastic	1.544	55.9	-70.11
11		5.990	ASP	0.149				
12	Lens 6	2.819	ASP	0.645	Plastic	1.515	56.5	-259.23
13		2.546	ASP	0.700				
14	IR-cut filter	Plano		0.250	Glass	1.517	64.2	—
15		Plano		0.773				
16	Image	Plano		—				

Reference wavelength is 587.6 nm (d-line).

TABLE 10

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-8.9909E-02	-9.0000E+01	8.3345E+01	-2.1332E+01	-3.2835E+01	-1.4393E+00
A4 =	-6.6574E-04	-6.6146E-03	-1.2073E-02	2.0724E-02	1.6149E-02	-3.1652E-02
A6 =	2.7861E-04	4.0674E-03	1.5078E-02	-7.7281E-03	-2.2924E-02	6.6435E-03
A8 =	-5.2968E-05	-1.5168E-03	-8.3139E-03	2.2333E-03	9.8462E-03	-2.3191E-03
A10 =	-1.0459E-06	3.2874E-04	2.5784E-03	-9.8359E-04	-3.2476E-03	6.7438E-04
A12 =	1.4349E-06	-2.5130E-05	-3.5902E-04	3.9053E-04	7.5414E-04	-8.2834E-05
A14 =			1.9496E-05	-4.9258E-05	-8.0613E-05	
Surface #	8	9	10	11	12	13
k =	-9.0000E+01	6.2064E+00	8.3938E+00	-2.5313E+00	-5.0185E+00	-6.3038E+00
A4 =	6.4163E-03	-3.9081E-02	-4.6801E-02	-3.2352E-02	-8.9009E-02	-5.3854E-02
A6 =	1.6797E-03	3.4729E-02	2.6586E-02	1.4353E-02	3.2895E-02	1.4282E-02
A8 =	-5.2033E-04	-1.6216E-02	-1.1113E-02	-2.2973E-03	-7.0218E-03	-3.1678E-03
A10 =	3.2549E-05	4.6003E-03	2.2242E-03	-2.8342E-04	9.3381E-04	4.2863E-04
A12 =	-1.5183E-06	-7.9100E-04	-2.4974E-04	1.3817E-04	-7.1750E-05	-3.0498E-05
A14 =		7.3489E-05	1.1779E-05	-1.7029E-05	2.9357E-06	1.9385E-06
A16 =		-2.8128E-06	3.0136E-09	7.2337E-07	-6.2611E-08	-1.0842E-07

In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following conditions:

5th Embodiment			
f [mm]	8.61	f/f3	-1.84
Fno	2.06	(f/R5) - (f/R6) + (f × CT3/(R5 × R6))	-2.43
HFOV [deg.]	18.0	Y11/Y62	0.78
Nmax	1.660	tan(2 × HFOV)	0.73
V3	20.4	SL/TL	0.94
CT1/CT2	2.61	BL/TD	0.21
(T34 + T56)/(T12 + T23 + T45)	2.14	f/EPD	2.06
R6/f	0.22	EPD/ImgH	1.44
(R5 + R6)/(R5 - R6)	2.05	TL/ImgH	3.49
R11/R12	1.11	ImgH/f	0.34
f/f1	1.13	TL/f	1.17

Furthermore, in the optical imaging lens assembly according to the 5th embodiment, when a focal length of the first lens element **510** is f1, a focal length of the second lens element **520** is f2, a focal length of the third lens element **530** is f3, a focal length of the fourth lens element **540** is f4, a focal length of the fifth lens element **550** is f5, and a focal length of the sixth lens element **560** is f6, the following conditions are satisfied: |f1|<|f2|; |f3|<|f2|; |f3|<|f4|; |f3|<|f5|; and |f3|<|f6|.

In the optical imaging lens assembly according to the 5th embodiment, when a central thickness of the first lens element **510** is CT1, a central thickness of the second lens element **520** is CT2, a central thickness of the third lens element **530** is CT3, a central thickness of the fourth lens element **540** is CT4, a central thickness of the fifth lens element **550** is CT5, and a central thickness of the sixth lens element **560** is CT6, CT1 is a maximum among CT1, CT2, CT3, CT4, CT5 and CT6.

In the optical imaging lens assembly according to the 5th embodiment, when the axial distance between the first lens

element **510** and the second lens element **520** is T12, the axial distance between the second lens element **520** and the third lens element **530** is T23, the axial distance between the third lens element **530** and the fourth lens element **540** is T34, the axial distance between the fourth lens element **540** and the fifth lens element **550** is T45, the axial distance between the fifth lens element **550** and the sixth lens element **560** is T56, and T34 is a maximum among T12, T23, T34, T45 and T56.

According to the 5th embodiment, a number of inflection points of each of the object-side surfaces (511, 521, 531, 541, 551, 561) and the image-side surfaces (512, 522, 532, 542, 552, 562) of the first lens element **510**, the second lens element **520**, the third lens element **530**, the fourth lens element **540**, the fifth lens element **550** and the sixth lens element **560** are listed in the following Table, wherein the number of the inflection points of each surface is a number of the inflection points located between an axial point of the surface and a position of a maximum effective radius of the surface.

Lens element	First	Second	Third
Surface	511	512	521
Number of	0	0	0
		522	531
		1	1
			532
			1

-continued

reflection point(s)						
Lens element	Fourth		Fifth		Sixth	
Surface	541	542	551	552	561	562
Number of reflection point(s)	1	0	1	1	3	2

6th Embodiment

FIG. 11 is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure. FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment. In FIG. 11, the image capturing device includes an optical imaging lens assembly (its reference numeral is omitted) and an image sensor 690. The optical imaging lens assembly includes, in order from an object side to an image side, a first lens element 610, an aperture stop 600, a second lens element 620, a third lens element 630, a fourth lens element 640, a fifth lens element 650, a sixth lens element 660, an IR-cut filter 670 and an image surface 680, wherein the image sensor 690 is disposed on the image surface 680 of the optical imaging lens assembly. The optical imaging lens assembly has a total of six lens elements (610-660), and there is an air space between every two lens elements of the first lens element 610, the second lens element 620, the third lens element 630, the fourth lens element 640, the fifth lens element 650 and the sixth lens element 660 that are adjacent to each other.

The first lens element 610 with positive refractive power has an object-side surface 611 being convex and an image-side surface 612 being convex. The first lens element 610 is made of a plastic material, and has the object-side surface 611 and the image-side surface 612 being both aspheric. Furthermore, the image-side surface 612 of the first lens element 610 includes at least one inflection point.

The second lens element 620 with positive refractive power has an object-side surface 621 being convex and an image-side surface 622 being concave. The second lens element 620 is made of a plastic material, and has the object-side surface 621 and the image-side surface 622 being both aspheric. Furthermore, both of the object-side

surface 621 and the image-side surface 622 of the second lens element 620 include at least one inflection point.

The third lens element 630 with negative refractive power has an object-side surface 631 being convex and an image-side surface 632 being concave. The third lens element 630 is made of a plastic material, and has the object-side surface 631 and the image-side surface 632 being both aspheric. Furthermore, the object-side surface 631 of the third lens element 630 includes at least one inflection point.

The fourth lens element 640 with negative refractive power has an object-side surface 641 being convex and an image-side surface 642 being concave. The fourth lens element 640 is made of a plastic material, and has the object-side surface 641 and the image-side surface 642 being both aspheric. Furthermore, the image-side surface 642 of the fourth lens element 640 includes at least one inflection point.

The fifth lens element 650 with positive refractive power has an object-side surface 651 being convex and an image-side surface 652 being concave. The fifth lens element 650 is made of a plastic material, and has the object-side surface 651 and the image-side surface 652 being both aspheric. Furthermore, both of the object-side surface 651 and the image-side surface 652 of the fifth lens element 650 include at least one inflection point.

The sixth lens element 660 with negative refractive power has an object-side surface 661 being concave in a paraxial region thereof and an image-side surface 662 being concave in a paraxial region thereof. The sixth lens element 660 is made of a plastic material, and has the object-side surface 661 and the image-side surface 662 being both aspheric. Furthermore, the object-side surface 661 of the sixth lens element 660 includes at least one convex shape in an off-axial region thereof, the image-side surface 662 of the sixth lens element 660 includes at least one convex shape in an off-axial region thereof, and both of the object-side surface 661 and the image-side surface 662 of the sixth lens element 660 include at least one inflection point.

The IR-cut filter 670 is made of a glass material and located between the sixth lens element 660 and the image surface 680, and will not affect the focal length of the optical imaging lens assembly.

The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

TABLE 11

6th Embodiment							
f = 8.58 mm, Fno = 2.30, HFOV = 17.5 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	3.099 ASP	1.069	Plastic	1.530	55.8	4.97
2		-15.536 ASP	-0.055				
3	Ape. Stop	Plano	0.292				
4	Lens 2	10.406 ASP	0.330	Plastic	1.614	25.6	26.81
5		27.933 ASP	0.160				
6	Lens 3	7.764 ASP	0.330	Plastic	1.639	23.3	-4.55
7		2.079 ASP	2.300				
8	Lens 4	6.387 ASP	1.200	Plastic	1.584	28.2	-68.93
9		5.131 ASP	0.197				
10	Lens 5	1.874 ASP	0.446	Plastic	1.530	55.8	8.17
11		3.034 ASP	0.986				
12	Lens 6	-8.322 ASP	0.686	Plastic	1.535	56.3	-9.29
13		12.687 ASP	0.500				
14	IR-cut filter	Plano	0.250	Glass	1.517	64.2	—

TABLE 11-continued

6th Embodiment						
f = 8.58 mm, Fno = 2.30, HFOV = 17.5 deg.						
Surface #		Curvature Radius	Thickness	Material	Index	Focal Length
15		Plano	0.504			
16	Image	Plano	—			

Reference wavelength is 587.6 nm (d-line).

TABLE 12

Aspheric Coefficients						
Surface #	1	2	4	5	6	7
k =	-4.5413E-02	2.5284E+01	-2.8352E+01	-1.8991E+01	1.4151E+01	-7.7307E-01
A4 =	2.4096E-05	-6.0473E-03	-1.7493E-02	3.4978E-02	2.3174E-02	-2.1146E-02
A6 =	-3.0386E-04	5.7351E-03	1.6078E-02	-8.7681E-03	-1.9431E-02	3.5710E-03
A8 =	1.2119E-04	-1.7062E-03	-8.2424E-03	7.8642E-04	8.6129E-03	3.9030E-04
A10 =	-2.0872E-05	3.0941E-04	2.3593E-03	-1.2627E-03	-3.7045E-03	-4.5292E-05
A12 =	6.0428E-06	-2.4950E-05	-4.5040E-04	3.8988E-04	8.8565E-04	6.0360E-06
A14 =			3.7254E-05	-3.4714E-05	-8.7489E-05	
Surface #	8	9	10	11	12	13
k =	5.1123E+00	-9.0000E+01	-7.6307E+00	-1.2795E+01	7.3461E+00	-9.0000E+01
A4 =	-8.6759E-04	-4.1392E-02	-5.4481E-02	-2.9021E-02	-5.1365E-02	-3.7644E-02
A6 =	2.0772E-03	3.7114E-02	3.4126E-02	1.4990E-02	3.0128E-02	1.6897E-02
A8 =	-4.9263E-04	-1.5954E-02	-1.1605E-02	-2.2885E-03	-6.9258E-03	-3.4739E-03
A10 =	8.1290E-05	4.5916E-03	2.2238E-03	-2.8294E-04	9.4152E-04	4.0721E-04
A12 =	-6.4926E-06	-7.8931E-04	-2.3688E-04	1.4003E-04	-7.2246E-05	-3.1173E-05
A14 =		7.4246E-05	1.2029E-05	-1.7144E-05	2.7857E-06	2.0688E-06
A16 =		-2.8043E-06	-1.8699E-07	6.0698E-07	-6.5799E-08	-9.6355E-08

In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following conditions:

6th Embodiment			
f [mm]	8.58	f/f3	-1.89
Fno	2.30	(f/R5) - (f/R6) + (f × CT3/(R5 × R6))	-2.85
HFOV [deg.]	17.5	Y11/Y62	0.78
Nmax	1.639	tan (2 × HFOV)	0.70
V3	23.3	SL/TL	0.89
CT1/CT2	3.24	BL/TD	0.16
(T34 + T56)/(T12 + T23 + T45)	5.53	f/EPD	2.30
R6/f	0.24	EPD/ImgH	1.34
(R5 + R6)/(R5 - R6)	1.73	TL/ImgH	3.29
R11/R12	0.66	ImgH/f	0.33
f/f1	1.72	TL/f	1.07

Furthermore, in the optical imaging lens assembly according to the 6th embodiment, when a focal length of the first lens element **610** is f1, a focal length of the second lens element **620** is f2, a focal length of the third lens element **630** is f3, a focal length of the fourth lens element **640** is f4, a focal length of the fifth lens element **650** is f5, and a focal length of the sixth lens element **660** is f6, the following

conditions are satisfied: |f1|<|f2|; |f3|<|f2|; |f3|<|f4|; |f3|<|f5|; and |f3|<|f6|.

In the optical imaging lens assembly according to the 6th embodiment, when the axial distance between the first lens element **610** and the second lens element **620** is T12, the axial distance between the second lens element **620** and the third lens element **630** is T23, the axial distance between the third lens element **630** and the fourth lens element **640** is

T34, the axial distance between the fourth lens element **640** and the fifth lens element **650** is T45, the axial distance between the fifth lens element **650** and the sixth lens element **660** is T56, and T34 is a maximum among T12, T23, T34, T45 and T56.

According to the 6th embodiment, a number of inflection points of each of the object-side surfaces (611, 621, 631,

37

641, 651, 661) and the image-side surfaces (612, 622, 632, 642, 652, 662) of the first lens element **610**, the second lens element **620**, the third lens element **630**, the fourth lens element **640**, the fifth lens element **650** and the sixth lens element **660** are listed in the following Table, wherein the number of the inflection points of each surface is a number of the inflection points located between an axial point of the surface and a position of a maximum effective radius of the surface.

Lens element	First		Second		Third	
Surface	611	612	621	622	631	632
Number of reflection point(s)	0	1	1	1	1	0

Lens element	Fourth		Fifth		Sixth	
Surface	641	642	651	652	661	662
Number of reflection point(s)	0	2	3	1	2	2

7th Embodiment

FIG. **13** is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure. FIG. **14** shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment. In FIG. **13**, the image capturing device includes an optical imaging lens assembly (its reference numeral is omitted) and an image sensor **790**. The optical imaging lens assembly includes, in order from an object side to an image side, an aperture stop **700**, a first lens element **710**, a second lens element **720**, a third lens element **730**, a fourth lens element **740**, a fifth lens element **750**, a sixth lens element **760**, an IR-cut filter **770** and an image surface **780**, wherein the image sensor **790** is disposed on the image surface **780** of the optical imaging lens assembly. The optical imaging lens assembly has a total of six lens elements (**710-760**), and there is an air space between every two lens elements of the first lens element **710**, the second lens element **720**, the third lens element **730**, the fourth lens element **740**, the fifth lens element **750** and the sixth lens element **760** that are adjacent to each other.

The first lens element **710** with positive refractive power has an object-side surface **711** being convex and an image-side surface **712** being convex. The first lens element **710** is made of a plastic material, and has the object-side surface **711** and the image-side surface **712** being both aspheric. Furthermore, the image-side surface **712** of the first lens element **710** includes at least one inflection point.

The second lens element **720** with positive refractive power has an object-side surface **721** being concave and an

38

image-side surface **722** being convex. The second lens element **720** is made of a plastic material, and has the object-side surface **721** and the image-side surface **722** being both aspheric. Furthermore, both of the object-side surface **721** and the image-side surface **722** of the second lens element **720** include at least one inflection point.

The third lens element **730** with negative refractive power has an object-side surface **731** being convex and an image-side surface **732** being concave. The third lens element **730** is made of a plastic material, and has the object-side surface **731** and the image-side surface **732** being both aspheric. Furthermore, the object-side surface **731** of the third lens element **730** includes at least one inflection point.

The fourth lens element **740** with positive refractive power has an object-side surface **741** being convex and an image-side surface **742** being concave. The fourth lens element **740** is made of a plastic material, and has the object-side surface **741** and the image-side surface **742** being both aspheric. Furthermore, the image-side surface **742** of the fourth lens element **740** includes at least one inflection point.

The fifth lens element **750** with positive refractive power has an object-side surface **751** being convex and an image-side surface **752** being concave. The fifth lens element **750** is made of a plastic material, and has the object-side surface **751** and the image-side surface **752** being both aspheric. Furthermore, both of the object-side surface **751** and the image-side surface **752** of the fifth lens element **750** include at least one inflection point.

The sixth lens element **760** with negative refractive power has an object-side surface **761** being concave in a paraxial region thereof and an image-side surface **762** being concave in a paraxial region thereof. The sixth lens element **760** is made of a plastic material, and has the object-side surface **761** and the image-side surface **762** being both aspheric. Furthermore, the object-side surface **761** of the sixth lens element **760** includes at least one convex shape in an off-axial region thereof, the image-side surface **762** of the sixth lens element **760** includes at least one convex shape in an off-axial region thereof, and both of the object-side surface **761** and the image-side surface **762** of the sixth lens element **760** include at least one inflection point.

The aperture stop **700** is located along an optical axis between a center of the object-side surface **711** of the first lens element **710** on the optical axis and a location of a maximum effective radius of the object-side surface **711** of the first lens element **710** projected on the optical axis.

The IR-cut filter **770** is made of a glass material and located between the sixth lens element **760** and the image surface **780**, and will not affect the focal length of the optical imaging lens assembly.

The detailed optical data of the 7th embodiment are shown in Table 13 and the aspheric surface data are shown in Table 14 below.

TABLE 13

7th Embodiment								
f = 8.38 mm, Fno = 2.06, HFOV = 18.5 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.650					
2	Lens 1	3.421 ASP	1.800	Plastic	1.544	55.9	5.68	

TABLE 13-continued

7th Embodiment f = 8.38 mm, Fno = 2.06, HFOV = 18.5 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
3		-26.000	ASP				
4	Lens 2	-7.315	ASP	Plastic	1.639	23.3	14.26
5		-4.197	ASP				
6	Lens 3	18.311	ASP	Plastic	1.660	20.4	-3.93
7		2.219	ASP				
8	Lens 4	27.971	ASP	Plastic	1.639	23.3	139.30
9		40.000	ASP				
10	Lens 5	3.984	ASP	Plastic	1.544	55.9	8.78
11		22.194	ASP				
12	Lens 6	-6.388	ASP	Plastic	1.515	56.5	-9.30
13		19.537	ASP				
14	IR-cut filter	Plano	0.250	Glass	1.517	64.2	—
15		Plano	0.243				
16	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line).

TABLE 14

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-6.6173E-02	2.3051E+01	-6.6201E+01	-2.2148E+01	-3.0520E+01	-1.1495E+00
A4 =	-1.7293E-04	-2.8444E-03	-1.3202E-02	2.7224E-02	2.9988E-02	-2.6743E-02
A6 =	-8.5768E-05	3.3015E-03	1.8720E-02	-5.7739E-03	-2.2067E-02	1.2529E-02
A8 =	1.0669E-05	-1.3521E-03	-9.0282E-03	2.3104E-03	1.0079E-02	-4.0927E-03
A10 =	-4.1118E-06	3.0000E-04	2.4405E-03	-1.2372E-03	-3.4264E-03	8.1521E-04
A12 =	4.0213E-07	-2.5505E-05	-3.3712E-04	3.4725E-04	6.7832E-04	-5.5433E-05
A14 =			1.8339E-05	-3.5372E-05	-5.6245E-05	
Surface #	8	9	10	11	12	13
k =	8.8645E+01	-4.9096E+01	-1.5629E+01	5.8378E+01	3.4663E+00	2.2356E+01
A4 =	-7.1880E-03	-4.9496E-02	-3.8354E-02	-2.3781E-02	-5.2045E-02	-5.0696E-02
A6 =	3.5086E-03	3.3758E-02	2.9031E-02	1.4459E-02	3.1064E-02	1.7755E-02
A8 =	-5.8003E-04	-1.5867E-02	-1.1171E-02	-2.4618E-03	-7.0262E-03	-3.2723E-03
A10 =	2.7623E-05	4.6573E-03	2.2055E-03	-2.8425E-04	9.3458E-04	3.8954E-04
A12 =	1.0164E-05	-7.9691E-04	-2.4091E-04	1.4083E-04	-7.2199E-05	-3.2369E-05
A14 =		7.0870E-05	1.3181E-05	-1.6885E-05	2.9199E-06	2.1307E-06
A16 =		-2.2984E-06	-2.6371E-07	6.8378E-07	-4.2038E-08	-7.9033E-08

In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following conditions:

Furthermore, in the optical imaging lens assembly according to the 7th embodiment, when a focal length of the first lens element **710** is f_1 , a focal length of the second lens element **720** is f_2 , a focal length of the third lens element **730** is f_3 , a focal length of the fourth lens element **740** is f_4 , a focal length of the fifth lens element **750** is f_5 , and a focal length of the sixth lens element **760** is f_6 , the following conditions are satisfied: $|f_1| < |f_2|$; $|f_3| < |f_2|$; $|f_3| < |f_4|$; $|f_3| < |f_5|$; and $|f_3| < |f_6|$.

7th Embodiment				
f [mm]	8.38	f/f3		-2.13
Fno	2.06	(f/R5) - (f/R6) + (f × CT3)/(R5 × R6)		-3.09
HFOV [deg.]	18.5	Y11/Y62		0.75
Nmax	1.660	tan(2 × HFOV)		0.75
V3	20.4	SL/TL		0.93
CT1/CT2	2.79	BL/TD		0.09
(T34 + T56)/(T12 + T23 + T45)	3.12	f/EPD		2.06
R6/f	0.26	EPD/ImgH		1.41
(R5 + R6)/(R5 - R6)	1.28	TL/ImgH		3.39
R11/R12	0.33	ImgH/f		0.35
f/f1	1.47	TL/f		1.17

In the optical imaging lens assembly according to the 7th embodiment, when a central thickness of the first lens element **710** is CT1, a central thickness of the second lens element **720** is CT2, a central thickness of the third lens element **730** is CT3, a central thickness of the fourth lens element **740** is CT4, a central thickness of the fifth lens element **750** is CT5, and a central thickness of the sixth lens element **760** is CT6, CT1 is a maximum among CT1, CT2, CT3, CT4, CT5 and CT6.

According to the 7th embodiment, a number of inflection points of each of the object-side surfaces (711, 721, 731, 741, 751, 761) and the image-side surfaces (712, 722, 732, 742, 752, 762) of the first lens element **710**, the second lens element **720**, the third lens element **730**, the fourth lens element **740**, the fifth lens element **750** and the sixth lens element **760** are listed in the following Table, wherein the number of the inflection points of each surface is a number of the inflection points located between an axial point of the surface and a position of a maximum effective radius of the surface.

Lens element	First		Second		Third	
Surface	711	712	721	722	731	732
Number of reflection point(s)	0	1	1	1	1	0

Lens element	Fourth		Fifth		Sixth	
Surface	741	742	751	752	761	762
Number of reflection point(s)	0	2	1	3	1	2

8th Embodiment

FIG. 15 is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure. FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment. In FIG. 15, the image capturing device includes an optical imaging lens assembly (its reference numeral is omitted) and an image sensor **890**. The optical imaging lens assembly includes, in order from an object side to an image side, a first lens element **810**, an aperture stop **800**, a second lens element **820**, a third lens element **830**, a fourth lens element **840**, a fifth lens element **850**, a sixth lens element **860**, an IR-cut filter **870** and an image surface **880**, wherein the image sensor **890** is disposed on the image surface **880** of the optical imaging lens assembly. The optical imaging lens assembly has a total of six lens elements (**810-860**), and there is an air space between every two lens elements of the first lens element **810**, the second lens element **820**, the third lens element **830**, the fourth lens element **840**, the fifth lens element **850** and the sixth lens element **860** that are adjacent to each other.

The first lens element **810** with positive refractive power has an object-side surface **811** being convex and an image-side surface **812** being concave. The first lens element **810** is made of a plastic material, and has the object-side surface **811** and the image-side surface **812** being both aspheric.

The second lens element **820** with positive refractive power has an object-side surface **821** being concave and an image-side surface **822** being convex. The second lens element **820** is made of a plastic material, and has the object-side surface **821** and the image-side surface **822** being both aspheric. Furthermore, both of the object-side surface **821** and the image-side surface **822** of the second lens element **820** include at least one inflection point.

The third lens element **830** with negative refractive power has an object-side surface **831** being convex and an image-side surface **832** being concave. The third lens element **830** is made of a plastic material, and has the object-side surface **831** and the image-side surface **832** being both aspheric. Furthermore, the object-side surface **831** of the third lens element **830** includes at least one inflection point.

The fourth lens element **840** with positive refractive power has an object-side surface **841** being convex and an image-side surface **842** being convex. The fourth lens element **840** is made of a plastic material, and has the object-side surface **841** and the image-side surface **842** being both aspheric. Furthermore, both of the object-side surface **841** and the image-side surface **842** of the fourth lens element **840** include at least one inflection point.

The fifth lens element **850** with positive refractive power has an object-side surface **851** being convex and an image-side surface **852** being convex. The fifth lens element **850** is made of a plastic material, and has the object-side surface **851** and the image-side surface **852** being both aspheric. Furthermore, both of the object-side surface **851** and the image-side surface **852** of the fifth lens element **850** include at least one inflection point.

The sixth lens element **860** with negative refractive power has an object-side surface **861** being concave in a paraxial region thereof and an image-side surface **862** being concave in a paraxial region thereof. The sixth lens element **860** is made of a plastic material, and has the object-side surface **861** and the image-side surface **862** being both aspheric. Furthermore, the object-side surface **861** of the sixth lens element **860** includes at least one convex shape in an off-axial region thereof, the image-side surface **862** of the sixth lens element **860** includes at least one convex shape in an off-axial region thereof, and both of the object-side surface **861** and the image-side surface **862** of the sixth lens element **860** include at least one inflection point.

The IR-cut filter **870** is made of a glass material and located between the sixth lens element **860** and the image surface **880**, and will not affect the focal length of the optical imaging lens assembly.

The detailed optical data of the 8th embodiment are shown in Table 15 and the aspheric surface data are shown in Table 16 below.

TABLE 15

8th Embodiment								
f = 8.39 mm, Fno = 2.10, HFOV = 18.5 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Lens 1	2.574	ASP	Plastic	1.535	55.8	4.99	
2		60.760	ASP					

TABLE 15-continued

8th Embodiment f = 8.39 mm, Fno = 2.10, HFOV = 18.5 deg.						
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #
3	Ape. Stop	Plano	0.081			
4	Lens 2	-9.490 ASP	0.359	Plastic	1.639	23.3
5		-4.391 ASP	0.138			
6	Lens 3	19.285 ASP	0.374	Plastic	1.660	20.4
7		2.647 ASP	2.765			
8	Lens 4	112.959 ASP	0.680	Plastic	1.639	23.3
9		-19.984 ASP	0.171			
10	Lens 5	6.954 ASP	0.323	Plastic	1.544	55.9
11		-434.189 ASP	0.349			
12	Lens 6	-6.271 ASP	0.350	Plastic	1.535	55.8
13		6.025 ASP	0.500			
14	IR-cut filter	Plano	0.250	Glass	1.517	64.2
15		Plano	0.693			
16	Image	Plano	—			

Reference wavelength is 587.6 nm (d-line).

TABLE 16

Aspheric Coefficients						
Surface #	1	2	4	5	6	7
k =	-3.8598E-02	9.0000E+01	-9.0000E+01	-2.2424E+01	-9.0000E+01	-4.2836E-01
A4 =	1.6885E-04	-4.0947E-03	-1.0201E-02	2.8291E-02	2.6952E-02	-2.2009E-02
A6 =	-9.0468E-06	3.5907E-03	1.9274E-02	-5.6340E-03	-2.2185E-02	1.1966E-02
A8 =	3.9299E-06	-1.2638E-03	-8.9724E-03	2.3857E-03	1.0092E-02	-3.8808E-03
A10 =	-3.2166E-06	3.0220E-04	2.4348E-03	-1.1879E-03	-3.3432E-03	1.1482E-03
A12 =	-6.2665E-07	-2.7291E-05	-3.3916E-04	3.4159E-04	7.7423E-04	2.9616E-06
A14 =			2.0581E-05	-3.2829E-05	-8.2621E-05	
Surface #	8	9	10	11	12	13
k =	-9.0000E+01	7.1232E+01	-9.0000E+01	9.0000E+01	4.6232E+00	-6.2040E+01
A4 =	-7.7227E-03	-5.3755E-02	-5.6413E-02	-2.3597E-02	-5.3452E-02	-3.7789E-02
A6 =	8.0785E-04	3.3510E-02	2.9369E-02	1.4146E-02	3.1652E-02	1.3903E-02
A8 =	-8.2678E-04	-1.5996E-02	-1.1402E-02	-2.3102E-03	-6.9972E-03	-3.2624E-03
A10 =	4.3941E-05	4.6225E-03	2.2174E-03	-2.6825E-04	9.3235E-04	4.0942E-04
A12 =	1.6866E-05	-7.9728E-04	-2.3878E-04	1.4224E-04	-7.2227E-05	-3.0726E-05
A14 =		7.1962E-05	1.2759E-05	-1.6909E-05	2.8833E-06	2.1033E-06
A16 =		-2.2200E-06	-1.4697E-07	6.1765E-07	-5.0826E-08	-1.1447E-07

In the 8th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 15 and Table 16 as the following values and satisfy the following conditions:

8th Embodiment			
f [mm]	8.39	f/f3	-1.79
Fno	2.10	(f/R5) - (f/R6) + (f × CT3)/(R5 × R6)	-2.67
HFOV [deg.]	18.5	Y11/Y62	0.80
Nmax	1.660	tan (2 × H FOV)	0.75
V3	20.4	SL/TL	0.84
CT1/CT2	3.47	BL/TD	0.21
(T34 + T56)/(T12 + T23 + T45)	6.78	f/EPD	2.10
R6/f	0.32	EPD/ImgH	1.38
(R5 + R6)/(R5 - R6)	1.32	TL/ImgH	2.88
R11/R12	1.04	ImgH/f	0.34
f/f1	1.68	TL/f	0.99

Furthermore, in the optical imaging lens assembly according to the 8th embodiment, when a focal length of the first lens element **810** is f1, a focal length of the second lens element **820** is f2, a focal length of the third lens element **830** is f3, a focal length of the fourth lens element **840** is f4, a focal length of the fifth lens element **850** is f5, and a focal length of the sixth lens element **860** is f6, the following conditions are satisfied: |f1|<|f2|; |f3|<|f2|; |f3|<|f4|; |f3|<|f5|; and |f3|<|f6|.

In the optical imaging lens assembly according to the 8th embodiment, when a central thickness of the first lens element **810** is CT1, a central thickness of the second lens element **820** is CT2, a central thickness of the third lens element **830** is CT3, a central thickness of the fourth lens element **840** is CT4, a central thickness of the fifth lens element **850** is CT5, and a central thickness of the sixth lens element **860** is CT6, CT1 is a maximum among CT1, CT2, CT3, CT4, CT5 and CT6.

In the optical imaging lens assembly according to the 8th embodiment, when the axial distance between the first lens element **810** and the second lens element **820** is T12, the axial distance between the second lens element **820** and the third lens element **830** is T23, the axial distance between the third lens element **830** and the fourth lens element **840** is T34, the axial distance between the fourth lens element **840** and the fifth lens element **850** is T45, the axial distance between the fifth lens element **850** and the sixth lens element **860** is T56, and T34 is a maximum among T12, T23, T34, T45 and T56.

According to the 8th embodiment, a number of inflection points of each of the object-side surfaces (811, 821, 831, 841, 851, 861) and the image-side surfaces (812, 822, 832, 842, 852, 862) of the first lens element **810**, the second lens element **820**, the third lens element **830**, the fourth lens element **840**, the fifth lens element **850** and the sixth lens element **860** are listed in the following Table, wherein the number of the inflection points of each surface is a number of the inflection points located between an axial point of the surface and a position of a maximum effective radius of the surface.

Lens element	First		Second		Third	
Surface	811	812	821	822	831	832
Number of reflection point(s)	0	0	1	1	1	0
Lens element	Fourth		Fifth		Sixth	
Surface	841	842	851	852	861	862
Number of reflection point(s)	2	1	1	2	2	1

9th Embodiment

FIG. 17 is a schematic view of an image capturing device according to the 9th embodiment of the present disclosure. FIG. 18 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 9th embodiment. In FIG. 17, the image capturing device includes an optical imaging lens assembly (its reference numeral is omitted) and an image sensor **990**. The optical imaging lens assembly includes, in order from an object side to an image side, a first lens element **910**, an aperture stop **900**, a second lens element **920**, a third lens element **930**, a fourth lens element **940**, a fifth lens element **950**, a sixth lens element **960**, an IR-cut filter **970** and an

image surface **980**, wherein the image sensor **990** is disposed on the image surface **980** of the optical imaging lens assembly. The optical imaging lens assembly has a total of six lens elements (**910-960**), and there is an air space between every two lens elements of the first lens element **910**, the second lens element **920**, the third lens element **930**, the fourth lens element **940**, the fifth lens element **950** and the sixth lens element **960** that are adjacent to each other.

The first lens element **910** with positive refractive power has an object-side surface **911** being convex and an image-side surface **912** being concave. The first lens element **910** is made of a plastic material, and has the object-side surface **911** and the image-side surface **912** being both aspheric.

The second lens element **920** with positive refractive power has an object-side surface **921** being concave and an image-side surface **922** being convex. The second lens element **920** is made of a plastic material, and has the object-side surface **921** and the image-side surface **922** being both aspheric. Furthermore, both of the object-side surface **921** and the image-side surface **922** of the second lens element **920** include at least one inflection point.

The third lens element **930** with negative refractive power has an object-side surface **931** being concave and an image-side surface **932** being concave. The third lens element **930** is made of a plastic material, and has the object-side surface **931** and the image-side surface **932** being both aspheric. Furthermore, the object-side surface **931** of the third lens element **930** includes at least one inflection point.

The fourth lens element **940** with positive refractive power has an object-side surface **941** being concave and an image-side surface **942** being convex. The fourth lens element **940** is made of a plastic material, and has the object-side surface **941** and the image-side surface **942** being both aspheric.

The fifth lens element **950** with positive refractive power has an object-side surface **951** being convex and an image-side surface **952** being convex. The fifth lens element **950** is made of a plastic material, and has the object-side surface **951** and the image-side surface **952** being both aspheric. Furthermore, both of the object-side surface **951** and the image-side surface **952** of the fifth lens element **950** include at least one inflection point.

The sixth lens element **960** with negative refractive power has an object-side surface **961** being concave in a paraxial region thereof and an image-side surface **962** being concave in a paraxial region thereof. The sixth lens element **960** is made of a plastic material, and has the object-side surface **961** and the image-side surface **962** being both aspheric. Furthermore, the object-side surface **961** of the sixth lens element **960** includes at least one convex shape in an off-axial region thereof, the image-side surface **962** of the sixth lens element **960** includes at least one convex shape in an off-axial region thereof, and both of the object-side surface **961** and the image-side surface **962** of the sixth lens element **960** include at least one inflection point.

The IR-cut filter **970** is made of a glass material and located between the sixth lens element **960** and the image surface **980**, and will not affect the focal length of the optical imaging lens assembly.

The detailed optical data of the 9th embodiment are shown in Table 17 and the aspheric surface data are shown in Table 18 below.

TABLE 17

9th Embodiment								
f = 10.14 mm, Fno = 2.20, HFOV = 15.4 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Lens 1	2.686 ASP	1.540	Plastic	1.535	55.8	5.36	
2		34.347 ASP	0.186					
3	Ape. Stop	Plano	0.163					
4	Lens 2	-11.215 ASP	0.363	Plastic	1.639	23.3	10.86	
5		-4.340 ASP	0.104					
6	Lens 3	-35.337 ASP	0.280	Plastic	1.660	20.4	-4.36	
7		3.142 ASP	2.773					
8	Lens 4	-26.193 ASP	0.732	Plastic	1.639	23.3	20.31	
9		-8.770 ASP	0.032					
10	Lens 5	9.855 ASP	0.342	Plastic	1.544	55.9	17.81	
11		-556.656 ASP	0.196					
12	Lens 6	-6.210 ASP	0.350	Plastic	1.535	55.8	-6.99	
13		9.575 ASP	0.600					
14	IR-cut filter	Plano	0.250	Glass	1.517	64.2	—	
15		Plano	1.583					
16	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

TABLE 18

Aspheric Coefficients						
Surface #	1	2	4	5	6	7
k =	-1.4656E-02	-9.0000E+01	-8.9898E+01	-2.1478E+01	9.0000E+01	-4.4080E-01
A4 =	-9.5998E-04	-3.6922E-03	-7.5437E-03	2.8669E-02	2.5433E-02	-2.2266E-02
A6 =	2.0883E-04	3.4614E-03	1.9483E-02	-5.5008E-03	-2.2715E-02	1.1786E-02
A8 =	-3.8823E-05	-1.3235E-03	-9.0520E-03	2.3326E-03	1.0174E-02	-3.4264E-03
A10 =	-1.4823E-05	2.8182E-04	2.4195E-03	-1.2158E-03	-3.2308E-03	1.0127E-03
A12 =	2.2329E-06	-2.1825E-05	-3.3174E-04	3.3090E-04	6.9692E-04	-2.4616E-05
A14 =			2.1139E-05	-2.5592E-05	-6.8153E-05	
Surface #	8	9	10	11	12	13
k =	-9.0000E+01	1.3249E+01	-9.0000E+01	9.0000E+01	5.0805E+00	-6.7335E+01
A4 =	-1.8865E-03	-5.6816E-02	-8.1014E-02	-3.1420E-02	-4.4107E-02	-3.3969E-02
A6 =	-1.3800E-03	3.5102E-02	3.2731E-02	1.5149E-02	3.1597E-02	1.3112E-02
A8 =	-7.4223E-04	-1.6051E-02	-1.1333E-02	-2.2372E-03	-6.9997E-03	-3.0612E-03
A10 =	7.9994E-05	4.5688E-03	2.2587E-03	-2.7710E-04	9.2855E-04	4.4093E-04
A12 =	-3.1540E-06	-7.9722E-04	-2.3317E-04	1.4028E-04	-7.3078E-05	-2.9220E-05
A14 =		7.3209E-05	1.2561E-05	-1.7116E-05	2.7954E-06	1.9056E-06
A16 =		-2.2712E-06	-1.8546E-07	6.4450E-07	1.0354E-08	-1.9328E-07

In the 9th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 9th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 17 and Table 18 as the following values and satisfy the following conditions:

9th Embodiment				
f [mm]	10.14	f/f3		-2.32
Fno	2.20	(f/R5) - (f/R6) + (f × CT3)/(R5 × R6)		-3.54
HFOV [deg.]	15.4	Y11/Y62		0.98
Nmax	1.660	tan(2 × HFOV)		0.60
V3	20.4	SL/TL		0.82
CT1/CT2	4.24	BL/TD		0.34

-continued

9th Embodiment			
$(T34 + T56)/(T12 + T23 + T45)$	6.12	f/EPD	2.20
$R6/f$	0.31	$EPD/ImgH$	1.59
$(R5 + R6)/(R5 - R6)$	0.84	$TL/ImgH$	3.28
$ R11/R12 $	0.65	$ImgH/f$	0.29
$f/f1$	1.89	TL/f	0.94

Furthermore, in the optical imaging lens assembly according to the 9th embodiment, when a focal length of the first lens element **910** is $f1$, a focal length of the second lens element **920** is $f2$, a focal length of the third lens element **930** is $f3$, a focal length of the fourth lens element **940** is $f4$, a focal length of the fifth lens element **950** is $f5$, and a focal length of the sixth lens element **960** is $f6$, the following conditions are satisfied: $|f1| < |f2|$; $|f3| < |f2|$; $|f3| < |f4|$; $|f3| < |f5|$; and $|f3| < |f6|$.

In the optical imaging lens assembly according to the 9th embodiment, when a central thickness of the first lens element **910** is $CT1$, a central thickness of the second lens element **920** is $CT2$, a central thickness of the third lens element **930** is $CT3$, a central thickness of the fourth lens element **940** is $CT4$, a central thickness of the fifth lens element **950** is $CT5$, and a central thickness of the sixth lens element **960** is $CT6$, $CT1$ is a maximum among $CT1$, $CT2$, $CT3$, $CT4$, $CT5$ and $CT6$.

In the optical imaging lens assembly according to the 9th embodiment, when the axial distance between the first lens element **910** and the second lens element **920** is $T12$, the axial distance between the second lens element **920** and the third lens element **930** is $T23$, the axial distance between the third lens element **930** and the fourth lens element **940** is $T34$, the axial distance between the fourth lens element **940** and the fifth lens element **950** is $T45$, the axial distance between the fifth lens element **950** and the sixth lens element **960** is $T56$, and $T34$ is a maximum among $T12$, $T23$, $T34$, $T45$ and $T56$.

According to the 9th embodiment, a number of inflection points of each of the object-side surfaces (911, 921, 931, 941, 951, 961) and the image-side surfaces (912, 922, 932, 942, 952, 962) of the first lens element **910**, the second lens element **920**, the third lens element **930**, the fourth lens element **940**, the fifth lens element **950** and the sixth lens element **960** are listed in the following Table, wherein the number of the inflection points of each surface is a number of the inflection points located between an axial point of the surface and a position of a maximum effective radius of the surface.

Lens element	First		Second		Third	
Surface	911	912	921	922	931	932
Number of reflection point(s)	0	0	1	1	2	0

Lens element	Fourth		Fifth		Sixth	
Surface	941	942	951	952	961	962
Number of reflection point(s)	0	0	2	2	2	2

10th Embodiment

FIG. **19** is a schematic view of an image capturing device according to the 10th embodiment of the present disclosure. FIG. **20** shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 10th embodiment. In FIG. **19**, the image capturing device includes an optical imaging lens assembly (its reference numeral is omitted) and an image sensor **1090**. The optical imaging lens assembly includes, in order from an object side to an image side, an aperture stop **1000**, a first lens element **1010**, a second lens element **1020**, a third lens element **1030**, a fourth lens element **1040**, a fifth lens element **1050**, a sixth lens element **1060**, an IR-cut filter **1070** and an image surface **1080**, wherein the image sensor **1090** is disposed on the image surface **1080** of the optical imaging lens assembly. The optical imaging lens assembly has a total of six lens elements (**1010-1060**), and there is an air space between every two lens elements of the first lens element **1010**, the second lens element **1020**, the third lens element **1030**, the fourth lens element **1040**, the fifth lens element **1050** and the sixth lens element **1060** that are adjacent to each other.

The first lens element **1010** with positive refractive power has an object-side surface **1011** being convex and an image-side surface **1012** being concave. The first lens element **1010** is made of a plastic material, and has the object-side surface **1011** and the image-side surface **1012** being both aspheric.

The second lens element **1020** with positive refractive power has an object-side surface **1021** being concave and an image-side surface **1022** being convex. The second lens element **1020** is made of a plastic material, and has the object-side surface **1021** and the image-side surface **1022** being both aspheric. Furthermore, both of the object-side surface **1021** and the image-side surface **1022** of the second lens element **1020** include at least one inflection point.

The third lens element **1030** with negative refractive power has an object-side surface **1031** being concave and an image-side surface **1032** being concave. The third lens element **1030** is made of a plastic material, and has the object-side surface **1031** and the image-side surface **1032** being both aspheric. Furthermore, the object-side surface **1031** of the third lens element **1030** includes at least one inflection point.

The fourth lens element **1040** with positive refractive power has an object-side surface **1041** being concave and an image-side surface **1042** being convex. The fourth lens element **1040** is made of a plastic material, and has the object-side surface **1041** and the image-side surface **1042** being both aspheric.

The fifth lens element **1050** with positive refractive power has an object-side surface **1051** being convex and an image-side surface **1052** being concave. The fifth lens element **1050** is made of a plastic material, and has the object-side surface **1051** and the image-side surface **1052** being both aspheric. Furthermore, both of the object-side surface **1051**

and the image-side surface **1052** of the fifth lens element **1050** include at least one inflection point.

The sixth lens element **1060** with negative refractive power has an object-side surface **1061** being concave in a paraxial region thereof and an image-side surface **1062** being concave in a paraxial region thereof. The sixth lens element **1060** is made of a plastic material, and has the object-side surface **1061** and the image-side surface **1062**

The IR-cut filter **1070** is made of a glass material and located between the sixth lens element **1060** and the image surface **1080**, and will not affect the focal length of the optical imaging lens assembly.

The detailed optical data of the 10th embodiment are shown in Table 19 and the aspheric surface data are shown in Table 20 below.

TABLE 19

10th Embodiment									
f = 9.90 mm, Fno = 2.20, HFOV = 15.8 deg.									
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length		
0	Object	Plano	Infinity						
1	Ape. Stop	Plano	-1.085						
2	Lens 1	2.829	ASP	1.466	Plastic	1.535	55.8	5.43	
3		91.666	ASP	0.327					
4	Lens 2	-9.642	ASP	0.434	Plastic	1.639	23.3	11.58	
5		-4.259	ASP	0.117					
6	Lens 3	-37.487	ASP	0.280	Plastic	1.660	20.4	-4.76	
7		3.437	ASP	3.057					
8	Lens 4	-30.034	ASP	0.741	Plastic	1.639	23.3	17.29	
9		-8.153	ASP	0.035					
10	Lens 5	10.003	ASP	0.341	Plastic	1.544	55.9	19.97	
11		125.777	ASP	0.181					
12	Lens 6	-6.533	ASP	0.350	Plastic	1.535	55.8	-6.57	
13		7.759	ASP	0.500					
14	IR-cut filter	Plano	0.250	Glass	1.517	64.2	—		
15		Plano	1.419						
16	Image	Plano	—						

Reference wavelength is 587.6 nm (d-line).

TABLE 20

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-1.1261E-02	9.0000E+01	-8.7265E+01	-2.1621E+01	9.0000E+01	-1.0540E+00
A4 =	-4.4236E-04	-3.1533E-03	-8.1415E-03	3.0204E-02	2.4999E-02	-2.4642E-02
A6 =	9.3935E-05	3.4924E-03	1.9421E-02	-5.5175E-03	-2.2801E-02	1.0609E-02
A8 =	-1.6688E-05	-1.3452E-03	-9.0688E-03	2.2667E-03	1.0097E-02	-3.5748E-03
A10 =	-1.0338E-05	2.7691E-04	2.4145E-03	-1.2173E-03	-3.2826E-03	1.0413E-03
A12 =	1.5212E-06	-2.0885E-05	-3.3085E-04	3.2569E-04	6.8995E-04	-7.7526E-05
A14 =			1.9879E-05	-2.7575E-05	-6.3017E-05	
Surface #	8	9	10	11	12	13
k =	-9.0000E+01	8.4137E+00	-9.0000E+01	-9.0000E+01	5.6752E+00	-2.7586E+01
A4 =	-2.4599E-03	-5.1550E-02	-8.0675E-02	-3.7059E-02	-4.3222E-02	-3.5597E-02
A6 =	-1.0512E-03	3.5086E-02	3.3099E-02	1.5955E-02	3.0988E-02	1.2841E-02
A8 =	-5.5244E-04	-1.6058E-02	-1.1625E-02	-2.1410E-03	-7.0067E-03	-3.1100E-03
A10 =	1.1862E-04	4.5382E-03	2.2467E-03	-2.7761E-04	9.3171E-04	4.3647E-04
A12 =	-1.9630E-05	-7.9934E-04	-2.3232E-04	1.3880E-04	-7.2415E-05	-2.9532E-05
A14 =		7.3368E-05	1.2659E-05	-1.7400E-05	2.8029E-06	1.9478E-06
A16 =		-2.3356E-06	-1.8213E-07	6.3379E-07	-2.5627E-08	-1.6909E-07

being both aspheric. Furthermore, the object-side surface **1061** of the sixth lens element **1060** includes at least one convex shape in an off-axial region thereof, the image-side surface **1062** of the sixth lens element **1060** includes at least one convex shape in an off-axial region thereof, and both of the object-side surface **1061** and the image-side surface **1062** of the sixth lens element **1060** include at least one inflection point.

The aperture stop **1000** is located along an optical axis between a center of the object-side surface **1011** of the first lens element **1010** on the optical axis and a location of a maximum effective radius of the object-side surface **1011** of the first lens element **1010** projected on the optical axis.

In the 10th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 10th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 19 and Table 20 as the following values and satisfy the following conditions:

10th Embodiment			
f [mm]	9.90	f/f3	-2.08
Fno	2.20	(f/R5) - (f/R6) + (f × CT3/(R5 × R6))	-3.17
HFOV [deg.]	15.8	Y11/Y62	0.90
Nmax	1.660	tan(2 × HFOV)	0.62
V3	20.4	SL/TL	0.89
CT1/CT2	3.38	BL/TD	0.30
(T34 + T56)/(T12 + T23 + T45)	6.76	f/EPD	2.20
R6/f	0.35	EPD/ImgH	1.56
(R5 + R6)/(R5 - R6)	0.83	TL/ImgH	3.28
R11/R12	0.84	ImgH/f	0.29
f/fl	1.82	TL/f	0.96

Furthermore, in the optical imaging lens assembly according to the 10th embodiment, when a focal length of the first lens element **1010** is f_1 , a focal length of the second lens element **1020** is f_2 , a focal length of the third lens element **1030** is f_3 , a focal length of the fourth lens element **1040** is f_4 , a focal length of the fifth lens element **1050** is f_5 , and a focal length of the sixth lens element **1060** is f_6 , the following conditions are satisfied: $|f_1| < |f_2|$; $|f_3| < |f_2|$; $|f_3| < |f_4|$; $|f_3| < |f_5|$; and $|f_3| < |f_6|$.

In the optical imaging lens assembly according to the 10th embodiment, when a central thickness of the first lens element **1010** is CT1, a central thickness of the second lens element **1020** is CT2, a central thickness of the third lens element **1030** is CT3, a central thickness of the fourth lens element **1040** is CT4, a central thickness of the fifth lens element **1050** is CT5, and a central thickness of the sixth lens element **1060** is CT6, CT1 is a maximum among CT1, CT2, CT3, CT4, CT5 and CT6.

In the optical imaging lens assembly according to the 10th embodiment, when the axial distance between the first lens element **1010** and the second lens element **1020** is T12, the axial distance between the second lens element **1020** and the third lens element **1030** is T23, the axial distance between the third lens element **1030** and the fourth lens element **1040** is T34, the axial distance between the fourth lens element **1040** and the fifth lens element **1050** is T45, the axial distance between the fifth lens element **1050** and the sixth lens element **1060** is T56, and T34 is a maximum among T12, T23, T34, T45 and T56.

According to the 10th embodiment, a number of inflection points of each of the object-side surfaces (1011, 1021, 1031, 1041, 1051, 1061) and the image-side surfaces (1012, 1022, 1032, 1042, 1052, 1062) of the first lens element **1010**, the second lens element **1020**, the third lens element **1030**, the fourth lens element **1040**, the fifth lens element **1050** and the sixth lens element **1060** are listed in the following Table, wherein the number of the inflection points of each surface is a number of the inflection points located between an axial point of the surface and a position of a maximum effective radius of the surface.

Lens element	First		Second		Third	
Surface	1011	1012	1021	1022	1031	1032
Number of	0	0	1	1	2	0

-continued

reflection point(s)						
Lens element	Fourth		Fifth		Sixth	
Surface	1041	1042	1051	1052	1061	1062
Number of reflection point(s)	0	0	2	3	2	3

11th Embodiment

FIG. **21** is a schematic view of an image capturing device according to the 11th embodiment of the present disclosure. FIG. **22** shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 11th embodiment. In FIG. **21**, the image capturing device includes an optical imaging lens assembly (its reference numeral is omitted) and an image sensor **1190**. The optical imaging lens assembly includes, in order from an object side to an image side, an aperture stop **1100**, a first lens element **1110**, a second lens element **1120**, a third lens element **1130**, a fourth lens element **1140**, a fifth lens element **1150**, a sixth lens element **1160**, an IR-cut filter **1170** and an image surface **1180**, wherein the image sensor **1190** is disposed on the image surface **1180** of the optical imaging lens assembly. The optical imaging lens assembly has a total of six lens elements (**1110-1160**), and there is an air space between every two lens elements of the first lens element **1110**, the second lens element **1120**, the third lens element **1130**, the fourth lens element **1140**, the fifth lens element **1150** and the sixth lens element **1160** that are adjacent to each other.

The first lens element **1110** with positive refractive power has an object-side surface **1111** being convex and an image-side surface **1112** being convex. The first lens element **1110** is made of a plastic material, and has the object-side surface **1111** and the image-side surface **1112** being both aspheric. Furthermore, the image-side surface **1112** of the first lens element **1110** includes at least one inflection point.

The second lens element **1120** with positive refractive power has an object-side surface **1121** being concave and an image-side surface **1122** being convex. The second lens element **1120** is made of a plastic material, and has the object-side surface **1121** and the image-side surface **1122** being both aspheric. Furthermore, both of the object-side

55

surface **1121** and the image-side surface **1122** of the second lens element **1120** include at least one inflection point.

The third lens element **1130** with negative refractive power has an object-side surface **1131** being convex and an image-side surface **1132** being concave. The third lens element **1130** is made of a plastic material, and has the object-side surface **1131** and the image-side surface **1132** being both aspheric. Furthermore, both of the object-side surface **1131** and the image-side surface **1132** of the third lens element **1130** include at least one inflection point.

The fourth lens element **1140** with positive refractive power has an object-side surface **1141** being concave and an image-side surface **1142** being convex. The fourth lens element **1140** is made of a plastic material, and has the object-side surface **1141** and the image-side surface **1142** being both aspheric.

The fifth lens element **1150** with positive refractive power has an object-side surface **1151** being convex and an image-side surface **1152** being convex. The fifth lens element **1150** is made of a plastic material, and has the object-side surface **1151** and the image-side surface **1152** being both aspheric. Furthermore, both of the object-side surface **1151** and the image-side surface **1152** of the fifth lens element **1150** include at least one inflection point.

56

The sixth lens element **1160** with negative refractive power has an object-side surface **1161** being concave in a paraxial region thereof and an image-side surface **1162** being concave in a paraxial region thereof. The sixth lens element **1160** is made of a plastic material, and has the object-side surface **1161** and the image-side surface **1162** being both aspheric. Furthermore, the object-side surface **1161** of the sixth lens element **1160** includes at least one convex shape in an off-axial region thereof, the image-side surface **1162** of the sixth lens element **1160** includes at least one convex shape in an off-axial region thereof, and both of the object-side surface **1161** and the image-side surface **1162** of the sixth lens element **1160** include at least one inflection point.

The aperture stop **1100** is located along an optical axis between a center of the object-side surface **1111** of the first lens element **1110** on the optical axis and a location of a maximum effective radius of the object-side surface **1111** of the first lens element **1110** projected on the optical axis.

The IR-cut filter **1170** is made of a glass material and located between the sixth lens element **1160** and the image surface **1180**, and will not affect the focal length of the optical imaging lens assembly.

The detailed optical data of the 11th embodiment are shown in Table 21 and the aspheric surface data are shown in Table 22 below.

TABLE 21

11th Embodiment								
f = 9.20 mm, Fno = 2.06, HFOV = 17.5 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.820					
2	Lens 1	3.504 ASP	1.600	Plastic	1.544	55.9	5.47	
3		-16.441 ASP	0.379					
4	Lens 2	-11.717 ASP	0.500	Plastic	1.639	23.3	9.21	
5		-3.980 ASP	0.258					
6	Lens 3	5.570 ASP	0.419	Plastic	1.660	20.4	-3.08	
7		1.444 ASP	2.304					
8	Lens 4	-9.874 ASP	0.822	Plastic	1.639	23.3	22.95	
9		-6.091 ASP	0.350					
10	Lens 5	18.701 ASP	0.785	Plastic	1.544	55.9	6.31	
11		-4.141 ASP	0.692					
12	Lens 6	-4.312 ASP	0.480	Plastic	1.515	56.5	-7.23	
13		28.171 ASP	0.500					
14	IR-cut filter	Plano	0.250	Glass	1.517	64.2	—	
15		Plano	0.958					
16	Image	Plano	—					

Reference wavelength is 587.6 nm (d-line).

TABLE 22

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	4.2068E-02	4.5253E+01	-8.9271E+01	-1.6265E+01	-7.9060E+00	-1.3194E+00
A4 =	4.0771E-04	5.2264E-03	-1.4365E-02	2.1959E-02	2.4214E-02	-3.0565E-02
A6 =	-5.1438E-05	2.7063E-03	1.8692E-02	-5.4763E-03	-2.2050E-02	1.4004E-02
A8 =	9.3272E-05	-1.3517E-03	-9.1805E-03	2.7664E-03	1.1203E-02	-3.4527E-03
A10 =	-2.3444E-05	2.6225E-04	2.5022E-03	-1.1933E-03	-3.2983E-03	7.9158E-04
A12 =	1.9715E-06	-1.6787E-05	-3.1397E-04	3.9624E-04	5.8383E-04	-1.1405E-04
A14 =			1.3406E-05	-4.5889E-05	-4.9120E-05	
Surface #	8	9	10	11	12	13
k =	-4.7133E+00	-6.4132E+01	3.2665E+01	9.4266E-01	7.7892E-01	8.3695E+00
A4 =	-1.3025E-02	-6.3230E-02	-3.9687E-02	-1.2164E-02	-4.3127E-02	-4.8918E-02

TABLE 22-continued

Aspheric Coefficients					
A6 =	2.1021E-03	3.5236E-02	2.9089E-02	1.5289E-02	3.1116E-02
A8 =	-6.7806E-05	-1.6113E-02	-1.1151E-02	-2.5450E-03	-7.1692E-03
A10 =	1.4425E-05	4.6547E-03	2.2019E-03	-2.8635E-04	9.3807E-04
A12 =	-3.2189E-06	-7.8802E-04	-2.3763E-04	1.4163E-04	-7.1278E-05
A14 =		7.2083E-05	1.3747E-05	-1.6511E-05	2.9419E-06
A16 =		-2.7634E-06	-3.3658E-07	6.7268E-07	-4.9891E-08
					-6.3087E-08

In the 11th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 11th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 21 and Table 22 as the following values and satisfy the following conditions:

11th Embodiment			
f [mm]	9.20	f/f3	-2.99
Fno	2.06	(f/R5) - (f/R6) + (f × CT3)/(R5 × R6)	-4.24
HFOV [deg.]	17.5	Y11/Y62	0.76
Nmax	1.660	tan(2 × HFOV)	0.70
V3	20.4	SL/TL	0.92
CT1/CT2	3.20	BL/TD	0.20
(T34 + T56)/(T12 + T23 + T45)	3.04	f/EPD	2.06
R6/f	0.16	EPD/ImgH	1.54
(R5 + R6)/(R5 - R6)	1.70	TL/ImgH	3.56
R11/R12	0.15	ImgH/f	0.31
f/fl	1.68	TL/f	1.12

Furthermore, in the optical imaging lens assembly according to the 11th embodiment, when a focal length of the first lens element **1110** is f1, a focal length of the second lens element **1120** is f2, a focal length of the third lens element **1130** is f3, a focal length of the fourth lens element **1140** is f4, a focal length of the fifth lens element **1150** is f5, and a focal length of the sixth lens element **1160** is f6, the following conditions are satisfied: |f1|<|f2|; |f3|<|f2|; |f3|<|f4|; |f3|<|f5|; and |f3|<|f6|.

In the optical imaging lens assembly according to the 11th embodiment, when a central thickness of the first lens element **1110** is CT1, a central thickness of the second lens element **1120** is CT2, a central thickness of the third lens element **1130** is CT3, a central thickness of the fourth lens element **1140** is CT4, a central thickness of the fifth lens element **1150** is CT5, and a central thickness of the sixth lens element **1160** is CT6, CT1 is a maximum among CT1, CT2, CT3, CT4, CT5 and CT6.

In the optical imaging lens assembly according to the 11th embodiment, when the axial distance between the first lens element **1110** and the second lens element **1120** is T12, the axial distance between the second lens element **1120** and the third lens element **1130** is T23, the axial distance between the third lens element **1130** and the fourth lens element **1140** is T34, the axial distance between the fourth lens element **1140** and the fifth lens element **1150** is T45, the axial distance between the fifth lens element **1150** and the sixth lens element **1160** is T56, and T34 is a maximum among T12, T23, T34, T45 and T56.

According to the 11th embodiment, a number of inflection points of each of the object-side surfaces (1111, 1121, 1131, 1141, 1151, 1161) and the image-side surfaces (1112, 1122,

1132, 1142, 1152, 1162) of the first lens element **1110**, the second lens element **1120**, the third lens element **1130**, the fourth lens element **1140**, the fifth lens element **1150** and the sixth lens element **1160** are listed in the following Table, wherein the number of the inflection points of each surface is a number of the inflection points located between an axial point of the surface and a position of a maximum effective radius of the surface.

Lens element	First		Second		Third	
Surface	1111	1112	1121	1122	1131	1132
Number of reflection point(s)	0	1	1	1	1	1

Lens element	Fourth		Fifth		Sixth	
Surface	1141	1142	1151	1152	1161	1162
Number of reflection point(s)	0	0	4	3	2	3

12th Embodiment

FIG. 23 is a schematic view of an image capturing device according to the 12th embodiment of the present disclosure. FIG. 24 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 12th embodiment. In FIG. 23, the image capturing device includes an optical imaging lens assembly (its reference numeral is omitted) and an image sensor **1290**. The optical imaging lens assembly includes, in order from an object side to an image side, an aperture stop **1200**, a first

59

lens element **1210**, a second lens element **1220**, a third lens element **1230**, a fourth lens element **1240**, a fifth lens element **1250**, a sixth lens element **1260**, an IR-cut filter **1270** and an image surface **1280**, wherein the image sensor **1290** is disposed on the image surface **1280** of the optical imaging lens assembly. The optical imaging lens assembly has a total of six lens elements (**1210-1260**), and there is an air space between every two lens elements of the first lens element **1210**, the second lens element **1220**, the third lens element **1230**, the fourth lens element **1240**, the fifth lens element **1250** and the sixth lens element **1260** that are adjacent to each other.

The first lens element **1210** with positive refractive power has an object-side surface **1211** being convex and an image-side surface **1212** being convex. The first lens element **1210** is made of a plastic material, and has the object-side surface **1211** and the image-side surface **1212** being both aspheric. Furthermore, the image-side surface **1212** of the first lens element **1210** includes at least one inflection point.

The second lens element **1220** with negative refractive power has an object-side surface **1221** being concave and an image-side surface **1222** being concave. The second lens element **1220** is made of a plastic material, and has the object-side surface **1221** and the image-side surface **1222** being both aspheric. Furthermore, the object-side surface **1221** of the second lens element **1220** includes at least one inflection point.

The third lens element **1230** with positive refractive power has an object-side surface **1231** being convex and an image-side surface **1232** being concave. The third lens element **1230** is made of a plastic material, and has the object-side surface **1231** and the image-side surface **1232** being both aspheric. Furthermore, both of the object-side surface **1231** and the image-side surface **1232** of the third lens element **1230** include at least one inflection point.

The fourth lens element **1240** with negative refractive power has an object-side surface **1241** being concave and an

60

image-side surface **1242** being convex. The fourth lens element **1240** is made of a plastic material, and has the object-side surface **1241** and the image-side surface **1242** being both aspheric. Furthermore, both of the object-side surface **1241** and the image-side surface **1242** of the fourth lens element **1240** include at least one inflection point.

The fifth lens element **1250** with positive refractive power has an object-side surface **1251** being convex and an image-side surface **1252** being convex. The fifth lens element **1250** is made of a plastic material, and has the object-side surface **1251** and the image-side surface **1252** being both aspheric. Furthermore, both of the object-side surface **1251** and the image-side surface **1252** of the fifth lens element **1250** include at least one inflection point.

The sixth lens element **1260** with negative refractive power has an object-side surface **1261** being concave in a paraxial region thereof and an image-side surface **1262** being convex in a paraxial region thereof. The sixth lens element **1260** is made of a plastic material, and has the object-side surface **1261** and the image-side surface **1262** being both aspheric. Furthermore, the object-side surface **1261** of the sixth lens element **1260** includes at least one convex shape in an off-axial region thereof, and both of the object-side surface **1261** and the image-side surface **1262** of the sixth lens element **1260** include at least one inflection point.

The aperture stop **1200** is located along an optical axis between a center of the object-side surface **1211** of the first lens element **1210** on the optical axis and a location of a maximum effective radius of the object-side surface **1211** of the first lens element **1210** projected on the optical axis.

The IR-cut filter **1270** is made of a glass material and located between the sixth lens element **1260** and the image surface **1280**, and will not affect the focal length of the optical imaging lens assembly.

The detailed optical data of the 12th embodiment are shown in Table 23 and the aspheric surface data are shown in Table 24 below.

TABLE 23

12th Embodiment									
f = 8.99 mm, Fno = 2.05, HFOV = 17.6 deg.									
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano		Infinity					
1	Ape. Stop	Plano		-0.850					
2	Lens 1	3.196	ASP	1.600	Plastic	1.544	55.9	4.81	
3		-11.877	ASP	0.141					
4	Lens 2	-4.104	ASP	0.400	Plastic	1.639	23.3	-4.76	
5		12.144	ASP	0.505					
6	Lens 3	1.963	ASP	0.709	Plastic	1.660	20.4	12.18	
7		2.225	ASP	2.155					
8	Lens 4	-1.308	ASP	0.500	Plastic	1.639	23.3	-10.31	
9		-1.875	ASP	0.100					
10	Lens 5	7.916	ASP	0.812	Plastic	1.544	55.9	4.33	
11		-3.229	ASP	0.419					
12	Lens 6	-3.021	ASP	0.983	Plastic	1.515	56.5	-8.80	
13		-10.066	ASP	1.000					
14	IR-cut filter	Plano		0.300	Glass	1.517	64.2	—	
15		Plano		0.859					
16	Image	Plano		—					

Reference wavelength is 587.6 (d-line).

TABLE 24

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-9.2116E-02	2.2242E+01	-2.2410E+01	-5.0774E+01	-3.5690E+00	-1.1116E+00
A4 =	3.0646E-04	1.2768E-02	2.6269E-02	3.2729E-02	1.3057E-02	-1.6668E-02
A6 =	-6.1711E-05	-9.0967E-04	-4.1586E-03	-2.3266E-03	-7.8569E-04	1.5712E-03
A8 =		-1.7983E-04	1.5079E-04	1.5746E-04	2.4389E-04	4.3100E-04
A10 =		4.0513E-05	1.9392E-05	-6.1928E-05	-1.5484E-04	-5.7035E-04
A12 =				1.6018E-05	1.6476E-05	7.9020E-05

Surface #	8	9	10	11	12	13
k =	-2.8738E+00	-4.7512E+00	8.2974E+00	1.5687E-01	-7.1699E-01	8.2870E+00
A4 =	-5.0856E-02	-4.5909E-02	-3.4282E-02	3.2353E-02	7.7633E-02	2.2151E-02
A6 =	2.2352E-02	2.2003E-02	9.6734E-03	1.3218E-03	-1.6924E-02	-7.3593E-03
A8 =	-1.1795E-02	-8.6563E-03	-2.7354E-03	-1.5282E-03	2.7254E-03	1.3807E-03
A10 =	3.3245E-03	2.0331E-03	3.3692E-04	1.9805E-04	-3.0249E-04	-1.4232E-04
A12 =	-3.0698E-04	-1.8002E-04	-1.5209E-05	-7.5355E-06	1.9304E-05	6.8176E-06
A14 =		5.1313E-06			-5.0715E-07	-8.2786E-08

In the 12th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 12th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 23 and Table 24 as the following values and satisfy the following conditions:

12th Embodiment			
f [mm]	8.99	f/f3	0.74
Fno	2.05	(f/R5) - (f/R6) + (f × CT3/(R5 × R6))	2.00
HFOV [deg.]	17.6	Y11/Y62	0.90
Nmax	1.660	tan(2 × HFOV)	0.70
V3	20.4	SL/TL	0.92
CT1/CT2	4.00	BL/TD	0.26
(T34 + T56)/(T12 + T23 + T45)	3.45	f/EPD	2.05
R6/f	0.25	EPD/ImgH	1.52
(R5 + R6)/(R5 - R6)	-16.01	TL/ImgH	3.62
R11/R12	0.30	ImgH/f	0.32
f/11	1.87	TL/f	1.17

In the optical imaging lens assembly according to the 12th embodiment, when a central thickness of the first lens element **1210** is CT1, a central thickness of the second lens element **1220** is CT2, a central thickness of the third lens element **1230** is CT3, a central thickness of the fourth lens element **1240** is CT4, a central thickness of the fifth lens element **1250** is CT5, and a central thickness of the sixth lens element **1260** is CT6, CT1 is a maximum among CT1, CT2, CT3, CT4, CT5 and CT6.

In the optical imaging lens assembly according to the 12th embodiment, when the axial distance between the first lens element **1210** and the second lens element **1220** is T12, the axial distance between the second lens element **1220** and the third lens element **1230** is T23, the axial distance between the third lens element **1230** and the fourth lens element **1240** is T34, the axial distance between the fourth lens element

20

1240 and the fifth lens element **1250** T45, the axial distance between the fifth lens element **1250** and the sixth lens element **1260** is T56, and T34 is a maximum among T12, T23, T34, T45 and T56.

According to the 12th embodiment, a number of inflection points of each of the object-side surfaces (1211, 1221, 1231, 1241, 1251, 1261) and the image-side surfaces (1212, 1222, 1232, 1242, 1252, 1262) of the first lens element **1210**, the second lens element **1220**, the third lens element **1230**, the fourth lens element **1240**, the fifth lens element **1250** and the

sixth lens element **1260** are listed in the following Table, wherein the number of the inflection points of each surface is a number of the inflection points located between an axial point of the surface and a position of a maximum effective radius of the surface.

Lens element	First		Second		Third	
Surface	1211	1212	1221	1222	1231	1232
Number of reflection point(s)	0	1	1	0	1	1

Lens element	Fourth		Fifth		Sixth	
Surface	1241	1242	1251	1252	1261	1262
Number of reflection point(s)	1	1	2	2	1	1

FIG. 25 is a schematic view of an image capturing device according to the 13th embodiment of the present disclosure. FIG. 26 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 13th embodiment. In FIG. 25, the image capturing device includes an optical imaging lens assembly (its reference numeral is omitted) and an image sensor 1390. The optical imaging lens assembly includes, in order from an object side to an image side, an aperture stop 1300, a first lens element 1310, a second lens element 1320, a third lens element 1330, a fourth lens element 1340, a fifth lens element 1350, a sixth lens element 1360, an IR-cut filter 1370 and an image surface 1380, wherein the image sensor 1390 is disposed on the image surface 1380 of the optical imaging lens assembly. The optical imaging lens assembly has a total of six lens elements (1310-1360), and there is an air space between every two lens elements of the first lens element 1310, the second lens element 1320, the third lens element 1330, the fourth lens element 1340, the fifth lens element 1350 and the sixth lens element 1360 that are adjacent to each other.

The first lens element 1310 with positive refractive power has an object-side surface 1311 being convex and an image-side surface 1312 being convex. The first lens element 1310 is made of a plastic material, and has the object-side surface 1311 and the image-side surface 1312 being both aspheric. Furthermore, the image-side surface 1312 of the first lens element 1310 includes at least one inflection point.

The second lens element 1320 with negative refractive power has an object-side surface 1321 being concave and an image-side surface 1322 being convex. The second lens element 1320 is made of a plastic material, and has the object-side surface 1321 and the image-side surface 1322 being both aspheric. Furthermore, both of the object-side surface 1321 and the image-side surface 1322 of the second lens element 1320 include at least one inflection point.

The third lens element 1330 with positive refractive power has an object-side surface 1331 being convex and an image-side surface 1332 being concave. The third lens element 1330 is made of a plastic material, and has the

object-side surface 1331 and the image-side surface 1332 being both aspheric. Furthermore, both of the object-side surface 1331 and the image-side surface 1332 of the third lens element 1330 include at least one inflection point.

The fourth lens element 1340 with negative refractive power has an object-side surface 1341 being concave and an image-side surface 1342 being convex. The fourth lens element 1340 is made of a plastic material, and has the object-side surface 1341 and the image-side surface 1342 being both aspheric. Furthermore, both of the object-side surface 1341 and the image-side surface 1342 of the fourth lens element 1340 include at least one inflection point.

The fifth lens element 1350 with positive refractive power has an object-side surface 1351 being convex and an image-side surface 1352 being convex. The fifth lens element 1350 is made of a plastic material, and has the object-side surface 1351 and the image-side surface 1352 being both aspheric. Furthermore, both of the object-side surface 1351 and the image-side surface 1352 of the fifth lens element 1350 include at least one inflection point.

The sixth lens element 1360 with negative refractive power has an object-side surface 1361 being concave in a paraxial region thereof and an image-side surface 1362 being convex in a paraxial region thereof. The sixth lens element 1360 is made of a plastic material, and has the object-side surface 1361 and the image-side surface 1362 being both aspheric. Furthermore, the object-side surface 1361 of the sixth lens element 1360 includes at least one convex shape in an off-axial region thereof, and the object-side surface 1361 of the sixth lens element 1360 includes at least one inflection point.

The aperture stop 1300 is located along an optical axis between a center of the object-side surface 1311 of the first lens element 1310 on the optical axis and a location of a maximum effective radius of the object-side surface 1311 of the first lens element 1310 projected on the optical axis.

The IR-cut filter 1370 is made of a glass material and located between the sixth lens element 1360 and the image surface 1380, and will not affect the focal length of the optical imaging lens assembly.

The detailed optical data of the 13th embodiment are shown in Table 25 and the aspheric surface data are shown in Table 26 below.

TABLE 25

13th Embodiment							
f = 9.00 mm, Fno = 2.05, HFOV = 17.6 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.720				
2	Lens 1	3.691 ASP	1.573	Plastic	1.544	55.9	4.73
3		-7.182 ASP	0.154				
4	Lens 2	-3.603 ASP	0.448	Plastic	1.639	23.3	-5.75
5		-200.000 ASP	0.124				
6	Lens 3	1.524 ASP	0.463	Plastic	1.660	20.4	29.79
7		1.453 ASP	2.436				
8	Lens 4	-1.539 ASP	0.500	Plastic	1.639	23.3	-14.90
9		-2.069 ASP	0.100				
10	Lens 5	9.785 ASP	0.539	Plastic	1.544	55.9	5.41
11		-4.121 ASP	0.511				
12	Lens 6	-4.129 ASP	1.312	Plastic	1.515	56.5	-11.53
13		-15.040 ASP	0.800				
14	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—
15		Plano	1.153				
16	Image	Plano	—				

Reference wavelength is 587.6 nm (d-line).

TABLE 26

Aspheric Coefficients						
Surface #	2	3	4	5	6	7
k =	-4.2435E-01	3.2757E+00	-3.1771E+01	9.0000E+01	-4.0292E+00	-1.3035E+00
A4 =	9.2705E-04	1.3260E-02	2.3493E-02	4.0523E-02	3.4304E-02	-3.5477E-02
A6 =	1.1417E-04	1.3973E-04	-3.1715E-03	-6.2854E-03	-7.4670E-03	1.9672E-02
A8 =		-9.1755E-04	-8.4211E-04	-6.4231E-04	1.3032E-03	-6.5866E-03
A10 =		2.2876E-04	3.3346E-04	4.2855E-04	-4.4667E-04	6.3761E-04
A12 =		-1.8101E-05	-3.1345E-05	-4.5280E-05	5.2218E-05	

Surface #	8	9	10	11	12	13
k =	-3.1567E+00	-5.1842E+00	1.8517E+01	5.0059E-01	7.2976E-02	2.2455E+01
A4 =	-1.6207E-02	-5.1491E-02	-3.6283E-02	5.0576E-02	6.7262E-02	6.0114E-03
A6 =	5.2855E-03	2.2041E-02	6.6139E-03	-8.7498E-03	-1.6392E-02	-3.8720E-03
A8 =	2.6418E-04	-5.6255E-03	-2.1053E-03	-6.0140E-04	2.9880E-03	8.2499E-04
A10 =	-4.8253E-04	1.0997E-03	-4.2499E-05	1.8082E-04	-4.0908E-04	-9.2664E-05
A12 =	7.6712E-05	-1.7626E-04	2.9891E-05	-6.0305E-06	3.3870E-05	4.1960E-06
A14 =		1.8973E-05			-1.1590E-06	

In the 13th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 13th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 25 and Table 26 as the following values and satisfy the following conditions:

13th Embodiment			
f [mm]	9.00	f/f3	0.30
Fno	2.05	(f/R5) - (f/R6) + (f × CT3/(R5 × R6))	1.59
HFOV [deg.]	17.6	Y11/Y62	0.90
Nmax	1.660	tan(2 × HFOV)	0.71
V3	20.4	SL/TL	0.93
CT1/CT2	3.51	BL/TD	0.28
(T34 + T56)/(T12 + T23 + T45)	7.80	f/EPD	2.05
R6/f	0.16	EPD/ImgH	1.52
(R5 + R6)/(R5 - R6)	41.71	TL/ImgH	3.60
R11/R12	0.27	ImgH/f	0.32
f/1	1.90	TL/f	1.16

Furthermore, in the optical imaging lens assembly according to the 13th embodiment, when a focal length of the first lens element **1310** is f1, a focal length of the second lens element **1320** is f2, the following condition is satisfied: |f1|<|f2|.

In the optical imaging lens assembly according to the 13th embodiment, when a central thickness of the first lens element **1310** is CT1, a central thickness of the second lens element **1320** is CT2, a central thickness of the third lens element **1330** is CT3, a central thickness of the fourth lens element **1340** is CT4, a central thickness of the fifth lens element **1350** is CT5, and a central thickness of the sixth lens element **1360** is CT6, CT1 is a maximum among CT1, CT2, CT3, CT4, CT5 and CT6.

In the optical imaging lens assembly according to the 13th embodiment, when the axial distance between the first lens element **1310** and the second lens element **1320** is T12, the

20

axial distance between the second lens element **1320** and the third lens element **1330** is T23, the axial distance between the third lens element **1330** and the fourth lens element **1340** is T34, the axial distance between the fourth lens element **1340** and the fifth lens element **1350** is T45, the axial distance between the fifth lens element **1350** and the sixth lens element **1360** is T56, and T34 is a maximum among T12, T23, T34, T45 and T56.

According to the 13th embodiment, a number of inflection points of each of the object-side surfaces (1311, 1321, 1331,

1341, 1351, 1361) and the image-side surfaces (1312, 1322, 1332, 1342, 1352, 1362) of the first lens element **1310**, the second lens element **1320**, the third lens element **1330**, the fourth lens element **1340**, the fifth lens element **1350** and the sixth lens element **1360** are listed in the following Table, wherein the number of the inflection points of each surface is a number of the inflection points located between an axial point of the surface and a position of a maximum effective radius of the surface.

Lens element	First		Second		Third	
Surface	1311	1312	1321	1322	1331	1332
Number of reflection point(s)	0	2	2	1	1	1

Lens element	Fourth		Fifth		Sixth	
Surface	1341	1342	1351	1352	1361	1362
Number of reflection point(s)	1	1	1	2	2	0

65

67

14th Embodiment

FIG. 27 is a schematic view of an electronic device 10 according to the 14th embodiment of the present disclosure. The electronic device 10 of the 14th embodiment is a smartphone, wherein the electronic device 10 includes an image capturing device 11. The image capturing device 11 includes an optical imaging lens assembly (its reference numeral is omitted) according to the present disclosure and an image sensor (its reference numeral is omitted), wherein the image sensor is disposed on an image surface of the optical imaging lens assembly.

15th Embodiment

FIG. 28 is a schematic view of an electronic device 20 according to the 15th embodiment of the present disclosure. The electronic device 20 of the 15th embodiment is a tablet personal computer, wherein the electronic device 20 includes an image capturing device 21. The image capturing device 21 includes an optical imaging lens assembly (its reference numeral is omitted) according to the present disclosure and an image sensor (its reference numeral is omitted), wherein the image sensor is disposed on an image surface of the optical imaging lens assembly.

16th Embodiment

FIG. 29 is a schematic view of an electronic device 30 according to the 16th embodiment of the present disclosure. The electronic device 30 of the 16th embodiment is a wearable device, wherein the electronic device 30 includes an image capturing device 31. The image capturing device 31 includes an optical imaging lens assembly (its reference numeral is omitted) according to the present disclosure and an image sensor (its reference numeral is omitted), wherein the image sensor is disposed on an image surface of the optical imaging lens assembly.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. It is to be noted that Tables 1-22 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. An optical imaging lens assembly comprising, in order from an object side to an image side:
 - a first lens element having positive refractive power;
 - a second lens element having positive refractive power;
 - a third lens element having negative refractive power;
 - a fourth lens element;
 - a fifth lens element; and
 - a sixth lens element having an image-side surface being concave, wherein at least one of an object-side surface and the image-side surface of the sixth lens element comprises at least one inflection point;
 wherein the optical imaging lens assembly has a total of six lens elements, a focal length of the first lens element

68

is f_1 , a focal length of the second lens element is f_2 , a focal length of the optical imaging lens assembly is f , a curvature radius of an object-side surface of the third lens element is R_5 , a curvature radius of an image-side surface of the third lens element is R_6 , a central thickness of the third lens element is CT_3 , an axial distance between the image-side surface of the sixth lens element and an image surface is BL , an axial distance between an object-side surface of the first lens element and the image-side surface of the sixth lens element is TD , a maximum image height of the optical imaging lens assembly is $ImgH$, and the following conditions are satisfied:

$$|f_1| < |f_2|;$$

$$-10.0 < (f/R_5) - (f/R_6) + (f \times CT_3 / (R_5 \times R_6)) < -1.7;$$

$$0 < BL/TD < 0.45; \text{ and}$$

$$0.10 < ImgH/f < 0.50.$$

2. The optical imaging lens assembly of claim 1, wherein at least one of the object-side surface and the image-side surface of the sixth lens element is concave in a paraxial region thereof and comprises at least one convex shape in an off-axial region thereof.

3. The optical imaging lens assembly of claim 1, wherein the object-side surface of the first lens element is convex, a central thickness of the first lens element is CT_1 , a central thickness of the second lens element is CT_2 , and the following condition is satisfied:

$$1.0 < CT_1/CT_2 < 7.0.$$

4. The optical imaging lens assembly of claim 1, further comprising:

an aperture stop disposed between an imaged object and the second lens element, wherein an axial distance between the aperture stop and the image surface is SL , an axial distance between the object-side surface of the first lens element and the image surface is TL , and the following condition is satisfied:

$$0.78 < SL/TL < 1.10.$$

5. The optical imaging lens assembly of claim 4, wherein the aperture stop is located along an optical axis between a center of the object-side surface of the first lens element on the optical axis and a location of a maximum effective radius of the object-side surface of the first lens element projected on the optical axis.

6. The optical imaging lens assembly of claim 1, wherein the sixth lens element has negative refractive power.

7. The optical imaging lens assembly of claim 1, wherein the fourth lens element has an object-side surface being concave and an image-side surface being convex.

8. The optical imaging lens assembly of claim 1, wherein an Abbe number of the third lens element is V_3 , and the following condition is satisfied:

$$V_3 < 25.0.$$

9. The optical imaging lens assembly of claim 1, wherein a half of a maximum field of view of the optical imaging lens assembly is HFOV, and the following condition is satisfied:

$$0.20 < \tan(2 \times HFOV) < 0.85.$$

10. The optical imaging lens assembly of claim 1, wherein an entrance pupil diameter of the optical imaging lens

69

assembly is EPD, the maximum image height of the optical imaging lens assembly is ImgH, and the following condition is satisfied:

$$1.25 < \text{EPD}/\text{ImgH} < 2.0.$$

11. The optical imaging lens assembly of claim 10, wherein the entrance pupil diameter of the optical imaging lens assembly is EPD, the maximum image height of the optical imaging lens assembly is ImgH, and the following condition is satisfied:

$$1.35 < \text{EPD}/\text{ImgH} < 2.0.$$

12. The optical imaging lens assembly of claim 1, wherein the curvature radius of the image-side surface of the third lens element is R6, the focal length of the optical imaging lens assembly is f, and the following condition is satisfied:

$$0 < R6/f < 0.45.$$

13. The optical imaging lens assembly of claim 1, wherein a curvature radius of the object-side surface of the sixth lens element is R11, a curvature radius of the image-side surface of the sixth lens element is R12, and the following condition is satisfied:

$$|R11/R12| < 1.30.$$

14. The optical imaging lens assembly of claim 1, wherein the curvature radius of the object-side surface of the third lens element is R5, the curvature radius of the image-side surface of the third lens element is R6, and the following condition is satisfied:

$$0.50 < (R5+R6)/(R5-R6) < 2.50.$$

15. The optical imaging lens assembly of claim 1, wherein at least three surfaces of object-side surfaces and image-side surfaces of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element comprise at least one inflection point thereon, the focal length of the optical imaging lens assembly is f, the focal length of the first lens element is f1, a focal length of the third lens element is f3, and the following conditions are satisfied:

$$-4.0 < f/f3 < -1.5; \text{ and}$$

$$1.0 < f/f1 < 3.0.$$

16. The optical imaging lens assembly of claim 1, wherein the axial distance between the image-side surface of the sixth lens element and the image surface is BL, the axial distance between the object-side surface of the first lens element and the image-side surface of the sixth lens element is TD, the maximum image height of the optical imaging lens assembly is ImgH, the focal length of the optical imaging lens assembly is f, the curvature radius of the object-side surface of the third lens element is R5, the curvature radius of the image-side surface of the third lens element is R6, the central thickness of the third lens element is CT3, and the following conditions are satisfied:

$$0 < BL/TD < 0.25;$$

$$0.10 < \text{ImgH}/f < 0.40; \text{ and}$$

$$-6.0 < (f/R5) - (f/R6) + (f \times CT3)/(R5 \times R6) < -2.3.$$

70

17. The optical imaging lens assembly of claim 1, wherein an axial distance between the first lens element and the second lens element is T12, an axial distance between the second lens element and the third lens element is T23, an axial distance between the third lens element and the fourth lens element is T34, an axial distance between the fourth lens element and the fifth lens element is T45, an axial distance between the fifth lens element and the sixth lens element is T56, and the following condition is satisfied:

$$2.0 < (T34+T56)/(T12+T23+T45) < 10.0.$$

18. The optical imaging lens assembly of claim 1, wherein a maximum effective radius of the object-side surface of the first lens element is Y11, a maximum effective radius of the image-side surface of the sixth lens element is Y62, and the following condition is satisfied:

$$0.70 < Y11/Y62 < 1.10.$$

19. The optical imaging lens assembly of claim 1, wherein an axial distance between the object-side surface of the first lens element and the image surface is TL, the focal length of the optical imaging lens assembly is f, and the following condition is satisfied:

$$0.70 < TL/f \leq 1.20.$$

20. The optical imaging lens assembly of claim 1, wherein the focal length of the second lens element is f2, a focal length of the third lens element is f3, a focal length of the fourth lens element is f4, a focal length of the fifth lens element is f5, a focal length of the sixth lens element is f6, and the following conditions are satisfied:

$$|f3| < |f2|;$$

$$|f3| < |f4|;$$

$$|f3| < |f5|; \text{ and}$$

$$|f3| < |f6|.$$

21. The optical imaging lens assembly of claim 1, wherein there is an air space between every two lens elements of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element that are adjacent to each other, a central thickness of the first lens element is CT1, a central thickness of the second lens element is CT2, the central thickness of the third lens element is CT3, a central thickness of the fourth lens element is CT4, a central thickness of the fifth lens element is CT5, a central thickness of the sixth lens element is CT6, CT1 is a maximum among CT1, CT2, CT3, CT4, CT5 and CT6, an axial distance between the object-side surface of the first lens element and the image surface is TL, the maximum image height of the optical imaging lens assembly is ImgH, and the following condition is satisfied:

$$2.2 < TL/\text{ImgH} < 4.0.$$

22. An image capturing device, comprising:
the optical imaging lens assembly of claim 1; and
an image sensor, wherein the image sensor is disposed on the image surface of the optical imaging lens assembly.

23. An electronic device, comprising:
the image capturing device of claim 22.

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